

Guide to the Atlantium Hydro-Optic UV System RZM Series

Model: RZM200-5

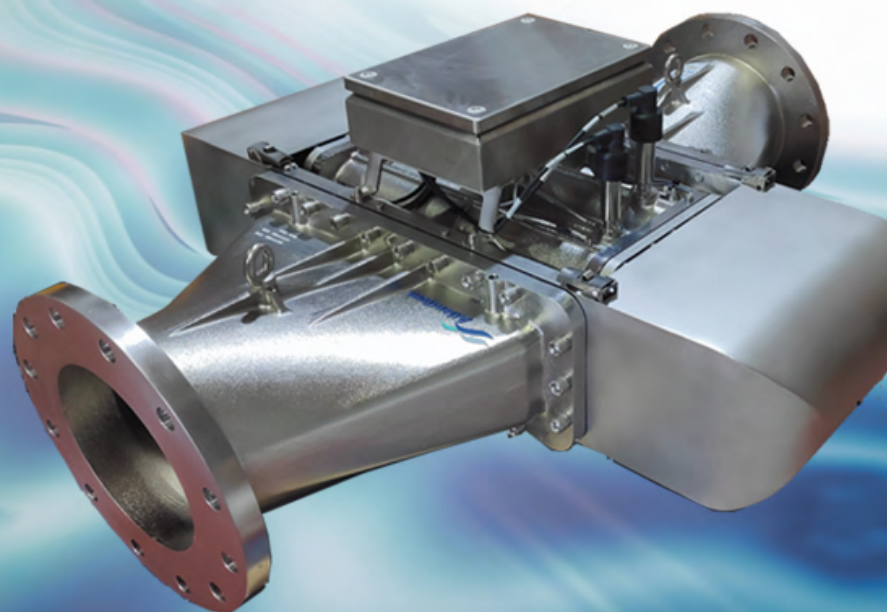
All-in-One Control Module

For General Applications

Supported Electrical Voltage:
400/440/480VAC

Document No. PM55A000E

January 2024



IMPORTANT NOTICE
It is strictly forbidden to alter/change the system hardware in any way.
Any change made without written permission from Atlantium shall void the warranty.



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Revision History

Revision #	Description	Date
001	Initial version, based on RZM350-8	Mar 2020
002	Style changes for US market. + spare parts list	Jul 2022
003	Modbus table updated	Jan 2024

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1 Introduction

Congratulations on purchasing an Atlantium **RZM200-5** Hydro-Optic UV Water Treatment System. This user guide provides you with clear guidelines for preparing the site for installation, correct installation and software configuration procedures, and a good understanding of all system functions for maximum benefit from your Atlantium Unit.

Atlantium systems are designed to provide applications for the following Industries:

- | | |
|--------------------|-----------------|
| ■ Aquaculture | ■ Bio-Pharma |
| ■ Fish Hatcheries | ■ Bottled Water |
| ■ Dairy | ■ Beverages |
| ■ Municipal | ■ Breweries |
| ■ Power Generation | ■ Aquariums |
| ■ Ultra Pure Water | ■ ...and more |
| ■ Marine | |

Patented Innovative Technology

- Features Hydro-Optic engineering that optimizes UV efficiency and enables uniform dose distribution.
- Medium Pressure High Intensity lamps provide a UV dose in the 200 to 454 nm spectrum which is much broader than that obtained from low pressure UV lamps (which yield a dose in the 254 nm range only). UV exposure is effective in cold & warm water.
- Sustainable operation provides the security and reliability that the industry demands.
- Uses "total internal reflection" of UV light within the reaction chamber, enabling more effective treatment at a lower dose.
- Some enclosures/components as noted on the Atlantium main website under <https://atlantium.com/about.html> include:
 - Nema 4x rating for indoor and outdoor use, protection against windblown dust and rain, splashing water and hose directed water.
 - IP56 rating protection from limited dust ingress. Protected from high pressure water jets from any direction

Real-time Monitoring and Control

- The sophisticated data logging system enables traceability over time of all monitored data.
- It maintains a database and provides detailed analysis and reporting functions that can be used for trend analysis or proof of compliance.

Controlled Dose Delivery

- Monitoring & control software ensures that required UV dose is delivered at all times.
- Dedicated UV Intensity Sensor (UVIS) for each lamp for real-time monitoring.
- Automatically adjusts lamp power based on real-time analyses of critical parameters such as water flow rate, lamp intensity and water Transmission to UV (UVT).
- A user-friendly interface displays the monitored data in real-time including the actual delivered dose as well as the other critical parameters, so you can verify that you are actually getting the dose that you need.

Lower Electrical Consumption

- Optimized light distribution, through Hydraulic and Fiber optic principles applied to the UV system design, enable:
- Optimized UV power and uniform dose distribution.
- Efficient use of power, with lower electrical consumption than low-pressure based UV systems.

Minimum Down Time

- True in-line system.
- Quick lamp replacement (4 minutes) and better safety.

Integrates with your Process Line

- Field proven installation.
- With no additional software, you can integrate the Atlantium system into your current process using Atlantium's customizable settings.
- Automatically generated reports at the push of a button for performance recording and traceability.

Efficiency and Easy Maintenance

- For maximum efficiency and dependable performance, the Atlantium system monitors the lamps for UV output, ensuring that a lamp is replaced only when absolutely needed.
- By following the easy preventive maintenance programs, you keep the system working at peak performance.
- The **Viewport** allows you to safely peek at what's going on inside.

For further information about Atlantium products and technologies, visit our website at <https://atlantium.com/>

1.1 Before You Begin

Before you begin using this product, or any installation or service operation, please read the following safety information:

- Attention to these warnings helps prevent personal injuries and damage.
- It is your responsibility to use the product in an appropriate manner.
- This product is designed for use solely indoors.
- You are responsible if the product is used for other than its designated purpose or in disregard of Atlantium instructions. Atlantium shall assume no responsibility in any way for such use of the product.
- The product must be used for its design purpose, based on its product documentation and within its performance limits.
- Using the product requires technical skills. It is therefore essential that only skilled and specialized staff or thoroughly trained personnel be allowed to use the product.
- Keep the basic safety instructions and the product documentation in a safe place and pass them on to the users.

- Applicable local or national safety regulations and rules for the prevention of accidents must be observed at all times.

1.2 Tags and Their Meaning

The following indicators are used in the product documentation to warn the reader about risks and dangers



DANGER!

Indicates a hazardous situation which, if not avoided, will result in death or serious injury.



WARNING!

Indicates a hazardous situation which, if not avoided, could result in death or serious injury.



CAUTION!

Indicates a hazardous situation which, if not avoided, could result in minor or moderate injury.



ATTENTION!

Indicates the possibility of incorrect operation which can result in damage to the product.



Indicates a hazardous situation involving electricity which, if not avoided, can result in death or serious injury.



Indicates a hazardous situation involving Electrostatic Discharge (ESD), which, if not avoided, can result in damage to the product.



Indicates a hazardous situation involving UV light which, if not avoided, can result in death or serious injury.



Indicates a hazardous situation involving a hot surface which, if not avoided, can result in death or serious injury.



Indicates a hazardous situation involving working in an enclosure with moving parts, which if not avoided, can result in serious injury to the hands.



Indicates a hazardous situation involving mercury, a poisonous material which, if not avoided, can result in death or serious injury.



Indicates a hazardous situation involving caustic chemicals which, if not avoided, can result in death or serious damage.



Indicates that components or equipment are heavy and care is to be taken to avoid lifting incorrectly. Incorrect lifting can be dangerous to the personnel lifting and may result in dropping and damaging the components or equipment.



Indicates that components or materials to be discarded are classified as hazardous waste and must be disposed of appropriately.



Indicates information related to installation, safety or use of system.



2 About the Atlantium RZM200-5 Systems



Information contained in this User Manual is for illustration purposes only. For updated specs of your particular system, **ALWAYS** refer to the accompanying Mechanical and Electrical ICD drawings.

2.1 The RZM200-5 Unit

This Atlantium **RZM200-5** System includes a chamber housing five Medium Pressure High Intensity (MPHI) UV lamps, an electrical cabinet housing ballasts for each of the lamps, and a **Control Module**. Optimized light distribution reflects UV light back into the water to uniformly distribute the UV dose. Water flows unimpeded, resulting in low head-loss.

Each specially designed short (Arc length: 400 mm /15.74 inch) Quick Connect MPHI lamp is housed within a thick quartz tube that significantly reduces risk of breakage and its consequences. A UV Intensity Sensor (UVIS) provides continuous measurement of each lamp's output. In addition, a UVT Analyzer reports and tracks the water quality.

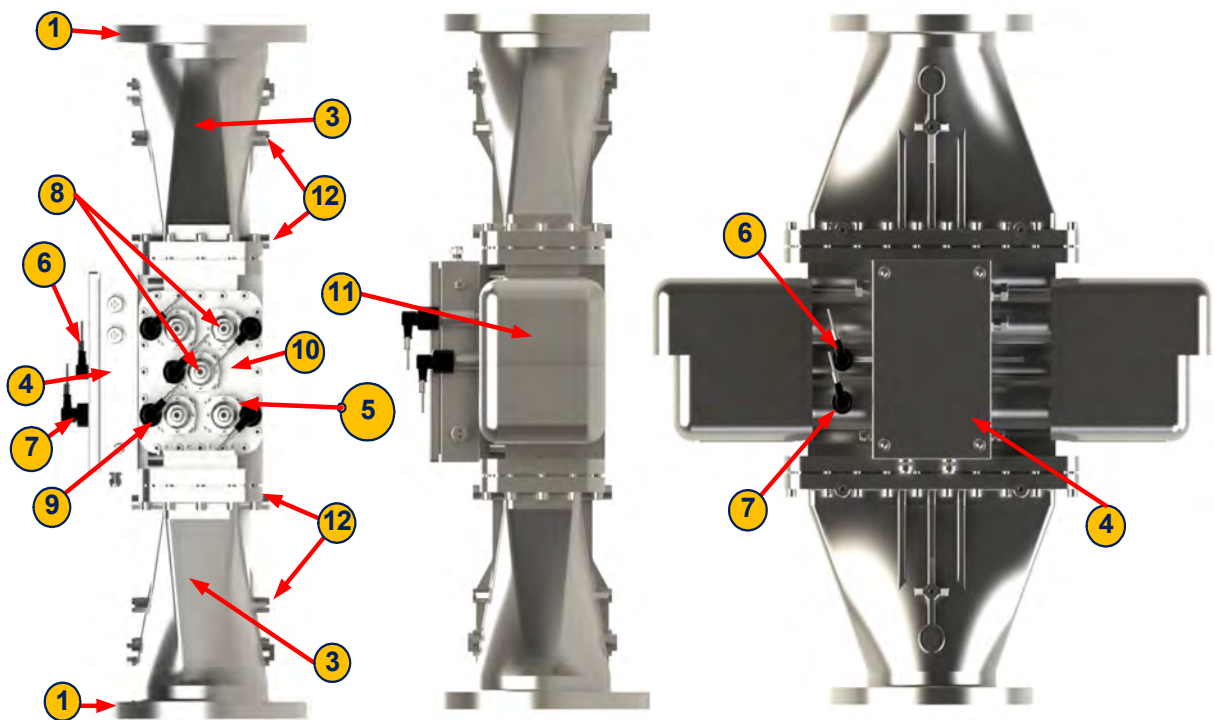


Figure 2-1: UV-1 **RZM200-5** UV Unit

Table 2-1: Atlantium **RZM200-5** UV Unit Diagram Key

#	Item	Description
1	Flange interface	The unit connects to the piping via a flange interface
2	Lifting Support	A lifting ring used when lifting the Unit.
3	UV Chamber	The chamber in which the UV exposure occurs
4	Connection Box	The box that sits on top of the lamp chamber and holds the I/O card for the sensors and the lamps.
5	Lamp Chamber	The chamber that holds the lamps
6 7	UVTS Sensor	The sensor that measures the lamp ultraviolet intensity through the water. Together with the UVIS sensor and the Main Control Application, it analyzes, tracks and reports the UV transmittance.
8	Lamps Arc length (mm / inch):	Medium Pressure High Intensity (MPHI) Lamps, each housed in its quartz sleeve assembly 400 / 15.74
9	UVIS	The sensor located next to each lamp. They measure the UV intensity of each individual lamp and feeds the real time dose calculation. On each side, a cover is over the group of sensors during regular operation.
10	Temperature	Measures the temperature of the lamp connectors.
11	Cover	The cover, which contains a cooling fan, protects the lamps and UV Intensity Sensors. There is an identical cover on the other side of the unit.
12	Support Mounts	Mounts that are used to support and balance a unit anchored to customer supplied supports or surface that hold up the unit (see UV Unit Interface Control Drawing accompanying this manual).

2.2 RZM200-5 Interface Control Drawings

For the ICDs of the your **RZM200-5** system, consult the accompanying ICD files.

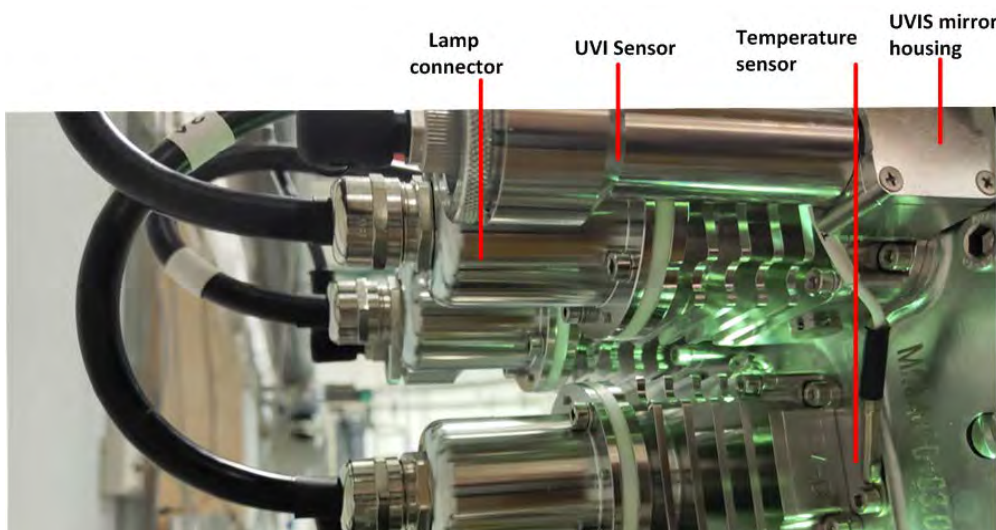


Figure 2-2: Lamp connectors, UVIS and temperature sensor

2.3 Atlantium System Physical Specifications

The Atlantium system has a small, compact Physical Specifications footprint that facilitates in-line installation in any process train.

Table 2-2: **RZM200-5** UV Unit Physical Specifications

Item	Detail*
Lamps	5 x 400mm 6.9kw long neck
Atlantium Unit Construction Materials: Casing:	ASTM A 890/A890M, 5A (CE3MN)
Standard Pipe Flange Interface Options:	DN350 DIN 2576
Unit Length (mm / inch]	1040/40.94
Flange Inner Ø Diameter (mm / inch]	200/7.87
Flange Outer Ø Diameter (mm / inch]	340/13.39
Width at Widest Point (mm / inch]	956/37.64
Bolt Circle Diameter (mm / inch]	295/11.61
Bolt Hole Diameter (mm / inch]	22 / 0.87
Number of Bolts	12
Unit Volume (Liter / Gallon)	21/5.5
Weight (without water) (Kg / lb)	177/390

Item	Detail*
Installation position	Vertical ONLY , with upward water flow
Service Clearance on each side (mm / inch) (Does not include human access space)	1550 / 61
Minimum ventilation space from wall (mm / inch) Minimum Height above Floor Level (mm / inch)	300 / 11/81 According to piping limitations
Maximum Working Pressure	7 Bars / 102 psi
Maximum Flow Rate	As per the Type Approval

**Values appearing in this table are for reference only and subject to change.*

2.4 Electrical Usage and Power Consumption

The table below detail the Atlantium Unit's electrical usage and power consumption.

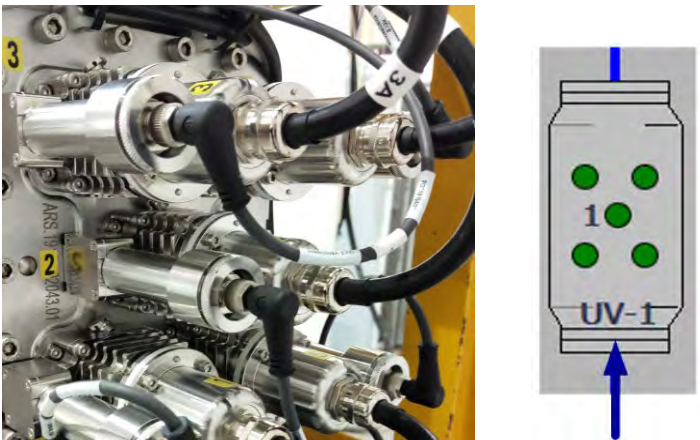
Table 2-3: **RZM200-5** power supply requirement

Voltage	Current
3 phase 400/440VAC 50/60Hz	55/50A
1 phase 220VAC	0.2A
24VDC	1A

2.4.1 MPHI LAMPS

Atlantium provides five specially-designed Quick Connect Medium Pressure High Intensity (MPHI) lamps that deliver a broad spectrum UV output. The thick quartz sleeve lamp housings significantly reduce the risk of breakage and its consequences.

Table 2-5: Lamp Details

Item	Description
5 Lamps Arc length (mm / inch):	Lamp encased in quartz sleeve assembly Medium Pressure High Intensity (MPHI) 400 /15.74
UV Lamp Number Labels on Actual Lamps and on Lamp Diagram in the PLC	 Figure 2-3: UV Lamp Number Labels on Actual Lamps and on Lamp Diagram in the PLC

2.4.2 UVIS Sensor

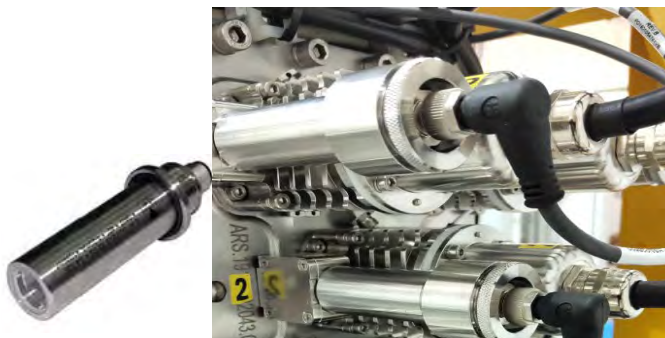
A UVIS Sensor monitors each light source (lamp). Atlantium uses a germicidal sensor that has been validated by German drinking water authorities according to DVGW Standard W294. The operating range of the sensor is 0–10 W/m² which sends a current of 4 -20 mA respectively. Operating range and type are labeled permanently on the sensor.

The lamp monitoring sensor is located in the dry lamp chamber so that the UV light intensity is unattenuated and the sensor can measure the UV accurately over time with minimal problems of drift or degradation.

Electronics are integrated in the sensors and are pre-calibrated. Each sensor will have its own dated QA verification certificate, serial number and lot number (3.1.B according to EN 10204) which is used to track it through its lifetime use and calibration.

For each lamp replacement, its associated UVIS Sensor should be re calibrated. In addition, all UVIS Sensors should be calibrated yearly against a Master Reference Sensor.

Table 2-6: UVIS Sensor Details

Detail	Description
Atlantium Part Number	OPB000100
Appearance and Position	 Figure 2-4: UVIS Sensors
Environmental Requirements	The sensor is connected via a sensor plug connector with IP56 protection inserted status.
Output Signal	4-20mA
Measuring Range	0 – 10 W/m ²

2.4.3 UVT Sensors

The UVIC Sensor tracks the UV transmittance through the waters of the UV Chamber. Atlantium uses a germicidal sensor that has been validated by German Drinking water authorities according to DVGW Standard W294. The monitoring sensor's operating range 0–10 W/m² and its output current to the PLC is 4 -20 mA. Operating range and type are labeled permanently on it.

Refer to the Reference Document: DOC000095.

Table 2-7: UVTS/UVIC Sensor Details

Detail	Description
Atlantium Part Number	OPB000100
Appearance and Position	 UVT Sensor
Environmental Requirements	The sensor is connected via a sensor plug connector with IP56 protection inserted status.
Output Signal	4-20mA
Measuring Range	0 – 10 W/m ²

2.4.4 RZM I/O Card

The RZM I/O card resides inside of the connection box, which is located over the lamp chamber of the **RZM200-5** UV Unit. See the table below for the list of connections on the card.

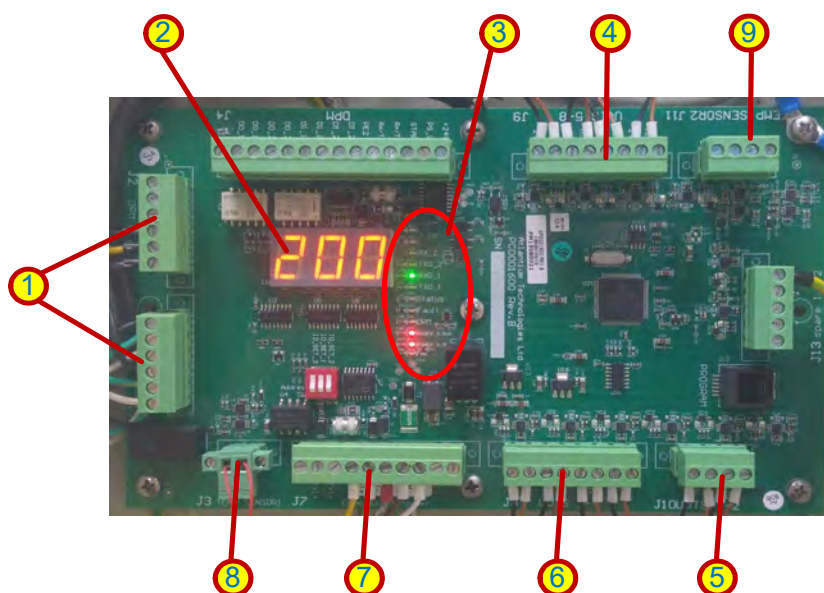


Figure 2-5: RZM I/O Card

Table 2-8: RZM I/O Card details

Item		Details
Atlantium Part Number:		AP0001600
1	Fans connectors	
2	Address display	
3	Indication LEDs:	
	RX_2	NA
	TXD_2	NA
	RX_1	Communication RX LED
	TXD_1	Communication TX LED
	Status	(System off / ignition / ready
	Fault	TS1
	DPM (NA)	TS2
	Power	Internal PS
	Power	Internal PS
	Fan	NA
4	Power + Communication Connector	
5	UVIS 5 Connector	
6	UVTS 1-2 Connector	
7	UVIS 1-4 Connector	
8 & 9	Temperature sensor connectors	

2.4.5 Ensuring Power Stability

The voltage of the electrical supply connected to every Atlantium system **must not vary** by more than $\pm 10\%$ from the correct value (sag, swell and surge) at the system inlet.

To comply with this requirement, you must protect the Atlantium system with a device that prevents electrical surges and fluctuations from causing damage. There are two types of surge protection:

- UPS (Uninterrupted Power Supply) has an internal battery that allows the equipment plugged into it to continue to run in the event of a power outage until power is restored. It also has the benefit of performing the same functionality of surge protectors, protecting against spikes and surges. Make sure to use a UPS that is capable of supplying sufficient current to the system and that the cable length is not more than 10m (33ft) in length. If needed, the UPS allows the system to be shut down cleanly if power is out for a lengthy period.
- Surge protection device protects your equipment from variations in electrical current, such as surges and spikes.



Surge protector devices /UPS must be inspected periodically and replaced if their protection capacity has expired.

2.4.6 Surge Protector Devices

Surge protection devices are designed to protect industrial communication networks. This device recommended by Atlantium uses a combination of 3-electrode gas discharge tubes and fast clamping diodes. Typical applications include industrial processing equipment, transmission systems, I/O cards, probes, actuators, and displays. Surge protection devices are available via Atlantium.

Table 2-9: Surge Protector Details

Atlantium PN	DATA SHEETS	Module PN	Description
EP0002300	DOC002300	8859660000	SURGE PROTECTOR 400V 3P 40KA TNC CONFIG
EP0002000	DOC002000	4983-DS277-404	SURGE PROTECTOR 440-480 3P 40KA
EP0002600	DOC002600	4983-DS120-402	SURGE PROTECTOR L + N, 115V
EP0002800	DOC002800	1352740000	SURGE PROTECTOR L + N, 230V

2.5 The Electrical Cabinet (EC)

For the **RZM200-5** system, an Electrical Cabinet (EC) is required to supply power to the lamps and other equipment.

For the ICD of the Electrical cabinet in the **RZM200-5** (PN SAM000050), consult the accompanying ICD files.

Item/Parameter	Details*
Dimensions (mm/inch):	
Width:	811/31.9
Depth:	536/21.1
Height with Lifting Element:	2064/82.3
Height without Lifting Element:	2004/78.9
Weight (Kg/lb):	250/551
Minimum radius for Opening Door (mm/inch):	105° 795/31.3
Clearance Radius:	
Side Clearance:	200/7.9
Protection Standard	IP 54

**Values appearing in this table are for reference only and subject to change.*

2.6 The Control Module

The Atlantium system is monitored and controlled from a **Control Module**. It can be mounted on a wall or suitable anchoring point up to 50 meters (164.04 ft.) from the Atlantium Unit. However, it should be located in close proximity to the Atlantium Unit in a place that it is easy to reach the touch screen and read the monitoring information.

For the ICD of the **Control Module** in the **RZM200-5**, consult the accompanying ICD files.

Table 2-11: Control Module Details

Physical	Details
Dimensions (W x D x H) (mm/inch):	400/15.75 x 300/11.82 x 144.4/6.12
Weight (Kg/Lb.):	9.8/21.6
Minimum Clearance for Electrical Cables Routing (mm/inch):	50/2
Electrical	
Maximum Power Consumption:	Maximum 60W
Input Voltage:	88 ~ 264VAC
Input Current:	1.4A/100-240VAC, 0.8A/230VAC
Frequency:	47 ~ 63 Hz
Protection Category:	IP 56.

2.7 Environmental Requirements

The Atlantium system should be situated indoors in an area that is adequately cooled and with sufficient air flow. Consult your Atlantium representative to choose the most suitable location.

** For water temperature above 70°C (160°F) consult with Atlantium service engineers*

Table 2-12: **RZM200-5** System Environmental Details

Environmental	Details
Maximum ambient air temperature:	45°C (113°F)
Maximum ambient air temperature in the Ballast Modules environment:	45°C (113°F)
Maximum atmospheric relative humidity:	90%
Water Temperature*: Maximum Water Temperature (lamps off) Maximum Water Temperature (lamps on) Minimum – cold temperature:	90°C (194°F)*90°C (194°F)* 60°C (140°F)* No limit - if the unit is fully drained of water before it freezes

** Any deviation from either of these values require consultation with Atlantium applications department and written approval by Atlantium.*

2.8 Regulatory Compliance

The Atlantium System complies with the following regulatory standards*:

- EMC directive 2014/30/EU
- European Low Voltage Directive (LVD), 2014/35/EU for electrical safety
- Council Directive 98 / 83 / EC of 3 November 1998 for the quality of water intended for human consumption
- ISO 9001-2015 2018-2021 Quality Management Standard - Development, Design, Production and Sales of Water Disinfection Systems
- UL 508, CSA C22.2 No, 14-13. By MET Labs (eurofins) NRTL certification
- IP54/IP56 & TYPE4X
- NSF 61. By NSF International.
- Hygienic certificate, National Institute of Public Health, Poland

**Note: Compliance is model dependent, for more details of specific UV Unit model. Consult your Atlantium representative.*

3 Getting Ready for Installation

Using the guidelines and information in this section, you can prepare the piping and electrical setup for installing the Atlantium Hydro Optic Water Treatment System. If you have any questions, consult with an Atlantium application engineer.

3.1 Planning

Consider where to locate the unit from the perspective of your process. Prepare your installation plan according to the ICD mechanical and electrical drawings provided with the system, the planning points below, as well as the other guidelines in this manual.



For preparing all electrical connections and mechanical installation, be sure to review the electrical and mechanical ICD drawings provided to you by Atlantium with this manual. If no electrical drawings were provided please contact your Atlantium representative.

3.1.1 Access

■ Installation Access

- Take into account that sufficient door space is required to transport the Atlantium Unit into the facility where it is to be placed.
- The Atlantium units are packed and transported in large wooden crates. Consider the need for a place near the installation area for unpacking of the units from the crates, prior to installation.

■ Service Access

- Ensure that sufficient space is required for maintenance personal to access the Atlantium Unit for periodic component replacement and maintenance.
- The installed units must also be accessible: for example if a unit is installed high above the ground, permanent stairs or walkways may be needed for convenient and safe service access
- Ensure sufficient space to access the back of the unit, to allow removal and replacement of lamps as well as to do maintenance on cleaning systems (where these are installed), connection boxes, etc.

■ Installation: vertical or horizontal

- Vertical installation of the Atlantium unit and piping, with an upwards direction of flow, is the preferred configuration for installation, where this is possible.
- If mounting the unit in a horizontal configuration, a slope of at least 5 % is recommended to minimize the risk of air pockets developing or remaining trapped in the Atlantium unit. The Customer is to supply the mechanical means for ensuring this slope.

3.1.2 Hydraulics

■ Piping

- Align the upstream and downstream piping with the designated location for the Atlantium unit.

- Prepare the flanges/ferrules and piping as specified according to *The Piping Infrastructure* on page 33. To avoid a step in the connection, make sure that the internal diameters where the Atlantium Unit and your piping match.
- **Do not** use sealing material that contains graphite when sealing the flanges/ferrules and gaskets as it accelerates the corrosive process on stainless-steel components, including those of the Atlantium system.
- To promote laminar flow of water through the Atlantium unit:
 - ◆ The piping connected to either end of the unit should be of the same diameter as the unit itself. Connection of narrower piping may reduce flow and increase water temperature and turbulence.
 - ◆ Allow a straight length of 10 pipe diameters (10D) before the Atlantium unit (upstream) and a straight length of 5 pipe diameters (5D) after the unit (downstream). In situations where less space is available, review with your Atlantium representative.
 - ◆ In order to minimize turbulence and air pocket formation in a vertical installation, install a concentric reducer/expander adapter to streamline the transition between piping dimension and the internal dimension of the unit.
 - ◆ For a horizontal installation, install an eccentric reducer/expander adapters, with downstream outlet higher than inlet.
- **Mechanics.**
 - Prepare the supports needed for the Atlantium Unit, ensuring that the minimum height above the floor is 750mm / 29.5 inch (between the floor and the bottom of the provided triangle support brackets).
 - Check that the piping on the inlet and outlet sides has adequate support structures so that there are no mechanical stresses that may endanger the Unit.
 - Check the upstream and downstream piping for vibrations. Make sure that the piping to be attached to the Atlantium Unit is anchored and reinforced to protect the Unit against vibration. A high vibration level can cause damage to any hydraulic system, including the Atlantium unit. High vibration can be caused by an unbalanced pump or by sudden valve close.
- **Hydrodynamics**
 - Water hammer can cause damage to any hydraulic system, including the Atlantium Unit. Make sure that this phenomenon does not occur in your facility's water line. To prevent water hammer, consult your system engineer and check your facility system procedures, including:
 - ◆ When starting the water flow when the lines are empty, keep valves open so that the air in the pipes has a simple release pathway.
 - ◆ Sudden valve closures can cause water hammer. Make sure that valves close gradually, enough to avoid this problem.
 - ◆ Ensure that all pumps that impact the Atlantium system employ soft start-up procedures.
 - ◆ The Atlantium Unit must be totally filled with water. Assure that no air can be trapped in the unit and that air can be released. Verify through the Unit's viewport after installation that there are no bubbles.

3.1.3 Electricity

- Evaluate the quality of your local mains power supply and the likelihood of blackout/brownouts, electrical surges or interruptions to determine if you need power surge protection and/or uninterrupted power supply.

- For systems utilizing saltwater, the Atlantium Unit must be grounded to minimize the corrosion potential of the treated water. Be sure to prepare a sufficient ground point next to the designated location for the Atlantium Unit.
- To verify that you order the correct lengths for cables (i.e., lamp cables, power cables, data cables, etc.), measure the distance between the proposed location of the relevant Atlantium components and between them and the Power source.
- Prepare cables of sufficient length to connect from the auxiliary equipment (i.e., the flow meter/flow switch, etc.) to the Atlantium system.

3.1.4 Flow Data

- Evaluate your flow trends to determine whether you require a flow meter or flow switch. If the flow varies, a **flow meter** is required to detect and report on the flow coming into the unit. Make sure that there are to be **no** branches on or off the pipes between the flow meter and the Unit inlet. If the flow rate is static, a **flow switch** can be used instead.

3.1.5 Application

- Verify that the dose-related parameters, such as the UVT and the flow, are the same as those defined in the Atlantium application scope of supply.
- Determine what a low flow situation means to your process and whether you need to shut down the Unit (low flow indicates upstream problems) or recirculate (low flow normal) when the flow goes below the normal range.

3.1.6 Feed Water Quality

If the water entering the Atlantium Unit contains debris or particles in suspension, pre-treatment (e.g. filtration) may be required to maximize the effectiveness of UV exposure and protect internal parts from blockage or damage. Consult with your Atlantium representative.

3.2 Typical Installation Set-Up

Use the diagram below as a basis for your installation drawings. Vertical installation with upward flow is preferred where it is possible and accessible, as it prevents the accumulation of air pockets. If horizontal installation is required, be sure to include a 5 degree slope (recommended/minimum) in the direction of flow to reduce the potential for air pockets. Be sure to include service access as you fit the Atlantium system into your facility (i.e., nearby pipes, electrical fixtures, support beams, etc.)

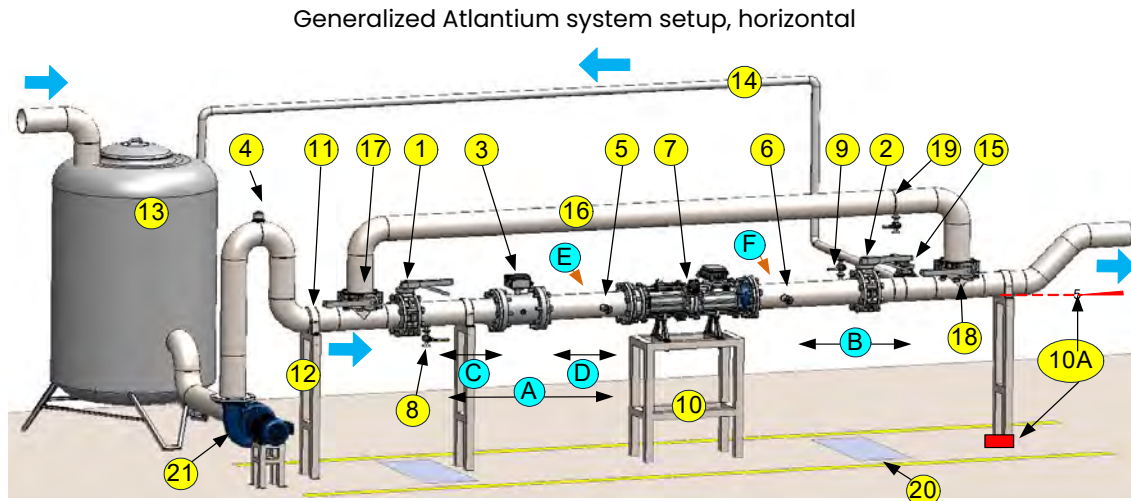


Figure 3-1: Generalized Atlantium system setup, horizontal (with bypass)

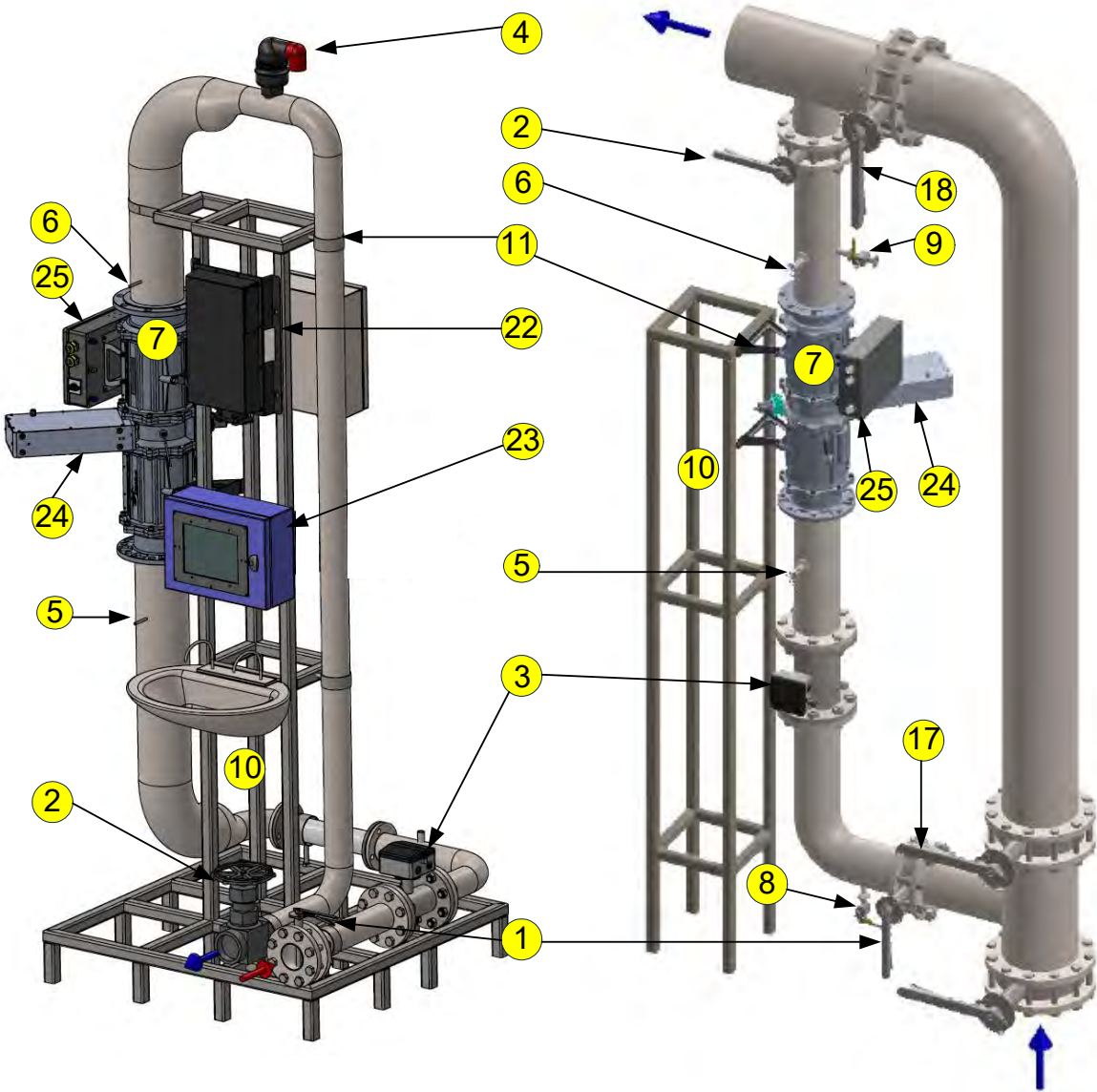


Figure 3-2: Generalized Atlantium system setups, vertical - : with bypass

Table 3-1: Typical Atlantium System Component Setup Key

#	Description		Atlantium P/N or recommended and available at Atlantium
	Direction of water flow in the pipes		-
1	Inlet isolation Valve ■ Isolating Valve on the Inlet side		-
2	Outlet isolation Valve ■ Isolating Valve on the Outlet side		-
3	■ Flow Meter flow measurement device with 4-20mA analogue output signal ■ To be installed on the same line as the Atlantium system, with no branching before or after the flow meter ■ Follow the manufacturer's Installation instructions		Available Flow Meters: KROHNE OPTIFLUX 6000 KROHNE OPTIFLUX 2300
4	Air Release valve – To prevent air bubbles where necessary ■ Automatic Air Release valve is used to release air bubbles trapped within the line ■ Install at a high position to allow air bubbles to accumulate and release from this point		-
5/6	Inlet/Outlet Sampling Points ■ Position aseptic sampling valves horizontally (90°) on the pipe center, within 50cm of the Unit ■ Aseptic sampling valve must be a dedicated valve for sampling only and resistant to alcohol and flame		■ Aseptic Sampling Valve Kit for Stainless Steel Pipe – KTB005700 ■ Aseptic Sampling Valve Kit for Plastic Pipe – KTB0011500
7	Generic Atlantium Unit		-

Table 3-1: Typical Atlantium System Component Setup Key

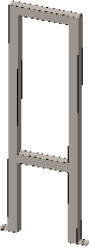
#	Description		Atlantium P/N or recommended and available at Atlantium
8	Draining/CIP Port - For performing CIP on the Atlantium Unit - The Draining port should be positioned close to the Inlet valve and between that and the Unit, on the bottom of the pipe - It is also used as one of the CIP ports and connects to a hose	 	- CIP Kit - SAB012900 - CIP Valve Kit for Stainless Steel - KT0011600 - CIP Valve Kit for Plastic - KT0011700 (the valve itself is not part of the kits) See CIP Ports and CIP Kits on page 34, for information on CIP equipment.
9	CIP Port - For performing CIP on the Atlantium Unit The CIP port should be positioned between the Unit and the Outlet valve, on top of the pipe, at the highest point before the isolation valve, in order to allow the pipes to be completely filled with solution		
10	System Supports Supports are required to assure no mechanical stress on the Atlantium Unit		-
10A	Shim, block or higher downstream support, to achieve 5° slope in horizontal installation		supplied by customer
11	Support brackets - Brackets are used to hold the Atlantium Unit and pipes attached to the support mounts - Is to include internal flexible material to absorb vibrations and assure that the pipes and Atlantium Unit are mounted tightly and securely		-
12	Pipe Supports Pipe Supports are required to assure no vibrations in the pipeline		-
13	Water tank		-
14	Recirculation Line >5m ³ /h (>20gpm (Optional)) A Recirculation line is highly recommended to keep water flowing constantly.		-

Table 3-1: Typical Atlantium System Component Setup Key

#	Description		Atlantium P/N or recommended and available at Atlantium
15	Recirculation Line Isolation valve (Optional)		-
16	By-Pass (Optional) ■ If the facility uses quartz corrosive CIP chemicals, such as caustic soda, use a By-Pass line to prevent damage to the Atlantium Unit		-
17	By-Pass Inlet isolation Valve (Optional)		-
18	By-Pass Outlet isolation Valve (Optional)		-
19	By-pass draining port (Optional) ■ The Draining port should be positioned on the bypass line, at the lowest point of the pipe		-
20	Drain ■ It is recommended to install a drain in the floor, close to the draining valve		-
22	Ballast Module (certain Atlantium systems)		
23	Control Module		
24	DPM cleaning system (certain Atlantium systems)		
25	Connection Box		

Table 3-2: Typical Atlantium System Distances Setup Key

#	Description
A	Position the Inlet isolation valve with a minimum 10D (3000mm/118inch) of straight horizontal pipe between it and the Atlantium Unit. (For sites with space constraints, consult an Atlantium Application Engineer to find appropriate solutions.)
B	Position the Outlet isolation valve with a minimum 5D (1500mm/59inch) of straight horizontal pipe between it and the Atlantium Unit. (For sites with space constraints, consult an Atlantium Application Engineer to find appropriate solutions.)
C	Generally, you should position the Flow Meter or Flow Switch with a minimum 5D (1500mm/59inch) of straight horizontal pipe immediately before it on the Inlet side and a minimum 2D (600mm/24inch) of straight horizontal pipe between it and the Atlantium Unit. Refer to the manufacturer's instructions.
D	Position the Inlet Sampling valve at a maximum distance of 0.5 meters/1.5ft from the Atlantium unit on the inlet side.
E	Position the Outlet Sampling valve at a maximum distance of 0.5 meters/1.5ft from the Atlantium unit on the outlet side.

3.3 Electrical Requirements

The electrical infrastructure is to be installed and in place prior to the system's delivery.

For information on the footprint of the components of the Atlantium system, see the section, . Use the Electrical ICD provided by Atlantium to help you plan the electrical infrastructure. See also **Power, Voltage and Grounding Requirements** on page 18 and **Connecting the Ballast Module Cables** on page 60.

Take into account:

For locations with 440Y/254 VAC 3~ +PE 50/60HZ
or 480Y/277 VAC 3~ +PE 50/60HZ



The Atlantium System requires a balanced voltage supply (WYE configuration). If you do not have such a connection, you must create an appropriate power feeding using a 1:1 isolation transformer (where the ground is connected to the middle point of the star configuration at the secondary side of the transformer).

- Evaluate the quality of your power and the likelihood of blackout/brownouts, electrical surges or interruptions. Determine if you need power surge protection and or uninterrupted power supply if any.
- To verify that you order the correct lengths for cables (i.e., Cable Harness, power cables, data cable, etc.), measure the distance between the proposed location of the relevant Atlantium components and between them and the Power source.
 - Planning the placement of the Ballast Modules, **Control Module** and the path of the various cables must take into consideration the length of the **Cable Harness** and the other cables. Cables must be placed in a path that does not interfere with passage ways.
 - The power mains connections must be in the vicinity of the Ballast Modules, Control Module.
 - The power cable and connecting plug are to be obtained locally.

- An extra-long, 10M (32' 9.7") cable is available for the RS104 system only.
- If your region experience periodic power outages, consider including uninterrupted surge protection devices. Plan to include surge protection and possibly an Over Voltage Protection Device (OVPD) (available from Atlantium) if needed. See Surge Protection on page 4-7.
- Prepare cables of sufficient length to connect from the auxiliary equipment (i.e., the flow meter/flow switch, etc.) to the Atlantium system.
- See *The Piping Infrastructure* on page 33.

3.3.1 Preparing for the Electrical Wiring to the Mains

You must prepare the elements listed below for connecting the Atlantium system to the mains.



For preparing all electrical connections, be sure to use the *Electrical ICD* drawing provided to you by Atlantium with this manual. If no electrical drawings were provided please contact your Atlantium representative.

- Mains system area circuit breaker according to the total systems' power consumption to be connected to this line. See *Electrical Usage* on page 17.
- For sites with 440/480VAC, plan to include an Over Voltage Protection Device (OVPD) (available from Atlantium) if needed. See on page 15.
- Install the Required Circuit Breakers, as defined in the Electrical ICD.
- If you intend to lengthen or replace the mains power cable connected to the Atlantium Unit with a longer cable, prepare the cables according to your location electrical distribution system requirements.
- For locations with 400 VAC: 5-wire system that contains 3 phases, Neutral and Ground (The Neutral and Ground are mandatory)
 - For locations with 440Y/254 VAC 3~ +PE 50/60HZ
or 480Y/277 VAC 3~ +PE 50/60HZ

The Atlantium System requires a balanced voltage supply (WYE configuration). If you do not have such a connection, you must create an appropriate power feeding using a 1:1 isolation transformer (where the ground is connected to the middle point of the star configuration at the secondary side of the transformer).

3.3.2 The Control Module's Mains Cable

- For sites with 400VAC 3 phase + Neutral, the Control Module's Mains feeding voltage is 230 VAC (L1 to N or L2 to N or L3 to N).
- For sites with 440/480VAC 3 phase, the Control Module **cannot** be fed from one phase of the 3 phase supply. The Control Module **must** be connected to a separate mains feeding, which is in the voltage range of 100-240 VAC.

Voltage option are:

- 1 phase x 120 - 240VAC +N +PE
- 2 phase x 120 - 240VAC +PE

3.4 The Piping Infrastructure

To review how to prepare the infrastructure, read through the typical installation section before you begin and refer to the section, *About the Atlantium RZM200-5 Systems* on page 11. Install the piping infrastructure before the Atlantium system arrives, but to avoid built-in mechanical stresses perform the final pipe adaptations and welds when the unit is actually situated in its permanent position.



- When sealing the flanges/ferrules and gaskets, **do not** use sealing material that contains graphite as it accelerates the corrosive process on stainless-steel components, including those of the Atlantium system.
- Be sure to align the piping on the inlet and outlet sides of the designated location for the Atlantium Unit with the unit to avoid mechanical stress on the Unit.
- Be sure to secure the piping around the designated location for the Atlantium Unit so that there is no vibration.
- Be sure to position adequate supports on the piping line needed to hold up and stabilize the process piping on the inlet and outlet side of the proposed location of the Atlantium unit.
- Determination of support requirements and design should be made by an on-site mechanical engineer.

3.5 Flow Measurement/Detection

The Atlantium unit requires a flow signal to calculate the dose and operate properly. The flow signal is a measurement of the actual flow or simply a flow/no-flow indicator. The flow signal is collected from either a flow meter or a flow switch.

- A flow meter is required in facilities where the flow rate is variable for any reason.
- A Flow switch can be used in facilities where the flow is at a fixed, steady rate.

The Flow meter or flow switch does not necessarily have to be installed in close proximity to the Atlantium Unit. It may be more remotely positioned, provided that it is located on the inlet side between it and the Unit and there is **no** branching or splits of the water flow.



The Atlantium system cannot be operated when there is no water or no water flow. Any damage caused due to operating the system without water or without water flow invalidates the Atlantium warranty.

Flow Meter

In most cases, a flow meter with a 4-20mA output is required to measure the flow and automatically adjust the output of the UV system. The flow meter must be able to measure the expected maximum water flow of the specified Atlantium unit. The power source for the flow meter must come from an external source according to the manufacturer's instructions.



Be sure to provide the power source for the flow meter as part of the preparations for installing the Atlantium Unit.

Install the flow meter in strict accordance with the instructions supplied by its manufacturer.

Flow meters are available from Atlantium, such as:

- **KROHNE OPTIFLUX 6000** flow meter for high-purity water applications
- **KROHNE OPTIFLUX 2300** flow meter – OPTIFLUX 2000 flow meter, includes **KROHNE IFC 300** signal converter

Consult your Atlantium representative.



Figure 3-3: KROHNE OPTIFLUX Flow Meters

Flow Switch

The flow switch tells the Atlantium Unit when water is or is not flowing. The recommended flow switch is Dry Contact with minimum flow sensing of 0.2m³/hr (1gal/min).

Install it in a vertical (12:00 o'clock) orientation on top of the pipe. Follow the manufacturer's instructions.



Figure 3-4: Examples of Flow Switch

3.6 Sampling Valves

If you need samples from before or after the Atlantium unit, position the sampling valves on the inlet and outlet sides at a maximum of 0.5meters/1.64ft. of the Unit. See **Planning** on page 23. for more information.

Valves should be able to be internally treated with alcohol as well as treated with flame. See **Taking Microbial Samples** on page 105. for more information. The Sampling valve should be located indoors in clean environment without away from direct contact with fan or A/C air flow.

- **For Stainless Steel. Pipes:**
 - Weld the 1 inch clamp fitting at a horizontal orientation (3 or 9 o'clock).
 - Perform a polishing process to achieve a smooth area on the external and internal part of the pipes.
 - Passivate the pipes to avoid corrosion.
- **For Plastic Pipes:**
 - Thread the clamp fitting to the port.
- On the inlet and outlet sides of the water line, connect the aseptic sampling valves, using the supplied 1" tri-clamp fitting, to any 1 inch tri-clamp fitted port.

3.7 Sampling Valve Components and Kits

Sampling valve kits are available from Atlantium. The table below details them. For information on installing the Sampling valves, see **Installing the Inlet/Outlet Sampling Points** on page 48.

One Accessory kit is required for each Sampling point.

Table 3-3: Sampling Valve Components and Kits

Description	Image	Catalog number	Kit for Stainless Steel Pipe KTB005700	Kit for Plastic Pipe KT0011500
Long welding ferrule – for welding the tri-clamp port to the pipe		FAC003700	X	-
Threaded Male Adapter 1/2" BSP – for connecting the tri clamp port to the plastic pipe		FAB008100	-	X
Aseptic sampling valve		COB000500	X	X
Gasket 1" – for sealing the valve connected to the pipe		FA0000046	X	X
Clamp 1.5" – for connecting the valve to the welded port		FA0000045	X	X
Gasket 1/2" – for sealing the valve's flushing port		FA0000048	X	X
Clamp 1/2" – for sealing the flushing port of the valve		FA0000047	X	X
End cap 1/2" – for covering the valve's flushing port		FA0000049	X	X
Round brush – for cleaning the valve during the CIP process		FA0008300	X	X

3.8 Draining Valves

It is important to plan for draining the system. Install one draining valve at the lowest point of the pipeline, before or after the Atlantium unit. If you plan to install CIP ports, one of them can double as a draining valve. If there is a Bypass line used occasionally to bypass the Unit in any circumstance, add an additional draining valve to ensure that the bypass line does become a potential source of standing water. The bypass line must be empty when not in use.

3.9 CIP Ports and CIP Kits

The CIP (Cleaning-In-Place) recirculation system is used for the periodic cleaning and sterilization process of the Atlantium unit's inner surfaces and quartz tube. The CIP Kit (Catalog number: SAB012900SAB012950,) includes the following items (see the diagram below):

- CIP Cart with all other items mounted on it
- Water reservoir and a pump, capable of pumping up to 4 cubic meters per hour
- Supply hose: internal diameter 19mm /0.75"- Length 5m/15ft
- Return hose: internal diameter 19mm /0.75"- Length 5m/15ft
- Pocket to hold the hoses.

The following diagram(**Figure 3.9.1**) shows the CIP ports (**3 +4**) connected to the cleaning kit, as well as the positions of the sampling valves. Detailed instructions are given later in Chapter on Maintenance.

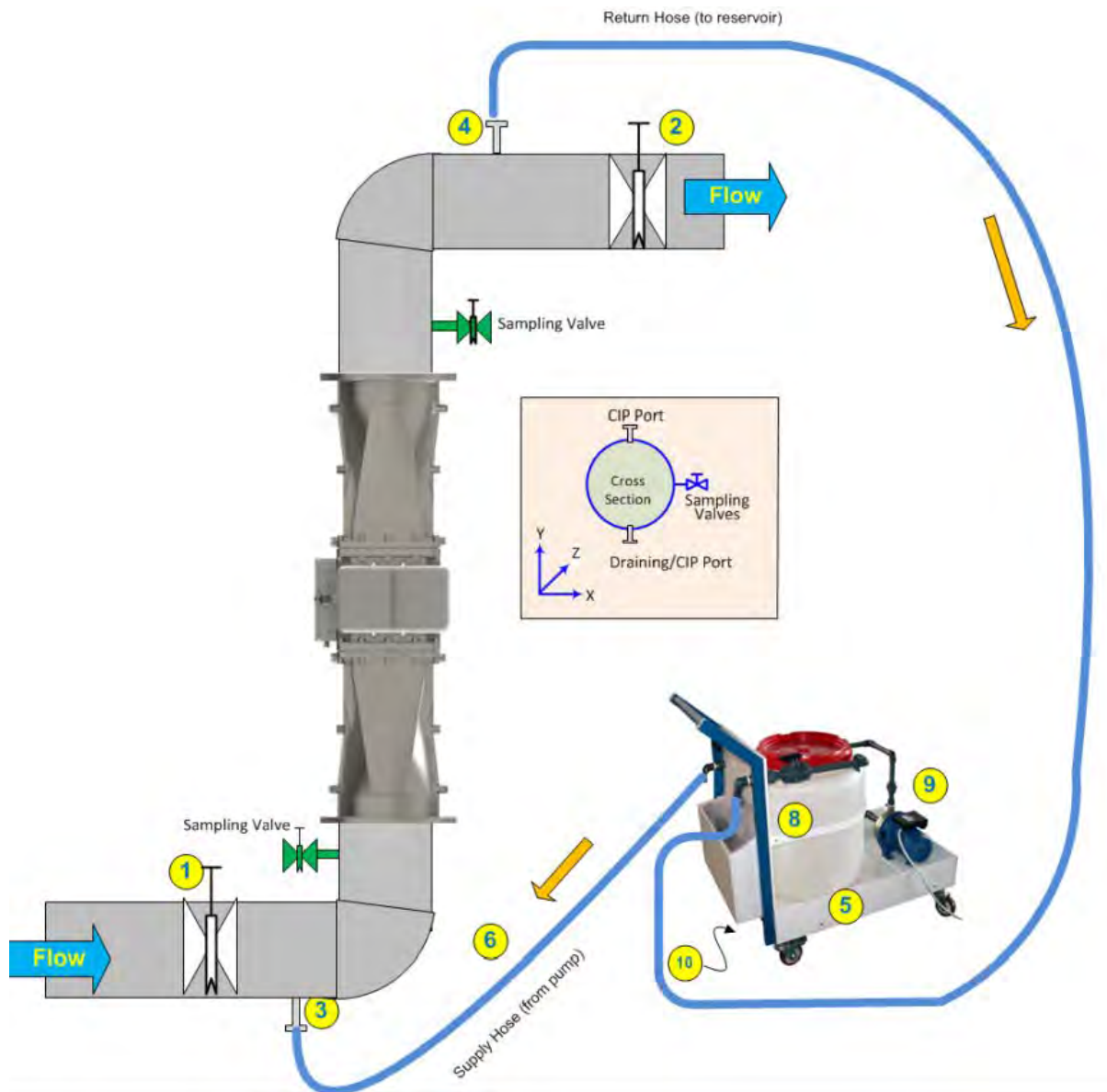


Figure 3-5: CIP ports (3 & 4)

3.9.1 CIP Accessories and Kits Available from Atlantium

One CIP Accessory kit is required for each CIP port.

Table 3-4: CIP Accessory Kits

Description	Image	Catalog number	Kit for Stainless Steel Pipe KT0011600	Kit for Plastic Pipe KT0011700
Long welding ferrule – for welding the tri-clamp port to the pipe		FAC003700	X	-
Threaded Male Adapter 1/2" BSP – for connecting the tri clamp port to the plastic pipe		FAB008100	-	X
Gasket 1" – for sealing the valve connected to the pipe		FA0000046	X	X
Hose Adaptor 1" – to attach the CIP hose		FAB005700	X	X
Clamp 1.5" – for connecting the valve to the welded port		FA0000045	X	X
End cap 1.5" – covering the CIP port while not in use		FAC004100	X	X

3.9.2 CIP Safety Devices

Required but not included in Atlantium's CIP Kits: protective devices such as goggles, clothing and gloves (chemical-safe) according to the specified safety standards at your facility.



- Atlantium does not supply chemicals – get them locally from commercial chemical suppliers.
- The chemicals selected and the concentrations used must be approved for use by the facility's quality assurance and safety officials, and must comply with the written protocols and procedures set in place by those departments.
- Use protective chemical-safe goggles, clothing and gloves
- Use and handle in strict accordance with their manufacturer's instructions, product information sheets, and material safety data sheets (MSDS).
- It is strictly forbidden to use of NaOH (Caustic Soda) as it causes irreversible damage to the quartz. Any use of this material requires an analysis of temperatures, concentration, frequency and duration and written approval from Atlantium's application engineer.

3.10 Connection to Facility Control

Options are available to set up control of the Atlantium system via your facility control system. Control setup can be made via any of the following:

- **MODBUS** – RTU on RS485 (network) serial connection directly to the Facility control (**Appendix Appendix A, *Modbus Communication Protocol***, on page **159**)
- **Signal control** – utilizing System Ready, External On/Off, and General Alarm functions, etc. (See ***Control Module Connections*** on page **56.**)

With any of the above control options, you can still utilize the function of the Control Module of the Atlantium system. Whichever control facility is utilized, the system communicates to the other control systems such that the last command issued determines the current status.

4 Safety Overview

The Atlantium System has been designed according to the highest safety standards, assuring the safety of operating personnel, the environment and the treated water.

DANGER!

Improper use of controls or adjustments or performance of procedures other than those specified herein could result in significant hazards.

Therefore, personnel operating or servicing this system must be thoroughly familiar with all safety requirements and operating procedures and are to adhere to them during use of this system.



To guard against injury, observe basic safety precautions, including the following:

All persons operating the Atlantium system must be aware of proper use and the potential hazards of violating safeguards. Be certain that all personnel carefully review the safety information and the procedures specified in this manual. Only authorized individuals with appropriate training and knowledge, including local regulations, should operate, assist in the operation of or provide maintenance to the system.

All service and repair of the Atlantium system must be performed by Atlantium field service technicians or other factory-authorized personnel **ONLY.**

4.1 UV Lamp Safety



UV Exposure: The system generates ultraviolet (UV) light within the Unit, which can cause serious eye damage or blindness if you stare at it directly when it is operating. Use the Viewport only. Do not look directly into the lamp enclosure during system operation, examining, or servicing the system's internal components during operation, or when lamp's breather caps are open, or if energized testing is required.

The Viewport is specifically designed with a UV filter to prevent UV light from escaping so it is safe to look inside. Nevertheless, personnel should not stare at it or any other brilliant light source for any but short intervals to check on Unit operations. The Viewport cover should also be preserved intact.

If the viewport is broken, DO NOT look directly into it. Be sure to request a replacement part immediately. Do not open the viewport's metal cover until the replacement is installed.

Persons potentially subject to UV light exposure – technicians who are servicing the Unit – must wear appropriate eye protection whenever the system is operating.

During maintenance procedures that are conducted with the UV lamps turned on:

- Use Caution floor signs at a distance of 1m/40inch from the Atlantium Unit to warn passersby against possible UV light exposure and not to approach without proper eye and skin protection.
- Wear white cotton gloves.
- Wear long sleeves rolled down to your wrist.
- Wear appropriate eye protection, such as containing polycarbonate lens that meet EN1 ultrasonic, CAN/CSA-Z94.3-02 and/or Z87.1 standards.
- **DO NOT look directly into openings that emit UV light.**

**Protect Hands:** Do not touch the UV lamp with bare hands.

- Wait at least 10 minutes until the lamp is cooled down before touching the lamp or starting the replacement procedure.
- Use appropriate protective gloves both to protect your hands and to avoid skin oils that leave fingerprints and/or harm the UV lamp.
- Lamps can reach a temperature of 1,000C° under operating conditions.
- Wait ten minutes to allow the lamp to cool down fully before replacing it.
- Keep materials that are sensitive to heat, or which contain solvents, a safe distance away from the lamp, Unit, and electrical connections.



Electric Shock: Before replacing a UV lamp or other components, and during any maintenance requiring lamps to be turned off, make sure the circuit breaker for the Connection Boxes of the relevant lamps are turned OFF and place a sign on the electrical cabinet alerting others NOT to touch the screen and circuit breakers during maintenance so that no one can inadvertently turn it on while maintenance is in progress.

ATTENTION!

Prevent Damage: Handle the UV lamp assembly by holding only the contact housings (white ceramic casing).

When handling the UV lamp, always place it on a flat surface or table, so that it is not accidentally damaged.

If the body of the UV lamp is touched accidentally, clean the fingerprints off with alcohol (at least 70%) and wipe dry with a soft, clean, lint-free cloth (usually provided with the lamp). Do not use cleaning rags or materials that can leave a residue.

Do not use a UV lamp that shows any scratches, cracks, or other damage.



Mercury Poisoning: Lamps contain mercury, which is a hazardous substance.

Always wear gloves and safety glasses when handling or replacing a UV lamp.

Inhalation of vaporized mercury compounds can be harmful to the lungs, kidneys, and nervous system. Mercury that penetrates the skin or is ingested can also be harmful.

If mercury is inhaled, penetrates the skin, or is ingested accidentally, seek emergency medical treatment immediately.

If a UV lamp breaks during handling and releases mercury, the following precautions should be observed to minimize the risk of exposure to mercury:

- Always wear gloves and safety glasses when handling or replacing a UV lamp.
- Leave the area immediately to avoid inhalation of the mercury vapor.
- Thoroughly ventilate the area for at least 30 minutes or until the mercury vapor concentration is in compliance with applicable federal and local health and safety regulations.
- After handling a broken UV lamp, carefully remove gloves, and then wash hands thoroughly with soap and water. Follow all applicable federal and local health, safety, and environmental regulations.
- Do not turn on or operate the system until all mercury contamination has been cleaned up and removed.
- Establish a Hazard Response Plan to deal with all related hazards and provide it to all personnel associated with the Atlantium system.

Refer the Chemical Safety information for mercury at: <http://www.inchem.org/documents/icsc/icsc/eics0056.htm>

ATTENTION!

Lamp Compatibility: Use original Atlantium UV lamps ONLY. Other lamps are incompatible. They void the warranty and may damage the system.



Proper Disposal of Lamps

Expired lamps are classified as hazardous waste and must be disposed of appropriately.

- A non-ruptured UV lamp that is no longer used must be disposed of according to specific local or federal environmental and hazardous waste regulations.
- Broken lamps and parts contaminated by the mercury have to be vacuum-packed and disposed of. Broken lamps, as well as contaminated packing material and other parts have to be considered as special waste, which may only be removed by authorized waste disposal companies.

4.2 Electrical Hazards and Safety Considerations

DANGER!

UNDER NO CIRCUMSTANCES

SHALL ELECTRICAL WORK BE CARRIED OUT WITHOUT VERIFYING THAT ALL MAIN CIRCUIT BREAKERS OF THE ELECTRICAL LINE FEEDING THE ATLANTIUM SYSTEM COMPONENTS ARE SET TO THE **OFF** POSITION AND THAT **NO** ELECTRICAL FEED IS LIVE.



DO NOT SET ANY MAIN CIRCUIT BREAKER OF THE ELECTRICAL LINES FEEDING THE ATLANTIUM SYSTEM COMPONENTS TO THE ON POSITION UNTIL **ALL** ELECTRICAL WIRING IS PROPERLY CONNECTED. UNCONNECTED WIRES/CABLES IN A LIVE ELECTRICAL FEED IS A HAZARDOUS SITUATION THAT CAN RESULT IN DEATH OR SERIOUS INJURY.



- **DO NOT DRILL INTO THE CONNECTION BOXES OR ALTER THEM IN ANY WAY!!!!**
- The Connection Box already contains entry glands for **ALL** of the cabling needs.
- Making holes in a Connection Box will destroy its waterproof seal, endangering the components inside.
- Do Not draw electricity from the Atlantium Unit's Connection Box as it will damage the Unit's electrical configuration.



The Atlantium Hydro-Optic system incorporates high-voltage internal components, which can cause serious injury or fatal electrical shock if not used or serviced properly. High voltage components can retain a charge for some period of time even after the system has been turned off.



- Never attempt to remove any system covers or to dismantle any parts, except when performing the maintenance procedures detailed in **System Maintenance** on page 108.
- No portion of the Atlantium system is to be opened or removed by anyone other than a trained and authorized technician.
- Do not spray or pour any type of fluid directly into the Unit or electrical components. Moisture causes damage to the equipment and electrical shock may result.
- Do not operate the system if the power cables or harnesses are frayed or otherwise damaged.

CAUTION!



DO NOT enter **Technician Mode** in the **Control Module** unless instructed to do so by Atlantium certified personnel or by an Atlantium System User Manual, for a specific task.

If you are in Technician Mode, **DO NOT WALK AWAY!**



NEVER LEAVE THE ATLANTIUM SYSTEM WHEN IT IS OPERATING IN TECHNICIAN MODE!!

A SYSTEM LEFT IN TECHNICIAN MODE WITHOUT SUPERVISION CAN RESULT IN PERMANENT DAMAGE!!



Utilize best practices regarding electrostatic discharge (ESD).

Avoid direct contact with the electronic circuit boards of the **Control Module**, which is sensitive to electrostatic discharge (ESD).

Wear an electrostatic discharge (ESD) Wrist Strap when performing procedures where contact with a circuit board is possible.

4.3 Keep the Unit Full of Water

ATTENTION!

The Atlantium system cannot be operated when there is no water. Any damage caused due to operating the system without water invalidates the Atlantium warranty.

4.4 Chemical Use for Cleaning in Place (CIP)

ATTENTION!



- The chemicals selected and the concentrations used must be approved by Atlantium's Application staff, and the ship's quality assurance and safety officials. The CIP process must comply with the written protocols and procedures set in place by those departments. Refer to the section, ***Selecting the Correct Chemicals for CIP*** on page **109**
- Use and handle all chemicals in strict accordance with their manufacturer's instructions, product information sheets, and Safety Data Sheets (SDS).
- NaOH (Caustic Soda) as it causes irreversible damage to quartz. It is strictly forbidden to use it without a specific analysis of temperatures, concentration, frequency and duration and written approval from Atlantium's application engineer.

Refer to the Chemical Safety information provided with the CIP chemicals which you choose to use, for example:

<https://portal.ecolab.com/servlet/PdfServlet?sid=915373&cntry=AU&langid=en-GB&langtype=RFC1766LangCode&locale=ja&pdfname=OXONIA+ACTIVE+150>

4.5 System Safety Features

The Atlantium Hydro-Optic system includes the following safety features:

4.5.1 UV Protective Seal



Each lamp is enclosed within a sealed lamp chamber that prevents the UV light from escaping. This means that with regular operation there is no ambient UV escaping from the Unit.

While servicing the UV Unit or checking the Intensity sensor, UV blocking safety glasses should be worn by service staff or others working in the area to prevent exposing the eyes to the UV light.

4.5.2 Software Safety Controls on Control Module



CAUTION!

When the system is started, the Atlantium software performs a series of internal self-testing routines prior to starting system operation.

If the system **does not** pass these tests satisfactorily, UV lamp operation is disabled until the problem is resolved.

If problems arise during normal operation, appropriate error messages with varying levels of severity are generated on the touch-screen of the Control Module.

Control Module screens that provide setup options and compliance reports are password protected.

ATTENTION!

Control Module has 3 levels of user permission hierarchy (**see Table 7-2**):

- **No User** - enables view of system status screens only
- **Operator** - enables general and maintenance actions
- **Manager** - enables full system settings

All changes in the system settings are logged and tracked in the Settings Report.

DO NOT leave the Control Module unattended, even for a short time, while setting change screens are accessible. An untrained person could damage the operation of the Atlantium system by making unauthorized setting changes.

4.6 Atlantium Training

As part of the Commissioning and Hand-Off process, Atlantium provides hands-on training on all aspects of the Atlantium system for your designated personnel. Should any component be replaced with upgraded features, Atlantium offers training as needed and is ready to provide additional training for new personnel or any other needs related to its systems.



5 Installing the Atlantium System



The installation instructions given in this manual must be followed. Failure to do so may render the warranty void.

Read through this typical installation section **before** you begin the installation process.

5.1 Installation overview:

Hardware setup to be implemented by your facility personnel:

- Phase 1** Unpacking and checking the contents of the Atlantium system packaging (See below)
- Phase 2** Installing Sampling valves and CIP ports (See page 49)
- Phase 3** Positioning the Atlantium Unit and connecting to the Piping Infrastructure (See page 54)
- Phase 4** Mounting the **Control Module**, Electrical Cabinet, etc. (See page 57)
- Phase 5** Cabling the **Control Module**, Electrical Cabinet, etc. (See page 66)

Commissioning to be implemented by Atlantium personnel:

- Phase 6** Starting up the system (See page 100)
- Phase 7** Configuring the **Control Module** (See page 71)
- Phase 8** Tuning the System (See page 132)
- Phase 9** First time System Activation (See page 95)

The Installation Process Flow diagram below maps out the installation process.:

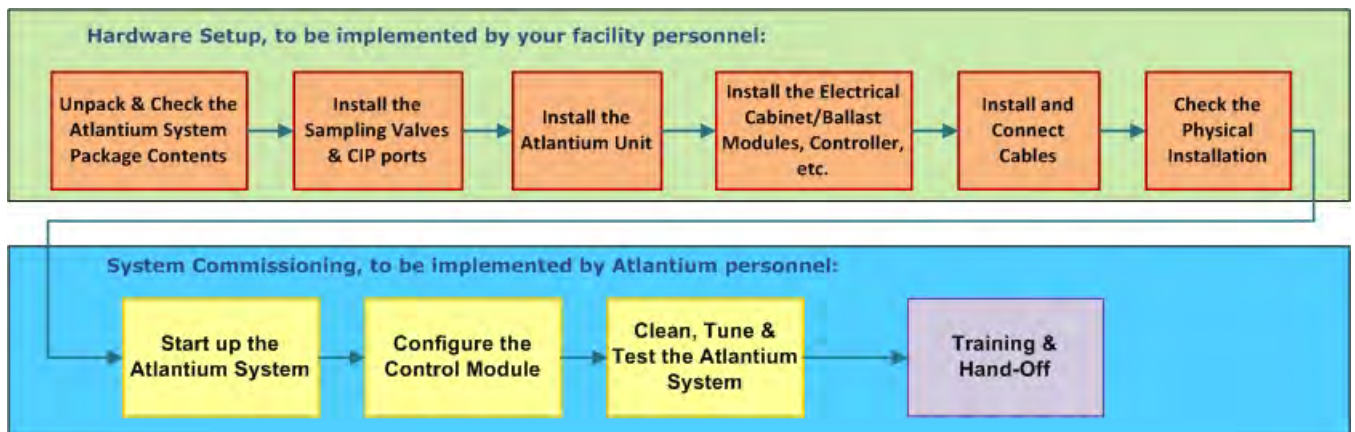


Figure 5-1: Installation Flow Diagram

5.2 Supply Conditions

5.2.1 Equipment Responsibility

Upon receipt of the equipment the system owner or installation yard automatically becomes fully responsible for the equipment, unless otherwise stated in the contract. This responsibility covers the storage period before installation, the actual installation, commissioning, and the period between the completion of the commissioning and the acceptance of the equipment by the end user (normally the owner of the vessel into which the equipment is to be installed).

5.3 Storage Guidelines

In some cases, it is inevitable that the system must be stored for a certain period of time. When this is necessary, make sure that system components' transport cases are stored somewhere dry and clean. A high humidity, salty air etc. may noticeably reduce the life time of equipment when not connected and operational and not maintained. We recommend not exceeding a storage time of four months.



Caution!

There is a risk that system components may be damaged due to improper storage.

The transport case must not be stored outdoors.

Protect the system from moisture and contamination during discharge.

Do not store the system in freezing conditions.

5.4 Unpacking And Checking Package Contents

The Atlantium system is packaged and shipped in specially packed crates that contain:

- The **RZM** Unit
- The Electrical Cabinets
- The Control Module
- Documentation
- Peripheral accessories as per your order and listed on your delivery documents are crated according to their size and weight. Small accessories may be crated together.

Each of the crates is connected to a forklift-friendly wooden pallet. The Atlantium crates must be lifted off the transport vehicle using a forklift or crane by a qualified operator.

⇒ To unload and check the system:

- Step 1.** Using the forklift or crane, lift each of the crates off the transport vehicle and lower them to a level ground surface.
- Step 2.** Walk around and inspect each crate for damage. Note if there is any separation in the joints or any other indication of any damage in transit. If there is any indication of damage, take a photograph of it and use extra caution when proceeding to the next step.
- Step 3.** Unclamp and remove the panels of the crates.
- Step 4.** As you remove the wrapping and shipping protections, verify that the contents of the crates are correct according to the delivery documents. The **RZM** Unit contains a plate with the model number and other product information. Verify that the correct product version is supplied.



- Immediately report to Atlantium and to the insurance carrier any damage to the crates or to the Atlantium system found before opening the crates or during unpacking, installing or testing of the system.
- If you ordered more than one Atlantium system, check the serial numbers against the delivery documents to verify which components comprise each individual system.

5.5 Installing Sampling Points and CIP/Draining Ports

Both the Sampling Points and CIP/Draining Ports must be installed on the inlet and outlet sides of the Atlantium Unit.

5.5.1 Installing the Inlet/Outlet Sampling Points

Aseptic sampling valves are to be located indoors in clean environment without away from direct contact with fan or A/C air flow. Each valve must be **dedicated for sampling only** and resistant to alcohol and flame. For information on the Sampling valve components, see **Sampling Valve Components and Kits** on page 34. For information on using the Sampling valves, See **Taking Microbial Samples** on page 105.

On the inlet and outlet sides of the water line, connect the aseptic sampling valves as follows:

➔ To install an inlet/outlet sampling point:

Step 1. On the Inlet side, install the **Long welding ferrule or threaded male adapter (1)**.

- Position aseptic sampling valves (**1 & 2** in **Figure** on page 51) **horizontally** (90^0) on the pipe center, within 50cm of the Unit.
- **For Stainless Steel. Pipes:**
 - Weld the 1 inch **ferrule (1)** clamp fitting at a horizontal orientation (3 or 9 o'clock).

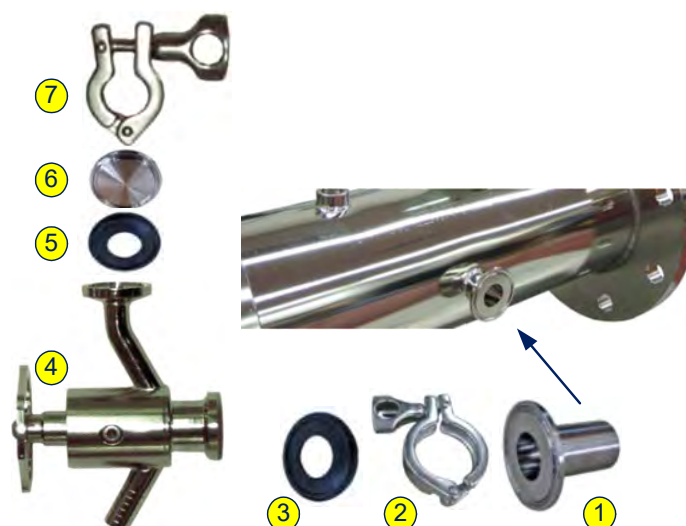


Figure 5-2: Sampling Valve Assembly, using Generic welded ferrule (Threaded male adapter is placed in the same position vis a vis the pipe)

- Perform a polishing process to achieve a smooth area on the external and internal part of the pipes.
- Passivate the pipes to avoid corrosion.
- **For Plastic Pipes:**
 - Thread the **male adapter (1)** clamp fitting to the port.
- Repeat for the Outlet side sampling point.
- Step 2.** Fit the gasket **(3)** into the **welding ferrule or threaded male adapter (1)**.
- Step 3.** Fit the aseptic **sampling valve (4)** on over the gasket **(3)**.
- Step 4.** Fit the clamp **(2)** around the aseptic **sampling valve (4)/gasket (2)/welding ferrule or threaded male adapter (1)** and close and lock it.



Figure 5-3: The clamp is placed around the aseptic **sampling valve /gasket /welding ferrule or threaded male adapter**

Closing Off the Aseptic Sampling Valve When Not in Use

- Step 5.** Into the spout of the sampling valve, fit the 1/2" gasket **(6)** and fit the cap **(7)** over it.
- Step 6.** Fit the 1/2" clamp **(5)** around the sampling valve spout/ gasket **(6)** /cap **(7)** and close and lock it.



Figure 5-4: The clamp is closed



Figure 5-5: The clamp is closed

5.5.2 Installing the CIP/Draining Ports

The CIP (Cleaning-In-Place) recirculation system is used for the periodic cleaning and sterilization process of the Atlantium unit's inner surfaces and quartz sleeve. For more information on performing CIP, see **Cleaning In Place (CIP)** on page 109. Details of the CIP kit are given in **CIP Ports and CIP Kits** on page 36.

Position the CIP/draining valves on both the inlet and outlet sides of the Atlantium unit, as indicated in **Figure 5-6**. To enable connection of the CIP kit, use high-quality, sanitary equipment, including a 1 inch tri-clamp/ferrule on both sides.

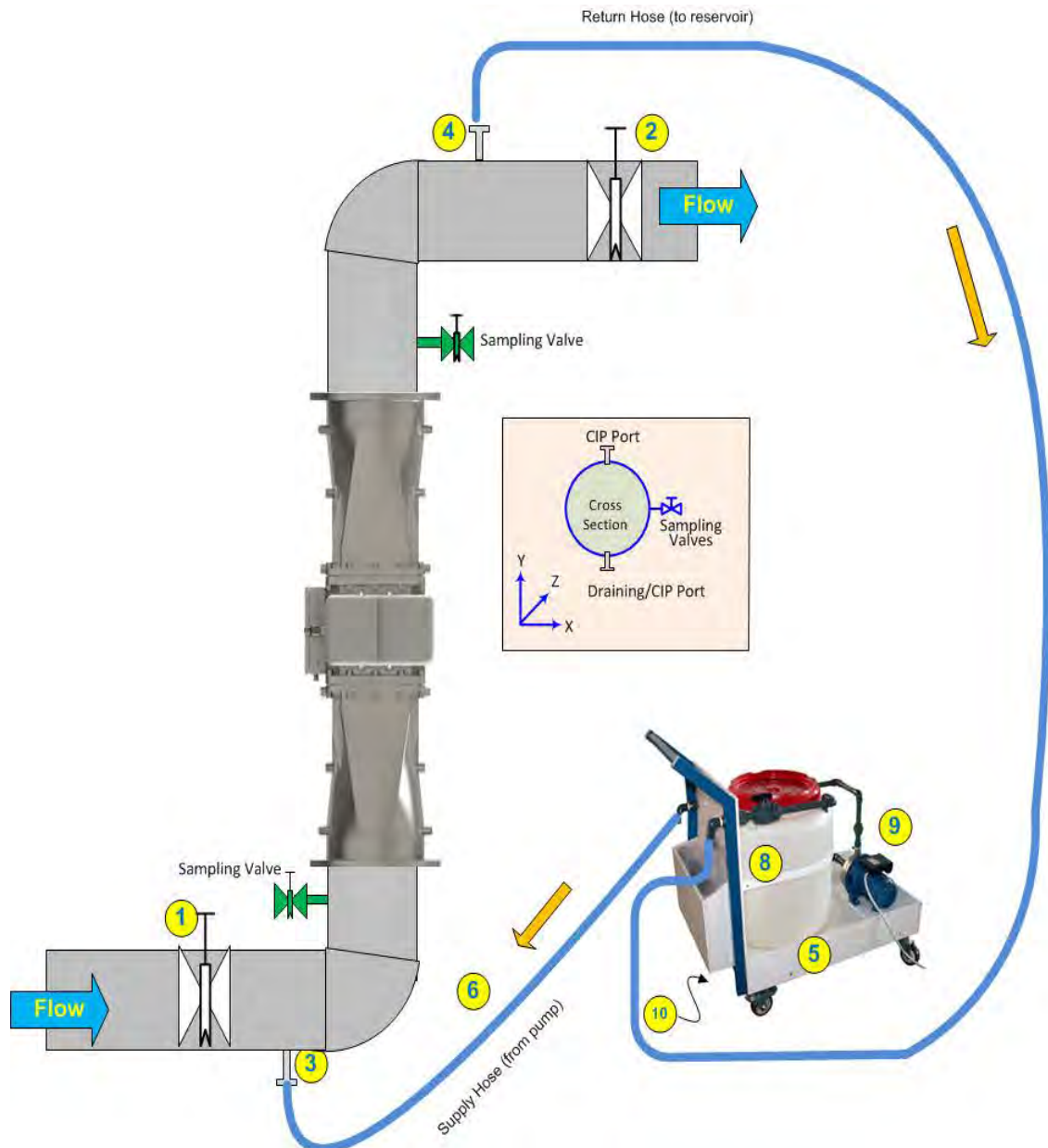


Figure 5-6: CIP Connections

One CIP/Draining valve 3 is to be located after the Inlet valve 1 and the other 4 before the Outlet valve 2 in **Figure 5-6**. When not using the CIP ports, close the valve outlets using the tri-clamp cover.

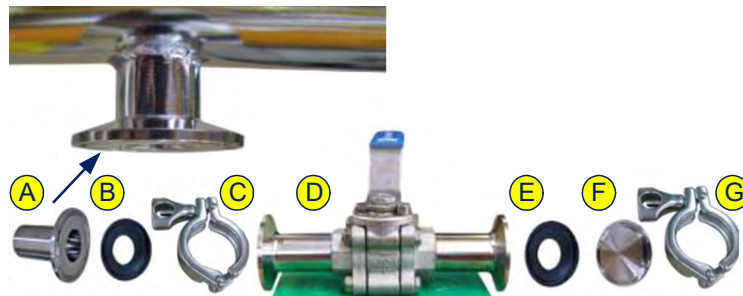


Figure 5-7: CIP Valve Kit

➔ To install the CIP ports:

Step 1. Weld or thread the 1 inch ferrule at a vertical orientation:

For municipal applications, prepare 1" screw ports for the valves and insert the valves:

- **Draining Valve:** at a 6:00 o'clock orientation on lower end, before the Atlantium unit.
- **CIP Valve:** at a 12:00 o'clock orientation on upper end, after the Unit.

For all application except municipal:

■ **For Stainless Steel. Pipes:**

Weld the 1 inch clamp fittings:

- **Draining port:** at a 6:00 o'clock orientation on one side of the Atlantium unit.
- **CIP port:** at a 12:00 o'clock orientation on other side of the Unit.
- Perform a polishing process to achieve a smooth area on the external and internal part of the pipes.
- Passivate the pipes to avoid corrosion.

■ **For Plastic Pipes:**

- Thread the clamp the valve fittings to the port..

Attaching the Valve to the ferrule

Step 2. Fit the gasket (B) into the **welding ferrule** or **threaded male adapter (A)**.

Step 3. Fit the **valve (D)** on over the gasket (B).

Step 4. Fit the clamp (C) around the **valve (D)/gasket (B)/welding ferrule** or **threaded male adapter (A)** and close and lock it.

➤ Closing Off the CIP/draining Ports When Not in Use

- Step 1.** Into the ferrule, fit the 1/2" gasket (2) and fit the cap (3) over it.
- Step 2.** Fit the 1/2" clamp (4) around the CIP port/ gasket (2) /cap (3) and close and lock it.



Figure 5-8: The clamp is closed



Figure 5-9: The clamp is placed around the CIP port /gasket /cap

➤ Preparing the CIP Hoses

- Step 1.** On one end of a hose (A), using an appropriate screwdriver, loosen the band (C).
- Step 2.** Connect the hose to one of the adapters on the cart (B). Push the hose up as far as you can.
- Step 3.** Using the screwdriver, tighten the band (C).
- Step 4.** On the other end of a hose (D), using an appropriate screwdriver, loosen the band (F).
- Step 5.** Connect the hose to one of the hose adapters (E). Push the hose up as far as you can.
- Step 6.** Using the screwdriver, tighten the band (F).
- Step 7.** Repeat with the second hose to the adapter on the other side of the cart.

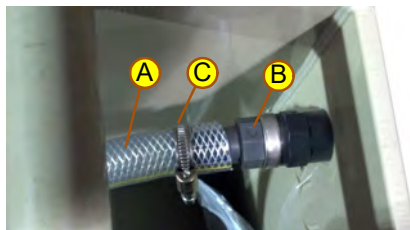


Figure 5-10: Hose Connection to One Side of Cart

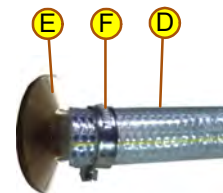


Figure 5-11: Hose Connection to Hose Adapter

5.6 Before You Start

Before you start the installation, take the following actions:

- Inform the supervisor / coordinator that the work is about to be carried out.
- Collect the required Installation Design Plan, drawings, and other documentation and read the applicable procedures before commencing work.
- Ensure that all power is switched off to the system.
- Label the ON / OFF switches, circuit breakers and fuses with warning notes clearly stating that work is being carried out on the system.



WARNING!

Do not attempt to run the system before the checks listed in the Atlantium BWMS check and verification procedure have been completed.

5.7 Placing the RZM Unit

Refer to **Figure 3-1** and **Figure 3-2** for generalized illustration of RZM setup.

Make sure to use your installation drawing during this procedure. Be aware of the **RZM** Unit's marked water-flow indicator. Prepare the required support brackets. To anticipate the correct height for transport, trace the path to be taken to transport the Unit to its designated location and clear it of all obstacles.



WARNING!

There is a risk of accidents from falling loads.

During installation of the system parts, accidents may occur due to sizes and weights.

- Use correctly proportioned and intact lifting accessories.
- Attach the separate lifting accessories properly to the components.
- Do not stand and do not work under raised loads.
- Wear the a appropriate safety equipment and clothing (helmet, protective gloves, safety shoes, etc.)

5.7.1 Required Equipment

The following tools and equipment are required:

- Lift or crane
- Lifting strap/shackles with latch hooks
- Supplied bolts, washers & lock washers appropriate to your piping adapter type
- Supports for inlet/outlet piping and for the Unit as shown on your installation plan



The **RZM** unit is heavy - refer to Table 2-2, **RZM200-5 UV Unit Physical Specifications**, in **Section 2**, page 13. Two or more people are required to get the **RZM** Unit to the designated installation location - a qualified lift or crane operator and at least one other person to guide the Unit as it is transported.

➞ To place the UV Unit:

- Note that the UV Unit is to be installed vertically (see **Figure 5-12**). Note the sticker **[C]** on lower part of Unit, indicating the direction of the water flow. Use this information to decide at which angle to lift the Unit for transport. This is to help you to prepare for placing the Unit at the location.
 - The Unit has 4 screw-in support points **[A]** & **[B]**.
- Step 1.** Ensure that lifting eyes **[B]** are screwed into the support points on either side of upper part of unit.
- Step 2.** To lift the Unit from its packing crate, attach lifting straps **[1 & 2]** with latch hooks through the lifting eyes **[B]** on either side of the upper part of the unit.
- Step 3.** Join the straps together with a third central strap **[3]** connected to the lifting beam **[4]** of the lift or crane.

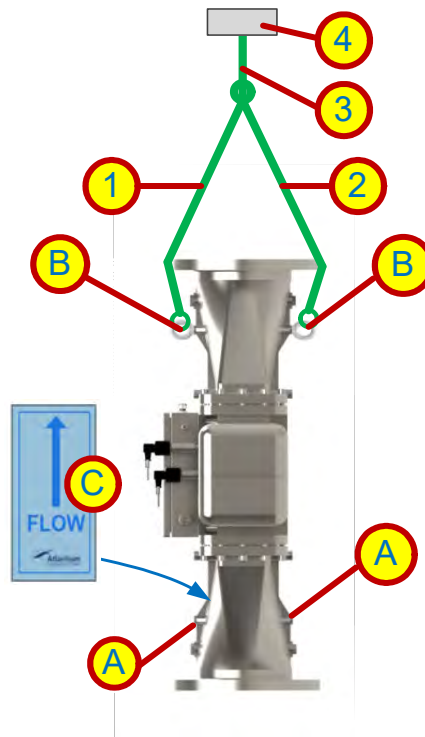


Figure 5-12: Configuration for lifting

- Step 4.** Ensure that all straps are connected securely to the Unit, to each other and to the lift/crane.
- Step 5.** Using the lift/crane, lift the Unit by the straps, out of the crate and position it for safe transport (near floor level is preferred).
- Step 6.** Using the lift/crane and the designated person(s) to guide, carry the Unit to its intended location.
- Step 7.** At the designated location, check the installation plan, including the distances between the bolt holes, etc.
- Step 8.** Position the Unit vertically so that the Flow label arrow is pointing upwards, on the supports constructed for this purpose. (Refer to the provided **RZM200-5 Interface Control Drawing**.)
- Step 9.** To connect the fitting on one side of the Unit to the fittings of the ship's pipes, Insert an o-ring between the Unit's and the ship's pipes. In some cases, a reducer or expander may be necessary

➤ To connect to facility piping

- Step 1.** To connect the fitting on one side of the Unit to the fittings of the facility pipes, Insert an o-ring between the Unit's and the facility's pipes. In some cases, a reducer or expander may be necessary.
- Step 2.** On one side of the Unit to the flanged fittings of the facility's pipes with eight appropriate (8) sets of bolts/nuts/flat and lock washers - do not tighten the hardware.
- Step 3.** Repeat the step above for the other side of the Unit.
- Step 4.** Tighten the fastening hardware so that the torque pressure around the fittings is even, and the Unit is firmly connected to the facility's pipes.
- Step 5.** To check for any leakage, open the Inlet and Outlet valves to allow water to flow through the newly installed Unit and its surrounding process piping.
- Step 6.** If you find any leakage, tighten the connections.
- Step 7.** Close the Inlet and Outlet valves and use the Draining valve, drain the piping from Inlet valve to Outlet valve.



Figure 5-13: Typical flange fitting to the water line

5.8 Power connections

5.8.1 Setting the Main Circuit Breakers to OFF Position

Before beginning work to connect the wires/cables of the Atlantium system components, turn the main circuit breakers of the electrical lines feeding the Atlantium system components to the **OFF** position. **Verify that no electrical feed to the Atlantium system components is live.**



WARNING!!!

Under no circumstances shall electrical work be carried out without verifying that **ALL** main circuit breakers of the electrical lines feeding the Atlantium system components are set to the **OFF** position and that **NO** electrical feed is live.



DO NOT set any main circuit breaker of the electrical lines feeding the atlantium system components to the on position until **ALL** electrical wiring is properly connected. Unconnected wires/cables in a live electrical feed create a hazardous situation that can result in death or serious injury.

5.9 Installing the Electrical Cabinets

The location of the Electrical Cabinets (EC) must be designated according to the length of the **Cables** of the **RZM** Unit and the designed cable paths. The standard length of **Cables** that connect the **RZM** Unit to its Electrical Cabinet is **10 meter (33 ft)**.

The Electrical Cabinet can be mounted against a wall, or on the floor using the supplied base.

5.9.1 Required Equipment

The following tools and equipment are required:

- Lift or crane
- Lifting strap



The Electrical Cabinet is heavy - refer to Table 2-10, **Electrical Cabinet Physical Details**, in Section 2, page 20. Two people are required to get the Electrical Cabinet to the designated installation location - a qualified lift or crane operator and at least one other person to guide the Electrical Cabinet as it is transported.



Do not position the Electrical Cabinet directly on the floor. Allow sufficient room to avoid contact with the floor, which may get wet.

The Electrical Cabinet includes fans and filters on the top and bottom of the door. There must be a clearance of at least 80cm (31.5 inches) in front of door for access and for proper ventilation. To review the complete clearance requirements of the Electrical Cabinet, see the mechanical ICD.

The power cables are to be obtained locally.

➔ To position the Electric Cabinet:

Step 1. Uncrate the Electrical Cabinet package.



The Electrical Cabinet of the Atlantium system is heavy! - It should be lifted and carried by mechanical means and by more than one person.

Step 2. Thread lifting straps through the lifting eye bolts on each upper corner of the Electrical Cabinet. Attach the lifting straps together and to the lift or crane.

Step 3. With the lift or crane, move the Electrical Cabinet to its location designated according to installation plan.

Step 4. For a floor-mount configuration, stand the Electrical Cabinet on its built-in legs.

Step 5. Lower the Electrical Cabinet to its designated location.

As appropriate, bolt the Electrical Cabinet to the wall or anchoring point.

5.10 Mounting and Connecting the Control Module



Information contained in this User Manual is for illustration purposes only. For detailed wiring diagrams of your particular system, ALWAYS refer to the supplied System Electrical Diagrams and Electrical ICD drawings.

The **Control Module** can be mounted on a wall or suitable anchoring point. The **Control Module** comes with an electric power cable 1.6M/5.25ft. long (1.5m/4.92ft. from the cable gland) and a Data cable. It can be placed up to 50M/ 164.04 ft. from the Atlantium Unit, in a place that it is easy to reach the touch screen and read the monitoring information. For Data Cable length options, please contact Atlantium.



ATTENTION!

- When mounting the **Control Module**, take into consideration the length of the cables.
- **Do not Stretch Electrical Cables - Avoid tension on the electrical cables. They are to be installed within conduits and must not be stretched.**

➞ To mount the Control Module on the wall:

- Step 1.** On the wall space designated for the **Control Module**, make four marks according to the spacing measurements for mounting orientation on the ICD drawing accompanying the documentation.
- Step 2.** The mounting holes **(1)** are 8.5mm (0.33") in diameter. Using the appropriate drill bit, drill the four holes.
- Step 3.** In each hole, insert a plastic anchor.
- Step 4.** Uncrate the **Control Module** package and carefully lift it out.
- Step 5.** Open the door of the **Control Module** and remove the plastic covers on the four holes.
- Step 6.** Position the **Control Module** so that its four holes fit over the drilled holes.
- Step 7.** Using appropriate screws and with a screwdriver, screw in the four screws in a diagonal pattern (one corner, then the opposite corner).
- Step 8.** To secure the **Control Module** firmly, return to each screw in the diagonal pattern and tighten the screws securely.

➞ To mount the Control Module on skids:

- Step 1.** At the location designated for the **Control Module** set up the skids with spacing that matches the spacing measurements for mounting orientation on the ICD drawing accompanying the documentation.
- Step 2.** The mounting hole **(1)** are 8.5mm (0.33") in diameter. Using the appropriate bolts and nuts, with a screwdriver, screw in the four bolts and add the nuts leaving around 5.6mm (0.20-0.24") sticking out.
- Step 3.** Uncrate the **Control Module** package and carefully lift it out.
- Step 4.** Open the door of the **Control Module** and remove the plastic covers on the four holes.
- Step 5.** Position the **Control Module** so that its four holes fit over the corresponding holes on the skids.
- Step 6.** Using appropriate bolts and nuts and with a screwdriver, screw in the four bolts adding the nuts in a diagonal pattern (one corner, then the opposite corner).

The **Control Module** contains a jumper marked **11** in the **Figure** below.

Table 5-1: **Control Module** Components

#	Item
1	Mounting Holes
2	Data Cable + Gland
3	Electrical Cable + Gland
4	Electrical Terminals
5	Cable glands for Peripheral Component Connections
6	Various Component Connectors (See Table 5-2 on page 60)
7	Circuit Breaker
8	Wireless Modem (optional)
9	Connector for Wireless Modem (optional with Modem)
10	Jumper
11	GND Wire
12	Screen Display
13	Two USB Ports (top side of the board)

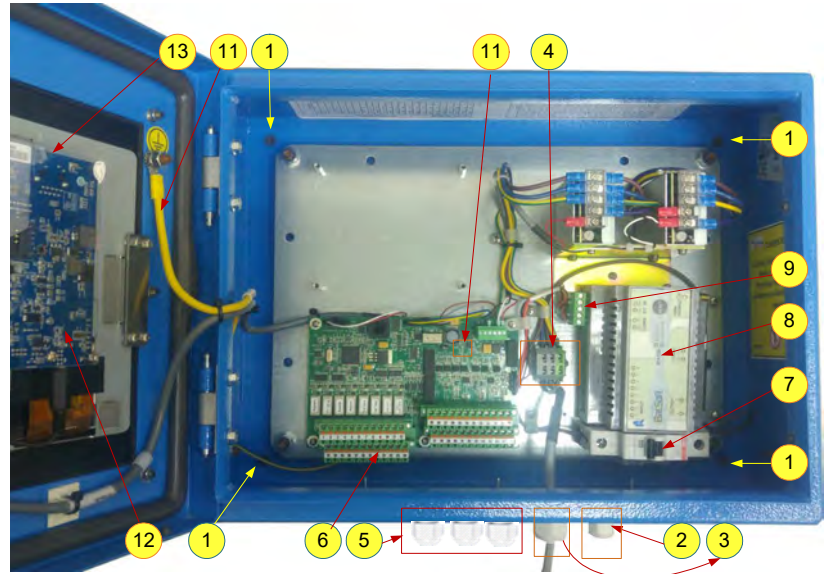


Figure 5-14: Control Module Interior

5.10.1 Control Module Connections

Communication connectivity can be via Digital/Analog I/O's hard wired Modbus RS485 cable connections (see below) or via (optional, purchased separately) wireless connection (see page 21).

You can connect signal wires for the following peripheral equipment to the Control Module's terminal block:

- | | |
|-----------------------------|-------------------------|
| ■ System Ready/Outlet Valve | ■ Bypass Valve Output |
| ■ General Alarm | ■ Flow Meter Input |
| ■ Shutdown Warning | ■ Flow Switch Input |
| ■ High Temperature | ■ External On/Off |
| ■ UVT Out | ■ Dose Output |
| ■ Low UVT | ■ Modbus communications |
| ■ Communication Loss | |
| ■ Inlet Valve Output | |

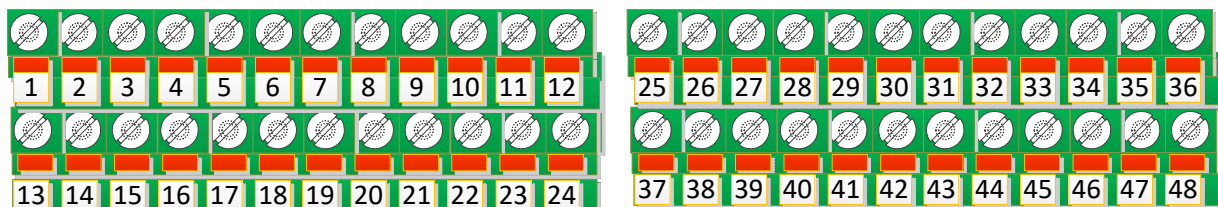


Figure 5-15: Control Module Terminal Connectors

Table 5-2: Control Module Terminal Block Connector Pins J35 (J3+J5)

	Function	Pin's Number	Values
Digital Outputs	System Ready/Outlet Valve COM	1	Dry Contacts Max. Voltage: 220VDC/250VAC Max. current: 2A Max. switching capacity: 60W
	System Ready/Outlet Valve N.O.	2	
	Alarm COM	3	
	Alarm N.O.	4	
	Shutdown Warning COM	5	
	Shutdown Warning N.O.	6	
	High Temperature COM	7	
	High Temperature N.O.	8	
Digital Outputs	Low UVT COM	9	
	Low UVT N.O.	10	
	Communication COM	11	
	Communication N.O.	12	
	System On/Inlet Valve COM	13	
	System On/Inlet Valve N.O.	14	
	Bypass Valve COM	15	
	Bypass Valve N.O.	16	
	Spare	17	
	Spare	18	
Digital Inputs	External On/Off (DI1) +	19	On: 10-48VDC (NA) Off: <1VDC See page 62
	External On/Off (DI1) -	20	
	Flow Switch (DI2) 1	21	
	Flow Switch (DI2) 2	22	
	Normal/Low (DI3)+	23	
	Normal/Low (DI3)-	24	

Table 5-2: Control Module Terminal Block Connector Pins J35 (J3+J5)

	Function	Pin's Number	Values
Analog Output	Dose Out+	25	4-20mADC, Max. Rsens= 200R
	Dose Out- (GND)	26	
	UVT Out+	27	
	UVT Out - (GND)	28	
Analog Input	Flow Meter Return 4-20mA	29	4-20mA
	Conductivity Meter Return 4-20mA	30	
	Spare	31	
	Spare	32	
	GND for 4-20mA (for Flow Meter-option 1)	33	
	GND for 4-20mA (for Connectivity Meter-option 1)	34	
	24V for 4-20mA (for Flow Meter-option 2)	35	
	24V for 4-20mA (for Connectivity Meter-option 2)	36	
	Spare	37	
Modbus Slave in	RS485 Rx / Tx+	38	Main Modbus connector. For RS485 physical connections, to make a Modbus daisy chain incoming connection from another Control Module. The jumper is required. (See item 11 in Figure 5-15 on page 59) For wireless modem connections, the jumper must be removed (LED is off).
	RS485 Rx / Tx-	39	
	RS485 PE	40	Earthing
Modbus Master	RS485 Rx / Tx+	41	Connector for the Atlantium Unit's Data Cable
	RS485 Rx / Tx-	42	
	RS485 PE	43	Earthing

Table 5-2: Control Module Terminal Block Connector Pins J35 (J3+J5)

	Function	Pin's Number	Values
Modbus Slave out	485 Rx / Tx+	44	To make a Modbus daisy chain outgoing connection to another Control Module
	485 Rx / Tx-	45	
	485 PE	46	Earthing
Slave RS485	Term 1	47	Termination connector
	Term 2	48	For the Control Module that is last in a daisy-chain, insert a jumper here

5.10.2 External On/Off - Additional Notes

For regular wiring:

- Positive voltage must be connected to pin 19 – IN+
- Negative (GND) must be connected to pin 20 – IN-

Table 5-3: External On/Off

Control Module Screen External On/Off	Physical Connection	Physical Connection	Lamp Control
	IN+	IN-	
Normally Open	0 VDC	0 VDC	Off
Normally Open	+24 VDC	0 VDC	On
Normally Closed	0 VDC	0 VDC	On
Normally Closed	+24 VDC	0 VDC	Off

5.11 Modbus Communication Connections

Modbus is an application layer messaging protocol for client/server communication between devices connected on different types of buses or networks. Modbus is often used to connect a supervisory computer with a remote terminal unit (RTU) in supervisory control and data acquisition systems.

The Atlantium system can be controlled directly from your facility control system via Modbus - RTU on RS485 (network) serial connection.

To accomplish this:

- The Modbus Settings must be configured in the **Control Module**. See **Configuring More Settings** on page 80.
- The Facility control system communication properties must be configured to match that of the Atlantium system. See **Setting Up the Facility System Communication Properties** on page 63.
- For the terminal block connectors for the Modbus in the **Control Module**, see Table 5-2 **Control Module Terminal Block Connector Pins J35 (J3+J5)** above.

- Signaling communication incoming to the facility control system is according to the Modbus Registers. **Appendix B, System Messages**, on page 147.



To ensure communications are operational, be sure to connect the cables **TX to RX** and **RZ to TX**.

5.11.1 Setting Up the Facility System Communication Properties

To properly configure communications between your Atlantium system and control network, set up the communication properties and **COM1** port settings at the facility control system as follows:

Node Address:	1 to 256 ***
Available Commands	Function codes
Function Codes: *	0x01, 0x02, 0x03, 0x04, 0x05, 0x06, 0xF, 0EC-10
Support IEEE Floating Point: **	NO (register/100)
Baud Rate:	4800, 9600, 19200, 38400, 57600, 115200 ***
2 Wire	
Parity:	None
# of Stop Bits:	1
# of Data Bits	8

(*) You can read all the registers in the system at once with **0x03** (includes coils, discreet coils, holding registers and input registers).

(**) Divide the desired parameter by 100 to calculate the real flow point.

(***) Must be the same in the Control Module setting

To view the Modbus Registers, which govern the signaling communication from the Atlantium system to the facility control system, see **Appendix B, System Messages**, on page 147.



Make sure the circuit breakers before the mains are set to the **Off** position **BEFORE** connecting the Atlantium system to the mains.



Avoid tension on the Electrical Cables: **Do not Stretch Them**. They are to be installed within conduits and must not be stretched.

5.11.2 Connecting the Control Module's Power Cable

The **Control Module** comes with a power cable of 1.6m/5.25ft. length. For the electrical specifications, See **The Control Module** on page 21. Prepare the appropriate plug for the electrical standards of your region. Considering the distance between the **Control Module** and your electrical box, you may want to replace the electrical cable with a longer one.

5.12 Electrical Cabinets: General View

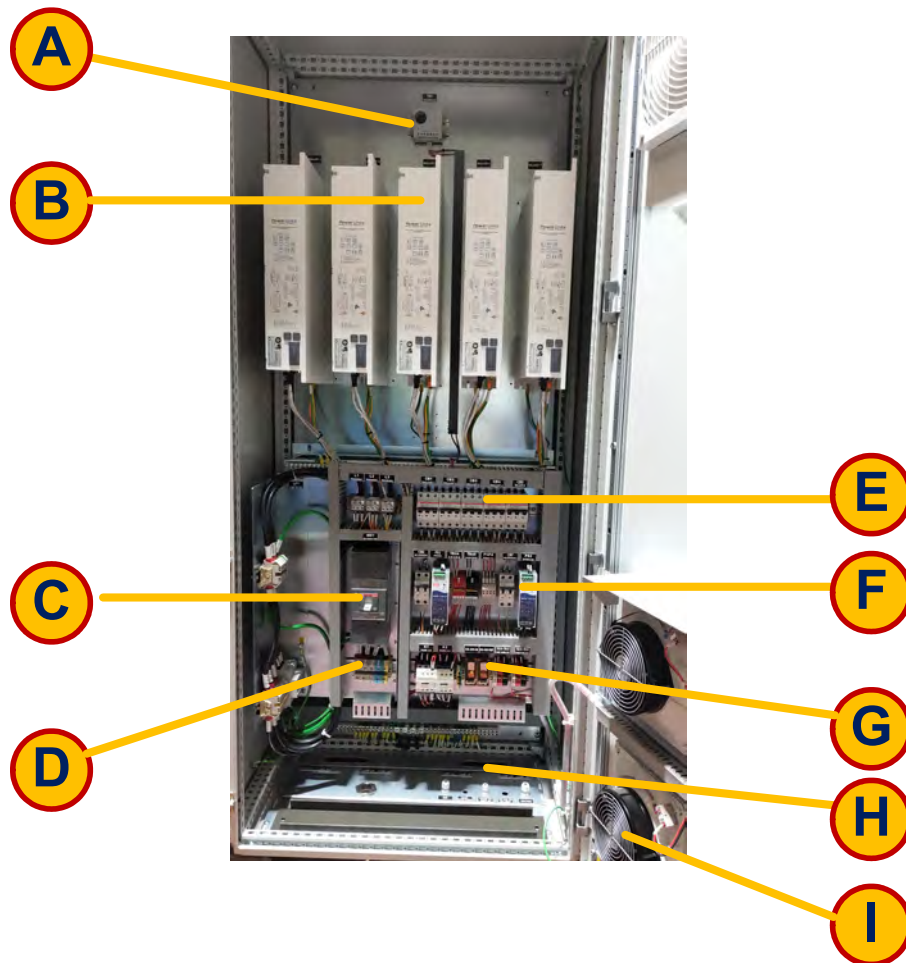


Figure 5-16: Electrical Cabinet general layout

Table 5-4: Electrical cabinet legend

#	Description
A	Thermostat
B	Ballast modules
C	Main CB
D	Terminal block
E	Circuit breakers
F	24V power supplies for fans
G	Connectors for 220VAC and 24VDC
H	Lamp cable holes, with brushes
I	Door fan

5.13 Connecting the Cables to the EC

The first Electrical Cabinet (EC-1) provides controlled power to lamps 1-5 of the Atlantium unit.



Make sure the circuit breakers before the mains are set to the **Off** position. **BEFORE** connecting the Atlantium system to the mains.



- **Do not Stretch Electrical Cables - Avoid tension on the electrical cables. They are to be installed within conduits and must not be stretched.**



- Refer to electrical diagrams supplied in the EC document pocket for detailed connections of all cables and wires
- The main power cable that you, the Customer, have prepared for this installation must be connected to the Electrical Cabinet.
- The Atlantium System requires a balanced voltage supply (WYE configuration). If you do not have such a connection, you must create an appropriate power feeding using a 1:1 isolation transformer (where the ground is connected to the middle point of the star configuration at the secondary side of the transformer).
- For other voltage supply options, a transformer is required. Contact Atlantium Customer Support.



WARNING!

- The electrical connections must be performed by a qualified electrician **ONLY**.
- **Do not** plug in the Unit's Power cable to an outlet until after the hardware installation is complete and the system is ready to be initialized. (See *Initializing the Atlantium System* on page 70.)

The **RZM** Unit comes with the **Cables** assemblies already connected to the Connection Box. You must position the **Cables** assemblies in a path that does not obstruct access to the **RZM** Unit or the Electrical Cabinet or any other access pathways. The **Cables** assemblies must be threaded through the designated openings on the underside of the Electrical Cabinet and the individual cables from the **Cables** assemblies must be attached to the appropriate connections inside the Electrical Cabinet. The following cables need to be connected:

- **Power Cable** from the Mains, 4 Wires (3L+PE), Customer supplied
 - 440Y/254 VAC, 34.8A
 - 480Y/277 VAC, 31.9A
 - A Neutral wire is not required, but if included in Power Cable connect it to Neutral terminal
- **Ground Cable** from the Body of the Unit, part of harness supplied by Atlantium - pre-connected to the UV Unit
- **Lamp cables**, up to 4 pairs, part of harness supplied by Atlantium - pre-connected to the UV Unit
- **Data Cable** to the **Control Module**, supplied by Atlantium - pre-connected to the EC
- **Data Cable** from the Connection Box on the Atlantium Unit, part of harness supplied by Atlantium - pre-connected to the UV Unit.

⇒ To connect the Lamp Cables from UV Unit to the Ballast PS in the EC:

The cable for each lamp is supplied pre-connected to the UV unit and the EC end is also pre-connected to a green plug.



Figure 5-17: Connect Lamp Cable to Ballast

- Step 1.** Take the 1st Lamp Cable from the UV unit and feed it through the entry (L) on the base plate of the EC.
- Step 2.** Feed the pre-connected green plug up along the rear panel of the EC and connect it as shown on the underside of the appropriate Ballast Power Supply, as shown in **Figure 5-17** above. The other cables shown are already pre-connected.
- Step 3.** Tighten the screws of the green connector to ensure secure connection.
- Step 4.** Repeat steps 1 to 3 above for each lamp cable in system.
- Step 5.** Fasten the lines to the vertical flange, using a cable tie.

5.14 Installation Check List

When you are ready to schedule the Atlantium technician to commission the Atlantium system,

- 1 Complete the checklist below - Table 5-5
- 2 Make digital photos of every angle of the installed system
- 3 Send the photos and the completed checklist to your Atlantium representative.

Table 5-5: Atlantium System Installation Check List

Item	Done
■ The System is installed: ☐ Vertically - Water is flowing upward ☐	
■ The RZM Unit, Electrical Cabinet and other components are located in their final position.	☐
■ All of the System components are securely installed and have sufficient service clearance on all sides (according to the ICDs).	☐
■ All power preparations are complete and Power connects are wired as instructed in this guide.	☐
■ Correct voltage was measured where the Power connects to the Electrical Cabinet Voltage rate _____ Number of phases _____	☐
■ The RZM Unit is connected to the water pipes according to the Installation plan developed by facility engineers and Atlantium personnel. Proper pipe lengths (Inlet - 10D, Outlet - 5D) are installed.	☐
■ Water flows through the RZM Unit - The Unit has been checked while full of water and there are no bubbles or leaks in the system. The system is balanced, supported and pressure stable (no vibration, etc.). No known water hammers or pressure above 10 bars are on the line.	☐
■ All piping accessories are installed and wired as instructed in this guide (e.g., Inlet/Outlet valves, sampling valves, draining valves, CIP valves, air release valve, etc.) ☐ Sampling Valves ☐ CIP Ports ☐ Draining Valves ☐ Air Release Valves	☐
■ A flow meter or flow switch is installed and the electrical and data connection are complete. ☐ If flow meter: 4 - 20 mA signal and Com wires are connected as instructed in this guide. ☐ If flow switch: 2 wires from the flow switch are connected as instructed in this guide. ☐ A 4 - 20 mA signal from the PLC is installed.	☐
■ The Control Module connections are completed. All supporting features (e.g., General Alarm, External On/Off, flow meter or flow switch, auto valves open/close, etc.) and wiring were installed according to the Installation plan based on the instructions in this guide.	☐
■ All Electrical Cabinet connections are completed.	☐

Table 5-5: Atlantium System Installation Check List

Item	Done
■ You are scheduling an electrician to be on site and available during the System commissioning (a must or the installation cannot be completed!).	☐
■ To be sent to Atlantium with this checklist, attach photos of the following: ☐ RZM Unit ☐ Electrical Cabinet ☐ Control Module and other components ☐ Accessories (Flow meter or flow switch, CIP ports, Sampling valves, etc.)	
Customer Approval Name of Customer's authorized representative: _____ Position: _____ Signature of Customer's authorized representative: _____ Date: _____	

6 Initializing the Atlantium System

The Atlantium system starts up with the initialization of the **Control Module** on the electrical cabinet.

➔ To start up the Atlantium system:

- Step 1.** Check that the power cables of the electrical cabinet are all connected to the mains power source.
- Step 2.** Connect the power cable of the **Control Module** to the main power source.
- Step 3.** Using the attached key, open the screen panel of the **Control Module**.
- Step 4.** Locate the circuit breaker [A] in Figure 6-1 and switch it to the **On** position.

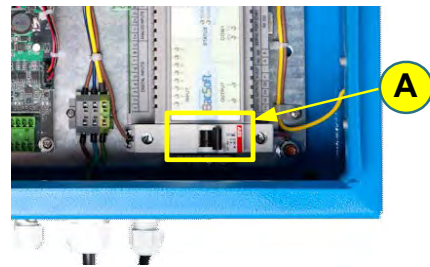


Figure 6-1: Control Module

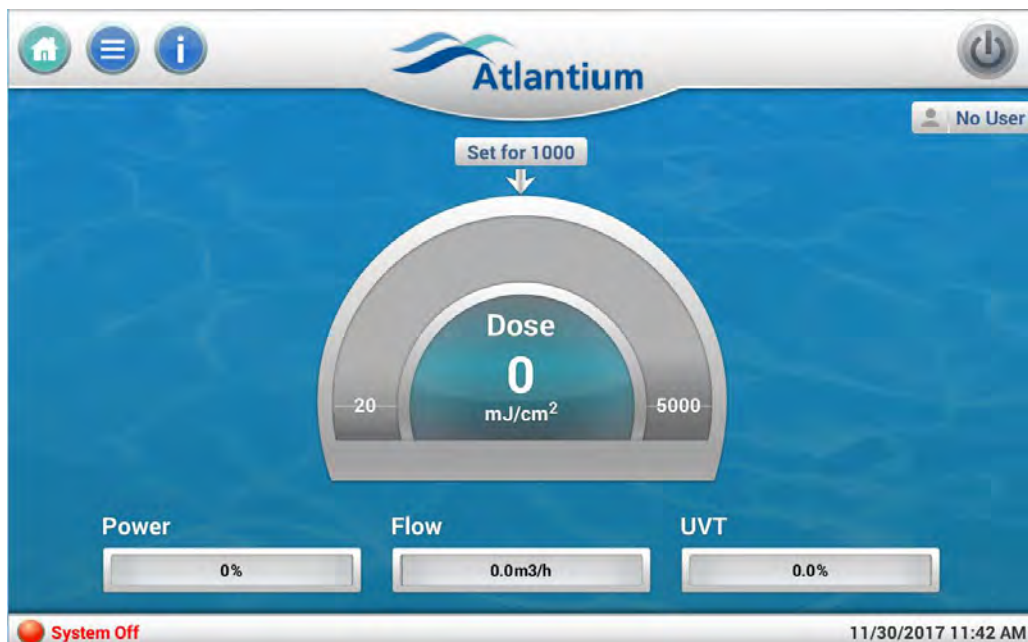


Figure 6-2: Main Operation screen

Figure 6-3: Press the Lamp Power button 

7 Configuring the Control Module

7.1 The Control Module HMI

The **Control Module** provides the Human Machine Interface (HMI) for control of operations and dose measurements of the Atlantium system.

The regulations applying to the type of application determine sizing as well as software defaults and operational control options. This is set up at the original manufacturing facility and completed on site by the Atlantium technician.

The **Control Module** provides a real-time display of operational data. A Ready indicator turns green when the system is ready for operation.

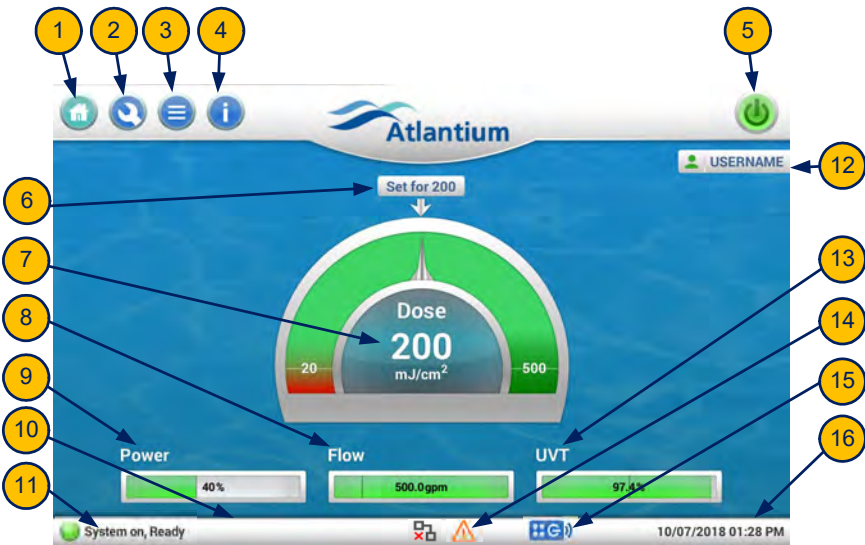


Figure 7-1: Initial Main Operation Screen*

Table 7-1: Control Module Main Screen Details

#	Item	Icon	Description
1	Return to Main screen		To return to the Main screen.
2	Settings Screen Access		To access the Settings screen
3	Event Log Access		To the Events screen.
4	System information Screen Access		Includes the Generic Info tab that displays general system information and lamp status and the Info Monitor tab that displays a summary of system status information.

Table 7-1: Control Module Main Screen Details

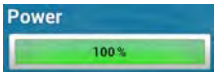
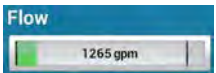
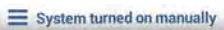
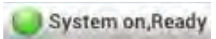
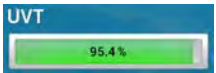
#	Item	Icon	Description
5	Main Unit Operation Button		ON/Off toggle button turns on/off the Atlantium unit components (lamp(s), UV power, etc.) in an orderly sequence.
6	Set For		The user should adjust the Set for dose to the desired operating dose. Confer with your Atlantium Representative, per the original sizing specifications. In some cases, this is 10-20% over the regulated or required dose. In other cases it depends on the variability of local conditions and the best strategy to ensure reliable and effective operation.
7	Dose Gauge		Represents the system dose range. The red area of the gauge represents off-spec low dose. The dark green area represents the user's set maximum allowable dose. The current dose (mJ/cm ²) is displayed in the middle of the gauge. The minimum and maximum allowable dosage levels are user-defined.
8	Power		Scale shows the current Power level.
9	Water Flow		Scale shows the current rate of water flowing through the Unit The scale color changes to red when the flow is either less than minimum flow setting or above the maximum flow setting. When there is no flow the scale goes gray.
10	Bottom Footer		Displays certain system events for a short period of time.
11	Ready LED		Displays the basic system status. When the system is ready for operation (all the lamps are ignited and the dose is above the minimum settings, the system displays the green Ready icon. If the system startup fails or the system is no longer working at or above the minimum dose, the system displays the red icon.
12	User Login		The Login button enables access to the login dialog box and User Management. See Adding Users on page 91.
13	UVT (Transmittance)		Scale shows the amount of UV light transmitted through the water. If the UVT percentage factor is high, the transparency of the water to UV light is high, and vice versa. When there are several UVT sensors in one system, this Scale displays the average of the UVT values measured by each sensor. The Scale color becomes red when the UVT is lower than minimum UVT setting. The color changes even if the UVT alarm is set to None.

Table 7-1: Control Module Main Screen Details

#	Item	Icon	Description
14	Event Icons		A relevant icon appears when a critical event occurs. It disappears after the event has been acknowledged in the log.
15	Remote On/Off indicator		The Remote-On/Off indicator appears if marked in the Signals tab settings.
16	Date and Time		Date and time* are displayed according USA or European conventions <i>*Time may require resetting if the Control Module is disconnected for a week or more and the internal battery runs low (see General Settings on page 88).</i>
17	Lost Network Communications Indicator		This icon appears when communication is lost.



The specific configuration for your Atlantium system is based on the Application type set for your industry and according to the parameters required for your facility's process.

Configuring the **Control Module** begins with logging in and accessing the Settings screen and follows the general order of the screen components.

7.2 Log In and Password Change

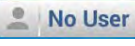
Users may log in, change their password, and reset their password if they forget it.

Users can login to the **Control Module** to access screens according to the permission level assigned to them. When logging on as the Facility Administrator for the first time, use the user name and initial password assigned to you during system commissioning.

Atlantium recommends that the initial Facility Administrator log on with this initial password, then change the password and make a record of the new password in a secure location, in the event that it is forgotten. The Facility Administrator can then define additional users according to facility needs.

For information on adding additional Users and permission levels, see *Adding Users* on page 91.

➔ To login:

- Step 1.** Tap the User Indicator . The Log In dialog box appears.
- Step 2.** From the **User Name** list, select your user name (for first-time use, the user name assigned to you during system commissioning).
- Step 3.** Tap the **Password** field and from the virtual keyboard that is displayed, enter your password (for first-time use, the password assigned to you during system commissioning).
- Step 4.** Tap **Log In**. Your user name appears in the User Indicator box on the upper-right of the main screen.

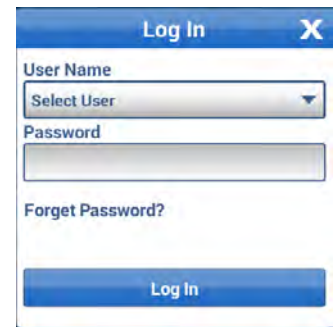


Figure 7-2: Login Dialog Box

➔ To change your password:

- Step 1.** Tap the User Indicator . The User Administration Tasks dialog box appears.
- Step 2.** Tap **Change Password**. The Change Password dialog box appears.
- Step 3.** In the **Password** field, enter your current password.
- Step 4.** In the **New Password** and **Confirm New Password** fields, enter the new password.
- Step 5.** Tap **Save New Password**. The password is changed.



Figure 7-3: User Administration Dialog Box



Figure 7-4: Change Password Dialog Box

7.2.1 Login Timeout

To prevent unauthorized system changes, the Control Unit reverts automatically to No User mode 10 minutes after the last tap on the screen by the current user. In the No User mode, system information may be viewed but no changes can be made (see Table 7-2). To make any changes, an authorized user must login again.

7.2.2 If You Forget Your Password

Generally, users with Manager Profile may configure other users and their passwords. If no user with Manager Profile is available, Atlantium Customer Support can help using the following procedure:

➔ To reset your password

- Step 1.** In the Login dialog box, tap **Forgot Password?**. The **Forgot Password** dialog box appears.
- Step 2.** In the **User Name** field, enter your user name.
- Step 3.** Tap **Generate Code**. A code appears in the field below.
- Step 4.** Send the User Name and Code to Atlantium Customer Support.
- Step 5.** In the **Confirmation Password** field, enter the **Confirm Password code** received from Atlantium Customer Support.
- Step 6.** Tap **Reset Password to Default**. The password is reset to the default **1234**. The user should then change it to a secure password.

Figure 7-5: Forgot Password Dialog Box

7.3 The Settings Configuration Screen

The settings screen and individual tabs are visible according to the permission level of the User and are usually accessed during the installation and setup of the Atlantium system and when there are hardware changes made in peripheral equipment.

During normal operations, an authorized facility technician may need to access certain functions, or view certain information, that appears on the some of these tabs. For information on which tabs are visible according to permission levels, see **Adding Users** on page 91.



WARNING!

Take care to ensure that the system configuration made via the Settings screen is not tampered with! Unauthorized changes may have a serious impact on the function of the Atlantium system. **Only trained and authorized** facility technicians are to access the Settings screen.



- The Control Module screen images shown in this User Manual are generic. The content of your system screens will vary according to the type of system, the number of lamps, cleaner type etc.
- Screens display values in metric or US units, according to customer localization.
- On completion of each screen, Tap Save to save settings.

- ➔ To access the **Settings** screen: at the top of the screen, tap 

Step 1. The main Settings screen is displayed with the Monitor tab open:

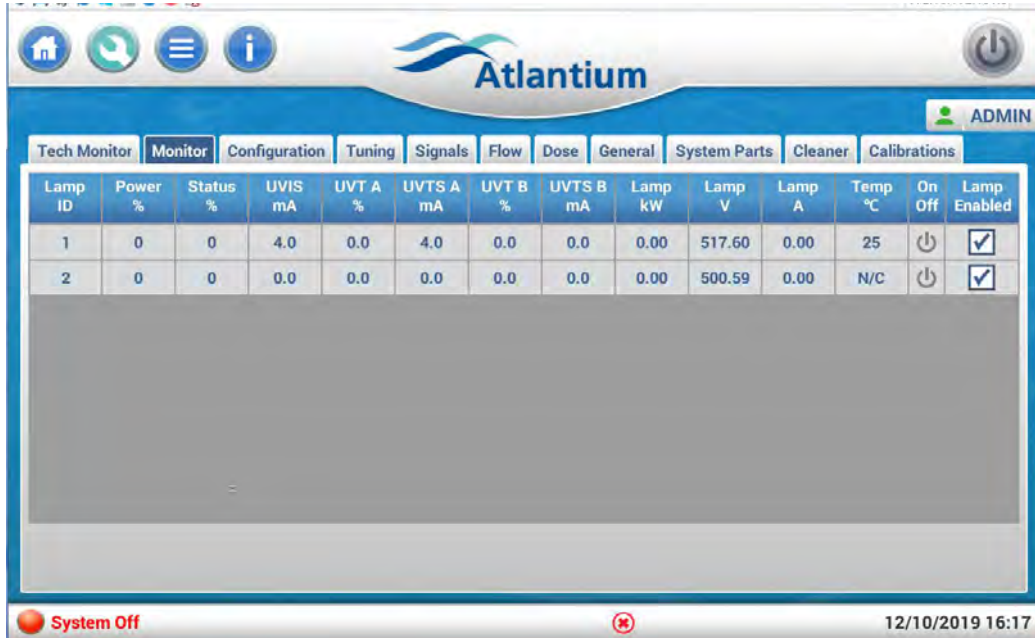


Figure 7-6: Settings Screen, showing Monitor Tab (generic example with 2 lamps)

Step 2. Tap the **Configuration** tab to open it.

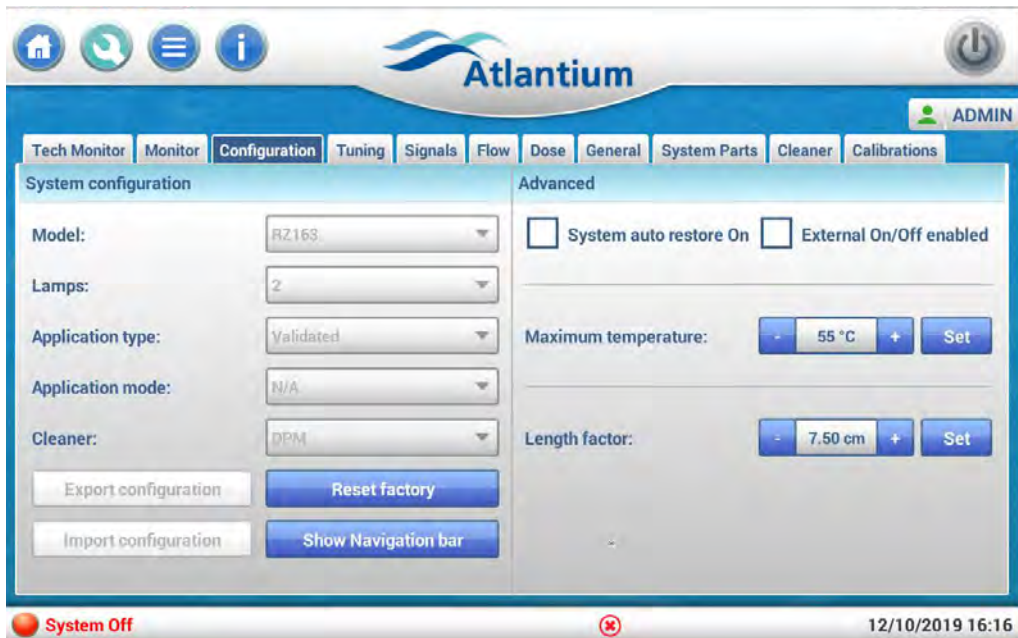


Figure 7-7: Settings Screen, showing Configuration Tab

Step 3. To enable the External On/Off, mark the **External On/Of Enabled** (see **Signal Inputs**: on page 78) checkbox on the R side (**Advanced Settings**).

- **System Auto Restore** - You can also set the Atlantium system to restart automatically when power is restored after a facility-wide power failure, so that it reactivates according to the configured **Auto Power On** settings.

Step 4. To enable System auto restore, mark the System auto restore checkbox.



The **System Configuration** panel on the Left side of the **Configuration** tab is used by the Atlantium technician during system setup to establish basic system parameters, such as Model, number of Lamps, Cleaner type etc. Most of the settings in this tab are not accessible with regular user or administrator privileges, and should **NOT** be changed.

- **Maximum temperature** enables the setup technician to set the maximum permitted temperature for the water in the UV chamber.
- **Length factor** enables the setup technician to set the distance between the UV lamp sleeve and the UVT sensor, through the water.

Step 5. Tap the **Signals** tab. The **Signals Tab** appears.



Figure 7-8: Signals Tab on Settings Screen

7.3.1 Configuring the Signal Settings

The Signals tab displays the settings for the system's dry-contact Input and Output connections: either **Normally Closed (NC)** or **Normally Open (NO)** see Figure 7-8 on page 77. The following is an explanation of Input and Output signals.

7.3.1.1 Signal Outputs

Inlet valve/System on - The facility's electronically-controlled **Inlet** and **Outlet valves**, that control the water supply to the Atlantium unit, as well as the **Bypass valve** (if a bypass is implemented), must each be configured to receive control signals from the **Control Module**. These valves affect to various facility processes relevant to the Atlantium system, such as:

- Managing high flow conditions
- Bypassing and streaming the water to a different place if the water quality has degraded
- Bypassing for cleaning chemicals that are not compatible with the quartz chambers
- When the system is activated (either manually via pressing the **Operation** button on the **Control Module**'s main screen, or via an external signal utilizing the **External On/Off**), the **Control Module** sends a signal, within 10 seconds, that opens the **Inlet valve**. This signal can also be used to provide feedback that the system is activated.
- When the all the lamps are ignited and ready to use, the **Control Module** sends a signal to the **Outlet valve** to open. This signal can also be used to notify that the system is **ready**. The **Control Module** also sends a signal to the **Bypass valve** to close.
- Similarly, when the system is deactivated, (either manually via pressing the **Operation** button on the **Control Module**'s, or via an external signal utilizing the **External On/Off**), the **Control Module** sends a signal that closes the **Inlet** and **Outlet valves**, and also opens the **Bypass valve**.

Signals to Alarms – To set off a General Alarm mechanism, such as a flashing light, alarm, bell, or a call or message to a telephone, a system alarm signal must be connected to a SCADA control system. See

General Alarm – requires connection of a General Alarm mechanism to receive a signal from the **Control Module**. Once configured, the alarm is triggered by the signal.

- **High Temperature** – requires connection of a General Alarm mechanism to receive a To connect a mechanism to receive a signal when system temperature has exceeded the configured maximum.
- **Low UVT** – To connect a mechanism to receive a signal when the system's UVT has fallen below the configured minimum.
- **Lost Comm** – When the **Control Module** AIO lost communication with UV unit (all cards disconnected).
- **Shutdown** – To connect a mechanism to receive a signal when an event causes the system to shutdown.

7.3.1.2 Signal Inputs:

- **Flow Switch** – If your system utilizes a Flow Switch, you must configure its electrical polarity in the **Control Module**. For the Flow Meter, see *Configuring the Flow Settings* on page 79.
- **External On/Off** – A switching mechanism to synchronize the Unit with the facility processes. This remote PLC turns the Atlantium Unit on/off. For details about the External On/Off signal wiring, see on page 159.

For example, the facility's SCADA (supervisory control and data acquisition) control system can send a signal to the Unit to start or stop its operation as needed. Using this setting, the Atlantium system recognizes this signal and turn the Unit on or off, as appropriate, without the need for manual intervention.



When the **External On/Off** sends a signal to activate the system, the Atlantium **Control Module** sends a response within **10 seconds** that the command was received. If no response is received, it is likely that the **External Operation Button** on the **Control Module** main screen is manually set to the **Off** position. Check this button and if needed manually tap it to reinstate the **On** position, so that the **External On/Off** can control the system activation.

For regular wiring, set according to the table in *External On/Off – Additional Notes* on page 59:

- For IN+ select **Normally Closed**
- For IN- select **Normally Open**
- **Low/Normal Mode** – for systems with an external signal for switching to Low Flow mode.

7.3.1.3 Other Settings

- **Maximum Temperature Warning** – The System shuts down automatically if the water temperature exceeds the maximum set value. You can set the maximum temperature threshold for sending a warning notification that alerts you to take action to prevent the water from reaching the maximum temperature that triggers shutdown. It is recommended to set this threshold to 5% below the maximum water temperature allowed for your system.
- **System Auto Power On** – You can also set the Atlantium system to restart automatically when power is restored after a facility-wide power failure, so that it reactivates according to the configured **Auto Power On** settings.

➔ To configure the Signals settings:

Step 1. At the top of the screen, tap . The **Monitor Tab** is displayed.

The states of the **Outputs** and **Inputs** are set via toggle switches that by default are set to **NC (Normally Closed)**. To change any of them to **NO (Normally Open)**, tap the corresponding toggle switch. It changes to **NO**. Tap again to change it back to NC.

Step 2. Under **Output**, set the electrical polarity for:

- **Inlet valve/System On**
- **Bypass Valve**
- **Outlet valve/Ready**
- **General Alarm**
- **High Temperature**
- **Low UVT**
- **Lost Communication** (not implemented in this version)
- **Shutdown**

Step 3. Under **Input**, set the electrical polarity for:

- **Flow Switch** (if installed)
- **External On/Off**

Step 4. To enable the External On/Off, mark the **External On/Of Enabled** checkbox on the Configuration tab (see 7.3. The Settings Configuration Screen, **Step 3**).

Step 5. Tap **Save** to save settings.

7.4 Configuring the Flow Settings

If a flow meter is utilized, the **Control Module** data must be configured to match the flow meter at the time the system is originally commissioned, but may need to be changed if a flow meter is changed, etc. The Max Flow setting provides a threshold to compare to the actual flow rate. If the actual flow rate exceeds this set Max Flow value, the system sends notification and triggers an alarm.

■ About the Low Flow Mode

Set the Atlantium system's response to a low water flow situation via the Low Flow mode. This unique Atlantium feature minimizes the on/off cycles by allowing the system to operate at an extremely low flow value. The Atlantium Application Engineer defines this value according to the specific application. In some cases, this value can be zero flow for a predefined period of time. The **Low Flow Mode** options are:

- ◆ Restart
- ◆ Shutdown
- **Restart** – (Default) This option assumes that low flow is not necessarily a sign of a fault and ensures that the system is ready to come up to full dose or power immediately when the flow rises.

Atlantium allows the system to continue operating at a low or even zero flow rate up to a max time interval, the **Flow Delay**, as defined by the Atlantium application engineer. With this setting, we reduce the amount of on/off and maintain the integrity of the UV operation, while preventing the flow of non-treated water into the line.

- As long as the flow rate stays below the **Min Flow** for less than the amount of time set in **Flow Delay**, the Unit continues to operate in standby mode, generally at the lowest possible power.
- However, if the water flow rate drops below the set **Min Flow** rate for a time that is **longer** than the **Flow Delay**, then:
 - ◆ The **Low Water Flow** warning is triggered
 - ◆ The lamps are automatically shut down
 - ◆ An event notification is written to the event screen and log file
- The **Restart** mode causes the system automatically restart when the water flow rate returns to a level above the **Min Flow**. The Unit automatically and instantly returns to full operation, without any operator intervention or connection to a higher control level, and without any delay or cool down/warm up period.

The default values for the minimum flow rate and maximum flow delay appropriate for **Restart** mode are preset by Atlantium. These values may be changed based on local site requirements and conditions, but **be sure to coordinate** with the Atlantium representative.

- **Shutdown** – In this mode, in the event of low flow, which may be a sign of an upstream malfunction, the system shuts itself down in an orderly fashion, but operator intervention is required to start it up again.
 - As long as the flow rate stays below the **Min Flow** for less than the amount of time set in **Flow Delay**, the Unit continues to operate in standby mode, generally at the lowest possible power.
 - However, if the water flow rate drops below the set **Min Flow** rate for a time that is **longer** than the **Flow Delay**, then:
 - ◆ The **Low Water Flow** warning is triggered
 - ◆ The lamps are automatically shut down
 - ◆ An event notification is written to the event screen and log file
- In the **Shutdown mode**, the operator is required to turn on the Atlantium system, once the water is back to the proper Flow rate.

➞ To configure the Flow Meter settings:

- Step 1.** At the top of the screen, tap . The **Monitor Tab** is displayed.
- Step 2.** Tap the **Flow tab**. The **Flow Settings** tab appears. The **Current Flow** indicator displays the current rate of water flowing through the Atlantium system in **GPM** (gal/min) or **H** (H) units. The **Main Flow Input** indicator displays the **Flow Meter (FM)** or **Flow Switch (FS)** in **GPM** or **H** (H) units.

For Flow Meter:

- **Flow Meter** appears as the **Main flow input** source,
- The **Current flow** is displayed on the left.

Step 3. Tap **Calibrate** to open the **Flow meter** calibration window

The 'Flow meter' calibration window displays three input fields with adjustment buttons (-/+):

- mA at 0 flow:** 4.00
- Flow x100 at 20mA:** 3.00
- Current flow:** 140.3 gpm

At the bottom are 'Cancel' and 'Save' buttons.

Figure 7-9: Flow meter calibration window

The 'Flow Setting Screen - Flow Meter' shows the following settings:

- Main flow input type:** Flow meter
- Main flow input source:** ISO controller
- Calibrate:** Button
- Current flow:** 51.8 m³/h
- Minimum flow:** 40 m³/h
- Flow delay:** 10 sec
- Low flow mode:** Shutdown
- Maximum flow:** 1000 m³/h
- Action:** None
- Save:** Button

At the bottom, there is a 'System Off' indicator and a timestamp: 12/10/2019 16:20.

Figure 7-10: Flow Setting Screen - Flow Meter

- Step 4.** For **mA at 0 Flow (Flow Factor)**, to check the calibration of the flow meter, turn off the flow and then check the value for Current Flow. If it is not zero, adjust the value, by using the -/+ buttons, until the Current Flow value is zero. Then resume the water flow.
- Step 5.** For **Flow x 100 at 20mA (Offset)**, on the flow meter's datasheet, find the meter's maximum-allowed flow rate.
- Step 6.** In the **Flow x 100 at 20mA** field, using the -/+ buttons, enter the meter's maximum-allowed flow rate divided by 100. (e.g. If the value given is 100 m³ /h, enter the value 1). Convert GPM values into m³ /h values to make this calculation¹.

1. If the meter's maximum-allowed flow rate is set to gpm (US units), multiply this value in gpm by 0.227 to get the flow rate in m³/h

Step 7. Tap **Save** to return to the **Flow meter** tab.

Step 8. For **Minimum flow**, using the **-/+** buttons, enter a value for the minimum water flow rate threshold. Below this threshold, the system enters **Low Flow Mode**.

Step 9. For **Flow delay**, using the **-/+** buttons, enter a value, in seconds.

When the flow rate falls below **Minimum flow**. The system then waits for the **Flow delay** period before entering **Low Flow Mode**.

Step 10. For **Low flow mode**, tap the arrow and from the drop-down list, select either **Restart** or **Shutdown**.

Step 11. The default value for maximum **Flow Delay** appropriate for **Restart** mode are preset by Atlantium. These values may be changed based on local site requirements and conditions, but be sure to coordinate with the Atlantium representative. If you need to change this value, set the desired value using the **-/+** buttons.

Step 12. For **Maximum flow**, using the **-/+** buttons, enter a value for the maximum water flow rate threshold. If the actual flow rate exceeds this threshold, the system delivers a notification message and triggers an alarm.

Step 13. To set the action to be performed if the water flow rate exceeds the value set in the **Maximum Flow** field. The options are:

- **None** - The event is not considered an error, so no corrective action is taken. No warning message is issued.
- **Caution** - One or more caution messages are issued.
- **Warning** - One or more warning messages are issued.
- **Shut-Down** - In case of a serious error or dangerous condition, the system shuts down and triggers a pop-up window message and a General Alarm.

Step 14. Tap **Save** to save settings.

For Flow Switch:

Flow Switch appears as the **Main Flow Input**.

The current flow is displayed on the left.



- A flow switch is used to determine when water flow goes above (Flow) or below (No Flow) a set value.
- Only install a flow switch (instead of a flow meter) in applications where the water flow through the Atlantium system is known to be constant, without significant fluctuations.
- Install a flow switch of recommended type (see **Flow Switch** on page 35).
- These adjustments are to be performed by a qualified Atlantium technician during installation and set-up

Step 15. Tap **Calibrate** to open the **Flow switch** calibration window

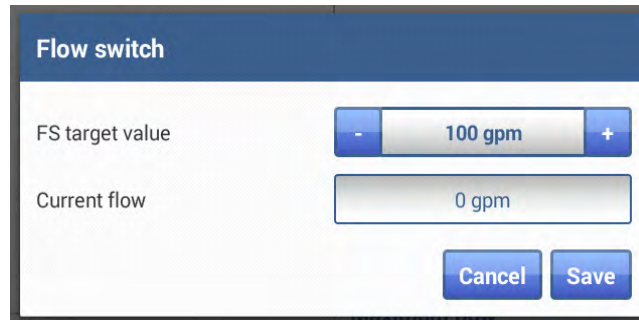


Figure 7-11: Flow switch calibration window

- Step 16.** For FS Target Val, using the -/+ buttons, enter the maximum (and steady) value expected for the flow.
- Step 17.** Tap **Save** to return to the **Flow switch** tab.



Figure 7-12: Flow Setting Screen – Flow Switch

- Step 18.** For **Min Flow**, using the -/+ buttons, enter a value for the minimum water flow rate threshold, which trips the flow switch mechanism so that it shows “closed”. Below this threshold, the system enters **Low Flow Mode**.
- Step 19.** For **Low Flow Mode**, tap the arrow and from the drop-down list, select either **Restart** or **Shutdown**.
- Step 20.** The default value for flow delay appropriate for **Restart** mode are preset by Atlantium. These values may be changed based on local site requirements and conditions, but be sure to coordinate with the Atlantium representative. If you require changing this value, for **Flow Delay**, using the -/+ buttons, set the desired value.
- Step 21.** Tap **Save** to save settings.

7.5 Configuring Dose Settings

The system is set at a particular dose. The Atlantium system maintains the designated dose as conditions allow.

➔ To configure the Dose settings:

- Step 1.** At the top of the screen, tap . The **Monitor Tab** is displayed.
- Step 2.** Tap the **Dose Settings** tab. The **Dose Settings** screen appears.

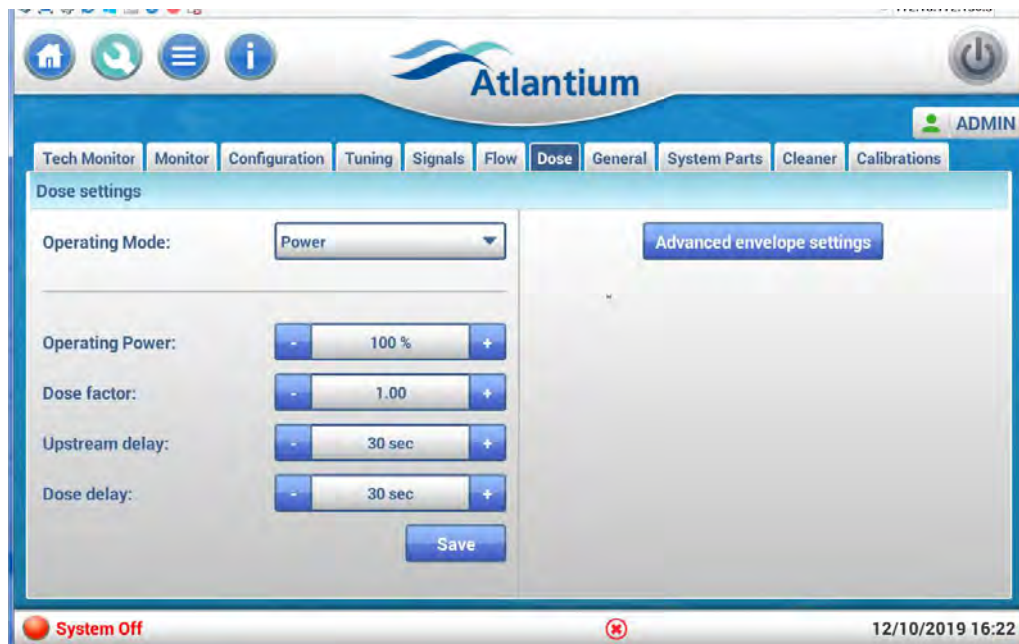


Figure 7-13: Dose Settings Screen - Power Mode

- Step 3.** For **Operation Mode**, tap the arrow and from the drop down list, select the desired mode:
- **Power mode**
 - **Dose mode**
- Step 4.** In Power mode, use the **-/+** buttons to set the desired power level. The system is set to operate at this power level as a constant. Most operational settings used in Power mode provide information or shut-down the system under specified conditions. The system operates at the prescribed power level and does not change power level to achieve dose settings.
- Step 5.** Tap **Save** to save settings.
- In **Dose mode**, continue with the rest of the settings:

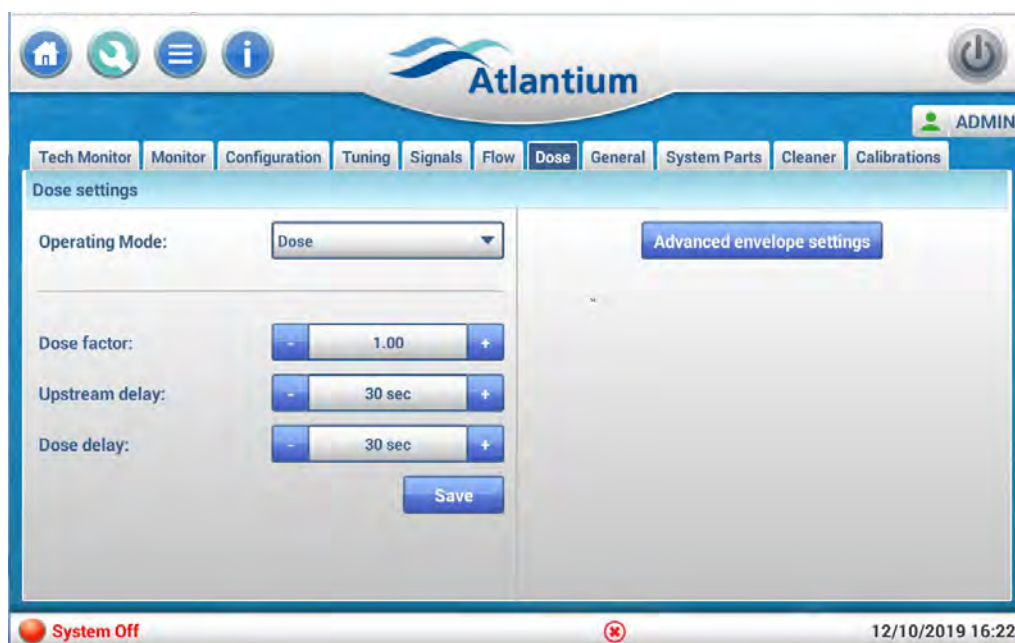


Figure 7-14: Dose Settings Tab- Dose Mode

- Step 6.** For **Minimum Dose Val**, using the $-/+$ buttons, set the lowest dose in mJ/cm^2 units under which you want the needle to move to the red zone to send a warning or shut down.
- Step 7.** For **UV Dose Set for**, using the $-/+$ buttons, set the general operating dose in mJ/cm^2 units. When the system is operating in that dose range the needle is in the center of the green zone.
- Step 8.** For **Maximum Dose**, using the $-/+$ buttons, set the maximum UV dose limit in mJ/cm^2 units, above which you want the system to send a warning or take other action. When the system operates above the maximum dose, the needle is in the dark-green zone.
- Step 9.** For **Action** corresponding to each of the above settings, tap the arrow and from the drop-down list, select the desired corresponding system response in the event that the set power level pushes the dose above or below the set values. The options are:
- **None** – The event is not considered an error, so no corrective action is taken. No warning message is issued.
 - **Caution** – One or more caution messages are issued.
 - **Shut-Down** – In case of a serious error or dangerous condition, the system shuts down and triggers a pop-up window message and a General Alarm.

The system responds as designated here when, under the current operating conditions and required power level, the dose exceeds or falls below the set values.

- Step 10.** For systems with bypass valve, for **Upstream Delay**, using the $-/+$ buttons, set the maximum time it takes for water to pass between the Atlantium system and the bypass valve + 20%. For systems with no bypass valve, set this value to 0.

7.5.1 Calibrations tab

The **Calibrations** screen is used for calibration of UVIS and UVT sensors.

- Step 1.** Tap the **Calibrations** tab. **UVT calibration** and **UVIS calibration** buttons appear.
- Step 2.** Tap either button to start the corresponding calibration procedure. The **Calibrations** tab opens

Step 3. Select the check box for each sensor you want to calibrate.

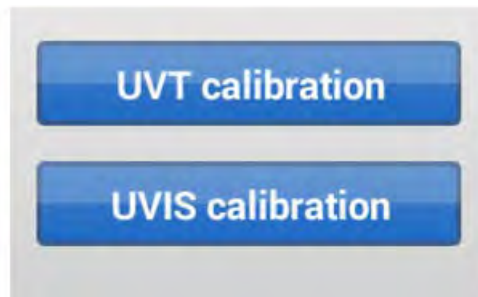


Figure 7-15: Calibrations tab

Step 4. For **UVT Bench**, enter the reading from the external calibration sensor (see *Checking/Replacing UVIS Sensors* on page 161 for these maintenance tasks)

Step 5. Press Start to start the automatic calibration procedure.

A time line displays the state of the calibration procedure

Once calibration is complete, results are displayed in the Calibrations tab.

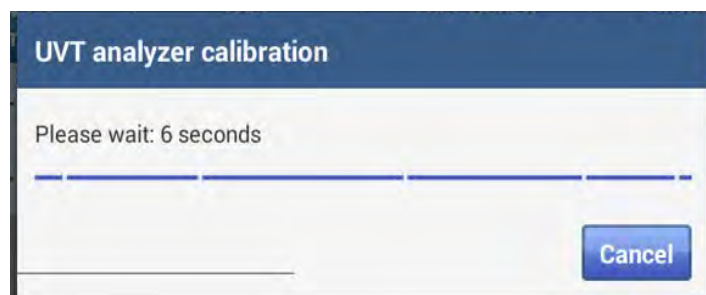




Figure 7-16: Calibrations results tab

7.6 Configuring the General & Connection Settings

You can set a number of general settings, as well as the connection settings, via the General tab.

The following General Settings are to be configured:

- Date and Time
- Format for units of measurement (US or Metric)
- Maximum Temperature Warning – Since the System shuts down automatically if the water temperature reaches the maximum allowed. You can set the maximum temperature threshold for sending a warning notification so that you can take action to prevent the water from reaching the maximum temperature that triggers shutdown. It is recommended to set this threshold to 5% under the maximum temperature allowed for your system.
- Automatic System Auto Power On in the event of restored power after power failure.
- Assign System Unit identification (PWSID (Public Water System IDentification), Plant, and Train)
- Minimum UVT threshold value for the UV transmission below which the system is to issue an alarm, warning of UVT degradation and the action to be taken accordingly.

The Connection Settings consist of the Modbus node address assigned to this system is configured in the **Control Module**, as well as the appropriate baud rate for this system.

➔ To configure the General & Connection settings:

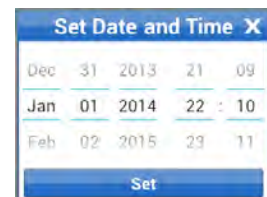
- Step 1.** At the top of the screen, tap . The **Monitor Tab** is displayed.
- Step 2.** Tap the **General tab**. The **General Settings** screen appears.



Figure 7-17: General tab

General Settings

- Step 3.** To set the date and time*, tap . The **Set Date And Time** dialog box appears.
- Step 4.** Swipe the date and time elements to toggle to the correct settings.
- Step 5.** Tap **Set**. The correct time appears on the **Output/ Input** Tab and is set for the system.



Depending on the format selected (see below) the Metric format is day first and a 24-hour clock and the US format is month first AM/PM. Figure 7-18: Set date And Time Dialog Box

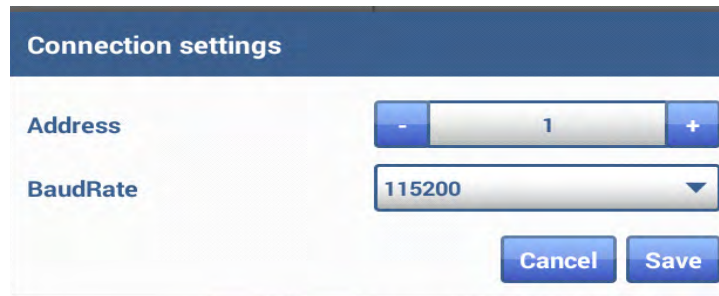
**Time may require resetting if the Control Module is disconnected for a week or more and the internal battery runs low.*

- Step 6.** To select the **Units format**, tap the arrow and from the drop down list and select the desired format, imperial for **US** units or **Metric** for metric units.
- Step 7.** When the maximum temperature threshold is reached, a warning notification is sent so that you can take action to prevent the water from reaching the temperature that triggers shutdown. For setting the maximum temperature:
- Set the system water flow to the minimum possible.
 - Wait a few minutes for the system to stabilize.
 - Check the **Control Module's** Monitor screen's temperature reading.
- Step 8.** Under **Maximum temperature**, using the **-/+** buttons, set the maximum temperature threshold to the temperature reading in **Step 7/c** above plus **5°C/41°F**. When this threshold is reached, a warning notification is sent.
- Step 9.** To enable the system to restart automatically when power is restored, mark **System auto restore On** checkbox.

- Step 10.** For each Unit defined in you system configuration, you can assign the PWSID, Plant, and Train identification. From the **Unit Number** drop down list, select the Unit number to be identified.
- Step 11.** For **PWSID**, **Plant**, and **Train**, tap the field. A virtual keyboard appears. Use it to enter the appropriate identification.

Connection Settings

- Step 12.** Tap Connection settings to open a settings window:



- Step 13.** For **Address**, using the **-/+** buttons, set the Modbus address node assigned to this system as a slave while the remote PLC is the Master.
- Step 14.** For **Baudrate**, select the appropriate baud rate for this system. The options are:
- 115200
 - 57600
 - 38400
 - 19200
 - 9600

Minimum UVT

- Step 15.** For **Minimum UVT**, using the **-/+** buttons, set the threshold value for the UV transmission below which the system is to issue an alarm, warning of UVT degradation.
- Step 16.** For **Action**, from the drop-down list, select the action to be performed if the factor falls below the Min UVT and becomes off-spec. The options are:
- **None** - The event is not considered an error, so no corrective action is taken. No warning message is issued.
 - **Caution** - The caution alerts you to an anomaly based on the user preferences.
 - **Warning** - One or more warning messages are issued. \
 - **Shut-Down** - In case of a serious error or dangerous condition, the system shuts down and triggers a pop-up window message and a General Alarm.

7.7 Importing/Exporting a System Configuration

After setting up a system configuration in the **Control Module**, you can export it to save as a backup or to use for configuring additional **Control Modules** by importing a saved system configuration.

7.7.1 Exporting a System Configuration

You can export a system configuration.

Prepare the following:

- USB Drive

➔ To export a system configuration:

- Step 1.** At the top of the screen, tap . The **Monitor Tab** is displayed.
- Step 2.** Tap the **General tab**. The **General Settings** tab appears.

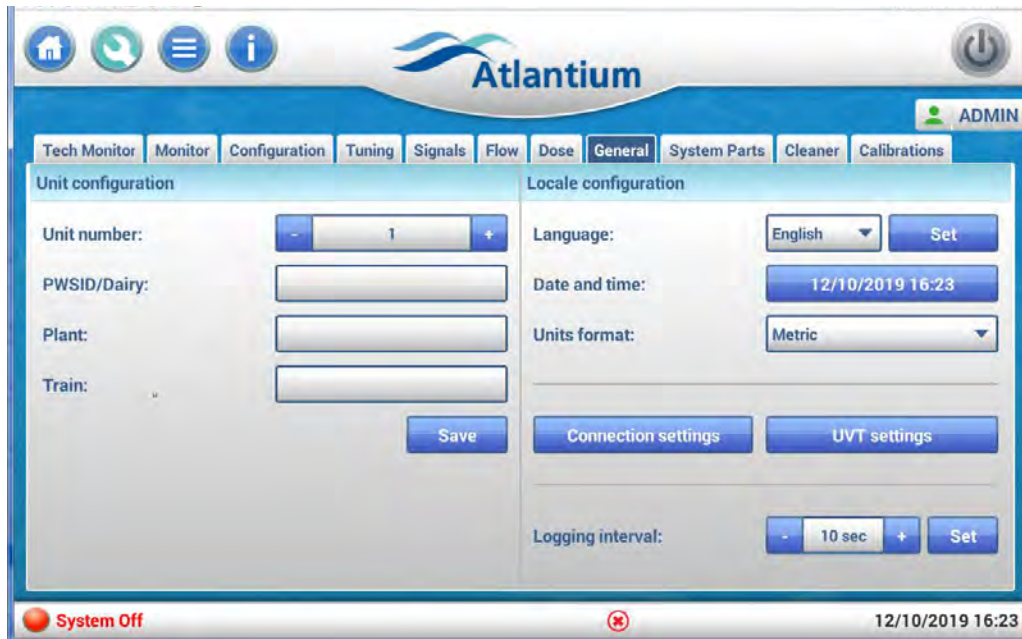


Figure 7-19: General tab

- Step 3.** Open the **Control Module** and on the back of the door, insert the USB drive into one of the USB ports at the top of the circuit board, as shown in the figure below.



Figure 7-20: Control Module USB Port Location

- Step 4.** On the **General tab**, tap **Export Configuration**. The system configuration is copied to the USB drive. confirmation appears.
- Step 5.** Remove the USB drive and keep it in a safe place.

7.7.2 Importing a System Configuration

You can import a system configuration.

Prepare the following:

- USB Drive containing a saved system configuration

➔ To import a system configuration:

- Step 1.** At the top of the screen, tap . The **Monitor Tab** is displayed.
- Step 2.** Tap the **General tab**. The **General Settings** tab appears.
- Step 3.** Open the **Control Module** and on the back of the door, insert the USB drive (containing the saved system configuration) into one of the USB ports at the top of the circuit board, as shown in Figure 7-20 above.
- Step 4.** On the **General tab**, tap **Import Configuration**. A popup appears asking you select the source.
- Step 5.** Tap **USB Flash**. The system configuration is copied to the **Control Module**.
- Step 6.** You are prompted to restart the **Control Module**. Tap **Yes**. The **Control Module** restarts with the imported system configuration.

7.8 Adding Users

Typically, the designated facility manager defines user names for the relevant personnel. However, each User can add addition Users with the same authorization type or in the hierarchy under him/her.

All-In-One screen access is according to the table below.

Table 7-2: User Hierarchy and Permissions

Section	Screen	Manager	Operator	No User Logged On
	Event Log	X	X	X
	System Info	X	X	X
	Monitor	X	X	X
	Lamps	X	X	
	Monitor	X	X	
	General	X	X	
	Cleaner	X	X	
	Tech Monitor	X		
	Tuning	X		
	Signals	X		

Table 7-2: User Hierarchy and Permissions

Flow Setting	X		
Dose Settings			
Settings System Parts	X		

➔ To add a User:

Step 1. Tap the User Indicator . The User Administration Tasks dialog box appears.

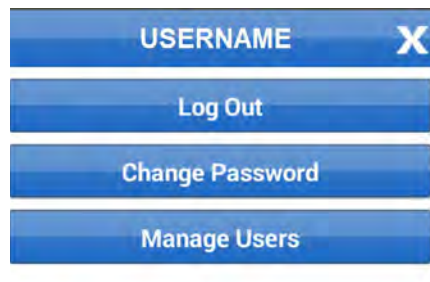


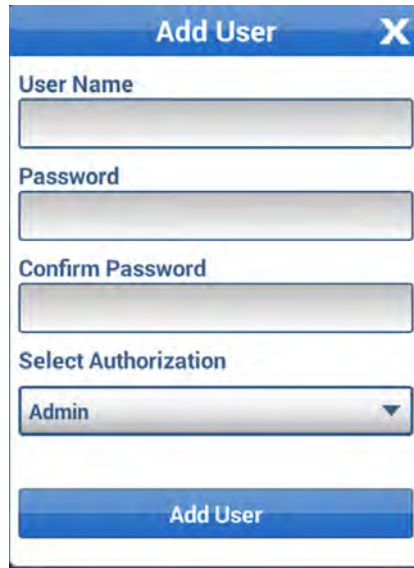
Figure 7-21: User Administration Tasks Dialog Box

Step 2. Tap **Manage Users**. The Manage Users dialog box appears.



Figure 7-22: Manage User Dialog Box

Step 3. Tap **Add User**. The Add User dialog box appears.



The 'Add User' dialog box has a blue header with the title 'Add User' and a close button 'X'. It contains four input fields: 'User Name', 'Password', and 'Confirm Password', each with a light blue border. Below these is a 'Select Authorization' dropdown menu with 'Admin' selected. At the bottom is a blue button labeled 'Add User'.

Figure 7-23: Add a User Dialog Box

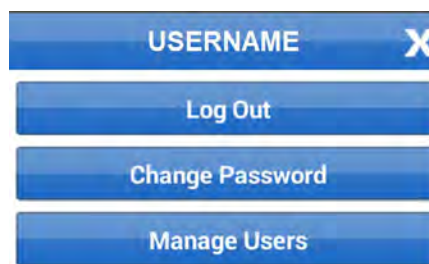
- Step 4.** Tap the **User Name** field. A virtual keyboard appears. Type in the user name.
- Step 5.** Tap the **Password** field. A virtual keyboard appears. Type in the password.
- Step 6.** Tap the **Confirm Password** field. A virtual keyboard appears. Type in the same password to confirm it.
- Step 7.** From the **Select Authorization** dropdown list, select the desired User type
- Step 8.** Tap **Add User**. The new User is added to the system and appears on the list of Users.

7.9 Editing/Deleting Users

Each User can reset the password to the default password **1234** of Users with the same authorization type or in the hierarchy under him/her or delete any of those Users.

➔ To edit/delete a User:

- Step 1.** Tap the User Indicator . The User Administration Tasks dialog box appears.



The 'User Administration' dialog box has a blue header with the title 'USERNAME' and a close button 'X'. It contains four blue buttons stacked vertically: 'Log Out', 'Change Password', and 'Manage Users'.

Figure 7-24: User Administration Dialog Box

- Step 2.** Tap **Manage Users**. The Manage Users dialog box appears.



Figure 7-25: Manage User Dialog Box

- Step 3.** From the **Select User** dropdown list, select the desired User. The Manage User dialog box appears displaying additional options.



Figure 7-26: Add a User Dialog Box

- Step 4.** To Reset the password of the selected User, tap **Reset Password To Default**. The password is reset to **1234**.
- Step 5.** To delete a User, tap **Delete User**. The User is deleted.

8 First Time System Activation

Following **Control Module** configuration, completing the system commissioning includes the following major steps:

- Cleaning the Atlantium Unit (See below)
- Filling unit with water (See below)
- Igniting the UV Lamps (See below)
- Initiating the Water Flow (See page 98)
- Adjusting the Auxiliary Equipment (See page 98)
- System Tuning (See page 98)
- Installation QA (See page 98)

8.1 Cleaning the Atlantium Unit

The inner surfaces of the Atlantium Unit are to be cleaned via the CIP process at installation, before starting up and proceeding with the commissioning phase. Follow the directions in ***Cleaning In Place (CIP)*** on page 109.

8.2 Filling unit with water

The unit must be filled with water before full activation.



Water hammer can cause damage to any hydraulic system, including the Atlantium Unit. **Make sure that water hammer is NOT present in your facility's water line.**

To prevent water hammer, consult your system engineer and check your facility system procedures, including:

- When starting the water flow when the lines are empty, keep valves open so that the air in the pipes has a simple release pathway.
- Sudden valve closures can cause water hammer. Make sure that valves close gradually enough to avoid this problem.
- Ensure that all pumps used with the Atlantium system employ soft start-up procedures.

➞ To introduce water:

- Step 1.** Open the **Inlet valve** and, using the **Viewport** on the Unit, check that the quartz chamber fills with water completely.

8.3 Igniting the Lamps

The lamps of the Atlantium Unit are ignited to commence live system operation.



- Temporarily, the breather caps on the lamps must be left open to allow any condensation water to evaporate. During this time, set up Caution floor signs at a distance of 1m/40inch from the Atlantium Unit to warn passersby against possible UV light exposure and not to approach without proper eye and skin protection.
- Wear white cotton gloves.
- Wear long sleeves rolled down to your wrist.
- Where appropriate eye protection, such as containing polycarbonate lens that meet EN166, CAN/CSA-Z94.3-02 and/or Z87.1 standards.
- DO NOT look directly into openings that emit UV light.

➔ To ignite the lamps:

Step 2. In the **Control Module**, log in and at the top of the screen, tap . The **Monitor Tab** is displayed.

Lamp ID	Power %	Status %	UVT %	UVIS mA	UVTS mA	Lamp kW	Lamp V	Lamp A	Temp °F	On Off	Lamp Enabled
1	100	100	97.0	17.8	18.0	1.70	220.00	7.72	77		☒
2	100	99	97.1	17.8	18.0	1.70	220.00	7.72	77		☒
3	100	99	0.0	17.8	0.0	1.70	220.00	7.72	77		☒
4	100	99	97.1	17.8	18.0	1.70	220.00	7.72	77		☒
5	100	99	0.0	17.8	0.0	1.70	220.00	7.72	N/C		☒
6	100	100	0.0	17.8	0.0	1.70	220.00	7.72	N/C		☒
7	100	99	0.0	17.8	0.0	1.70	220.00	7.72	N/C		☒
8	100	99	0.0	17.8	0.0	1.70	220.00	7.72	N/C		☒

Figure 8-1: Monitor Screen*

** This screen (as well as all screens shown in this manual) is a generic example. The number of lamps shown on the actual screen are configured according to the number of lamps in your Atlantium Unit.*

Verify that all of the lamps in your system are enabled - check box appears filled .

- Step 3.** Post temporary Caution signs on the ground 1 m away from the Atlantium unit to warn bystanders of possible UV light exposure and require the use of appropriate skin and eye protection to approach.
- Step 4.** Open each of the lamp vent caps (2 on each side of the lamp). For the location of the lamp vent caps, see Figure 10-18 on page 149.
- Step 5.** Check that the Main screen shows that system is working properly according to spec (see Figure 8-2)

- Step 6.**  Let the system operate for at least one hour.
- Step 7.** Continue with Section 8.5 below while the lamps complete the first hour of operation.
- Step 8.** After that first hour, close all of the lamps' breather caps and remove the Caution floor signs.

8.4 Initiating the Water Flow

When initiating the water flow, the water may be set to drain out or collect in a tank until the system is fully operational. Once the entire installation and QA testing is complete, the water is set to flow normally as part of the facility operation.

➤ To initiate the water flow:

- Step 1.** Open the relevant valve that allows water to flow through the Atlantium unit. This may be the draining valve or bypass valve.
- Step 2.** Open the viewport and verify that the Unit is completely filled with water and that no air gaps and no air bubbles exist.

On the **Control Module**, verify that the Flow reading on the right is within normal range. If it is not, check the connections between the Atlantium system and the Flow meter. See **Control Module Connections** on page 56.

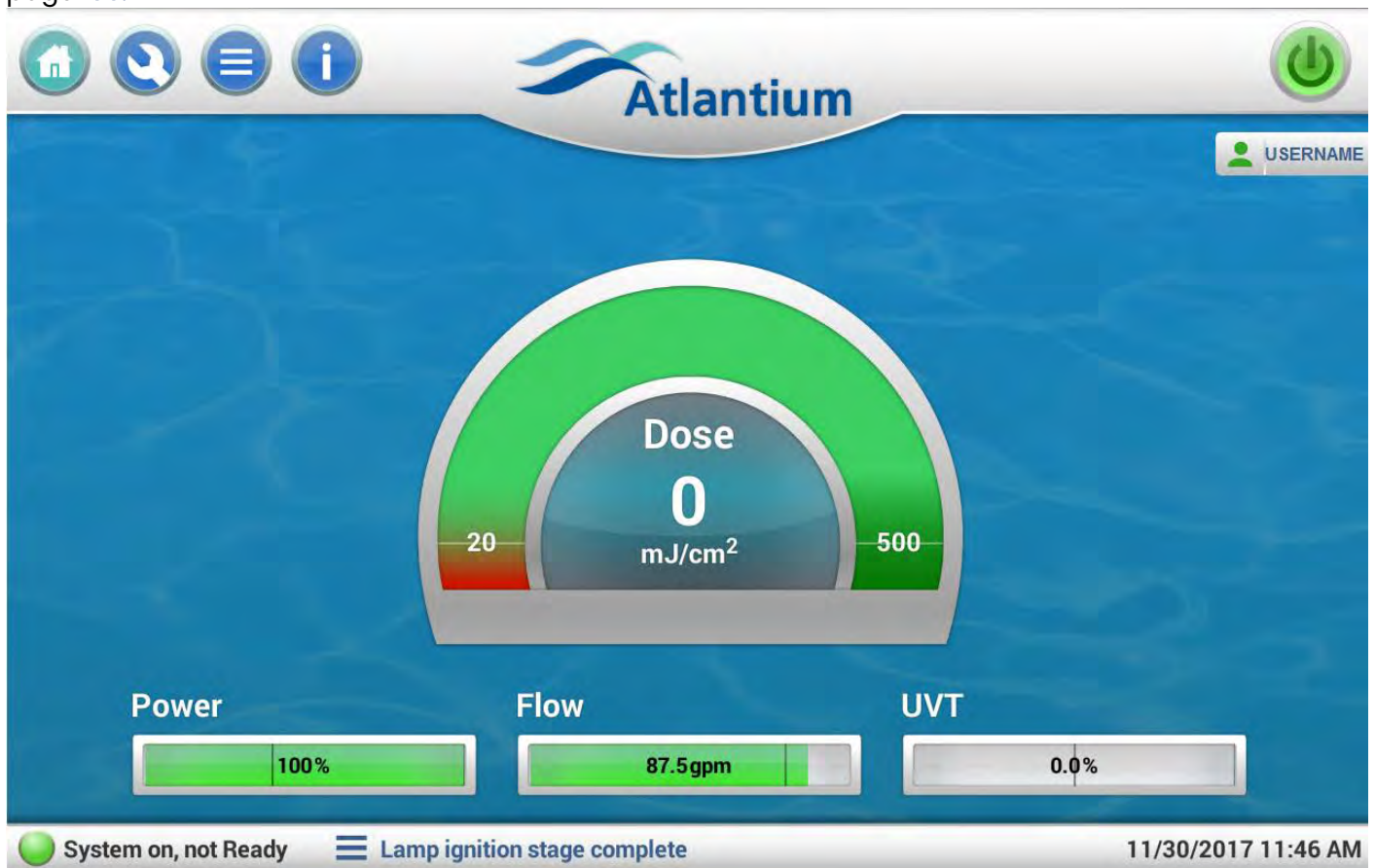


Figure 8-2: Main screen showing on-spec system (generic)

8.5 Adjusting the Auxiliary Equipment

All relevant auxiliary equipment, i.e., flow meter, automatic valves, PLC, etc., must be adjusted at this time. Refer to the manufacturers' user documentation of the relevant equipment.

If your configuration includes a flow meter, check that the value of the flow displayed on the **Control Module** matches the reading on the flow meter. For flow meter calibration, see *Configuring the Flow Settings* on page 79.

8.6 System Tuning

The Atlantium system must be tuned via the **Control Module**. Follow the instructions in *Replacing/Checking a UVT Analyzer or UVIS Sensor* on page 128.

8.7 Installation QA

Atlantium personnel performs a quality assurance procedure (see *Installation Check List* on page 82) following which the Atlantium system is ready for regular facility operation.

9 Operating the Atlantium System

This section details how to operate the system and handle basic and periodic routines to be carried out by Atlantium- trained the local staff.



WARNING!

- Unauthorized servicing or modification of this system in a manner not specified in this manual could expose personnel to potential electrical or other hazards.
- Improper use or adjustment of this system may invalidate the service warranty agreement.

9.1 Operational Guidelines

Follow these guidelines for smooth Atlantium System operations:

- **Incoming Water Supply** – Be alert about incoming water supply. If you do not have pre-treatment, be alert to reports of contamination by upstream slugs of mud, particles, oil, etc. If the water source was not previously in operation or the line is new, – flush the line prior to introducing water into the Atlantium system
- **Avoid chemicals that are harmful to UV systems** – Avoid caustic soda and other corrosive chemicals that can degrade gaskets and O-rings. For information on approved chemicals for CIP use, see *Selecting the Correct Chemicals for CIP* on page 128
- **Prevent Water Hammer** – Water hammer can cause damage to any hydraulic system, including the Atlantium unit. Take steps to ensure that this phenomenon does not occur in your facility's water line. To prevent water hammer, consult your system engineer and check you facility system procedures for following:
 - Avoid filling empty lines with water without keeping valves open for releasing the air from the pipes and the system.
 - Avoid rapid closure of water lines by sudden closure of valves.
 - Before introducing the water, ensure that all pumps that impact the Atlantium system employ soft start-up procedures.
- **Avoid vibrations on upstream and downstream piping** – Make sure that the piping to be attached to the Atlantium Unit is reinforced to protect the Unit against vibration.
 - A high vibration level can cause damage to any hydraulic system, including the Atlantium unit.
 - High vibration can be caused by an unbalanced pump or by sudden valve close.

9.2 Basic Operational Tasks

- Starting up the Atlantium system – See page 100
- Taking microbial samples – See page 107

9.2.1 Shutting Down the Atlantium System

In a non-emergency situation, you can shut down the Atlantium system from the **Control Module**. Be sure to enable the **Auto Power On** function in the **Control Module** if you want the system to save your settings. (See *Igniting the Lamps* on page 96.)

9.2.2 Starting up the Atlantium System

You can start up the Unit via the **Control Module** and then check the water and system parameters. See page 70

➔ To start up the Atlantium system:

Step 1. Check that the electrical cabinet's power cable is connected to the main power source.

Step 2. On the Main screen of the **Control Module**, tap .

Step 3. Wait 10 minutes for lamps to turn on and the system to stabilize.

Step 4. Open the viewport and check that there are no air bubbles in the water flow. If necessary, release all air bubbles from within the system.

Step 5. Check the main screen of the **Control Module** and verify that the Atlantium system parameters for **Power**, **Flow**, and **UVT** are within wanted acceptable range (use the table below to enter your expected values).

Notify the executive maintenance engineer in case the values are out of range.

Table 9-1: Abbreviation of Table in **Appendix C. Checking the System Parameters**

	Parameter	Expected Value	
		Minimum	Maximum
1	Power		
2	Flow Rate		
3	UVT		
4	UV Dose		
5	Lamp Status		



For periodic checks on Atlantium system parameters, a convenient check list is supplied in **Appendix C**

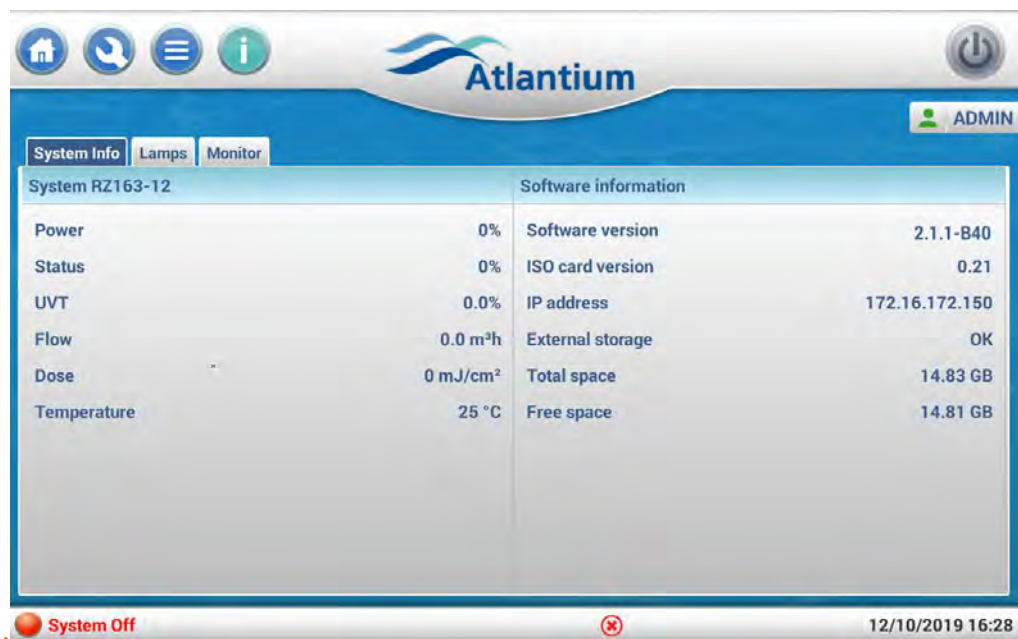
9.2.3 .Viewing System Information

You can view general information about the system in the **System Info** screen.

➔ To view generic System information:

At the top of the screen, tap . The **System Info** screen is displayed, containing the following information:

- Power in % of maximum power
- Status in % (Average lamp status, relative to output of a new lamp)
- UVT in %
- Flow in m³/h
- Temperature
- Dose in gpm or h³/m
- Software version (of the **Control Module**)
- ISO version (of the **Control Module** card)
- IP Address (Optional for connecting to the Control Module via remote PC to generate history Excel files)
- External Storage (SD card for backing up system data, i.e., event log, system configuration, system parts info, history)



Total Space GB.  System Off  12/10/2019 16:28

Figure 9-1: System Info Screen

9.3 Viewing Lamp Information

You can view information about the lamps in the **Lamps** screen.

➔ To view information about the Lamps:

Step 1. At the top of the screen, tap . The **System Info** screen.

Step 2. Tap the **Lamps** tab. The Lamps screen is displayed, containing the following information:

- ◆ Lamp ID
- ◆ Status in % (lamp status, relative to output of a new lamp)
- ◆ Age in hours
- ◆ Number of ignitions
- ◆ State: On, Ignition, or Off



9.4 Lamps Screen Viewing the System Monitor

You can view information about the system status in the **Monitor** screen.

➔ To view the System Monitor:

Step 1. At the top of the screen, tap . The **System Info** screen.

Step 2. Tap the **Monitor** tab. The Monitor screen is displayed, containing the following information:

- Lamp ID – The ID number of the lamp.
- LM Card SW – The software version of the Connection Box card to which the lamp is connected.
- Power % – The Power supply's power in percentage.
- Status % – The Real time efficiency status of the lamp in percentage.
- UVT% – The quality of the UV transmittance through the water in percentage.
- UVIS mA – The output current of the UV Intensity Sensor adjacent to the lamp
- UVTS mA – The output current of the UV transmittance sensor through the water
- UVT A, UVT B % – Quality of transmittance measured by two separate sensors (RZM systems)
- UVTS A, UVTS B mA – The output current of the 2 separate UV transmittance sensors (RZM systems) Lamp kW –

- Lamp V -
- Lamp A -
- Temp - The lamp's temperature reading in either °C or °F.

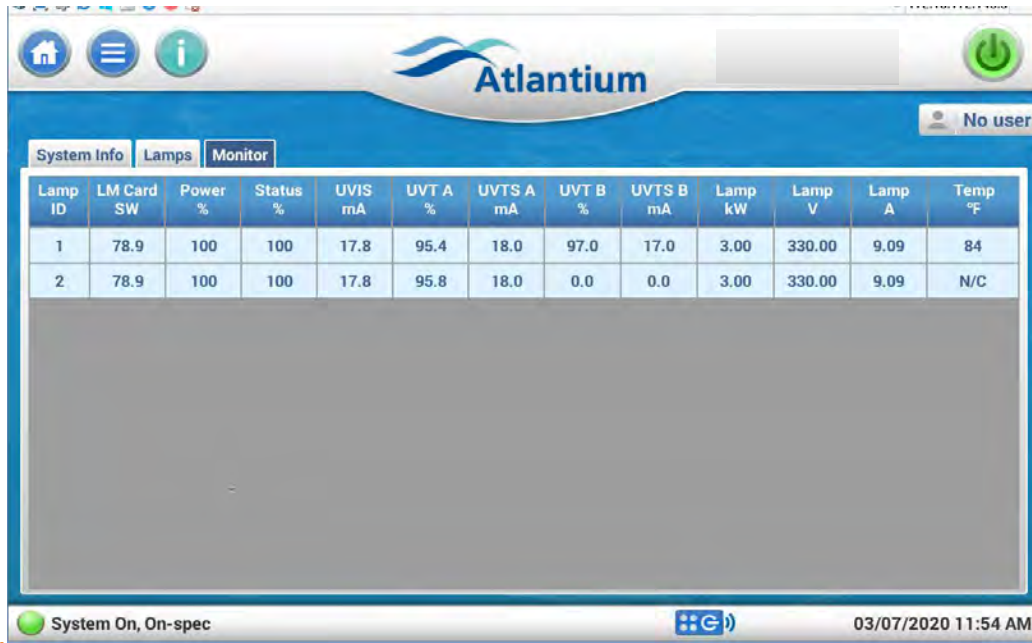


Figure 9-2: Monitor Screen

9.5 Viewing the Alarm Log

Every defined event /alarm is written to the Alarm Log of the **Control Module**. alarms with alerts also trigger an alert icon at the bottom of the screen. You are can acknowledge each alarm or acknowledge all together.

➔ To view the alarm log and acknowledging alarms:

At the top of the screen, tap  the **Alarm Log** screen is displayed.



Figure 9-3: Event Log Screen

To view more information about a single alarm, tap the alarm. The alarm item expands to display the full information, including reading for System Power, Dose, Flow, Status, UVT, and Temperature at the time the alarm occurred. When expanding an alarm with an alert, the alert is acknowledged, and relevant icons at the bottom of the screen disappear. To acknowledge all alerts, tap the **Acknowledge all** button. The relevant icons at the bottom of the screen



disappear.

Figure 9-4: Log Screen – Event Information Examples

9.6 Taking Microbial Samples

The designated aseptic sampling valves (i.e., EGMO ESV valves) are the **Inlet** and **Outlet Sampling Points** located on the sides of the Atlantium Unit. Before taking a microbial sample, a valve is sterilized internally with Ethanol 70%, as well as with flame.

Required Equipment

■ Sampling containers that are clean, laboratory-grade, sampling bottles with a volume no less than 100ml (3.38oz).	■ Permanent marker
■ Ethanol 100-70%	■ Sterile gloves
■ Bunsen Burner and lighter	■ Proper Sampler clothing, according to local laboratory protocol



CAUTION!



- Use gloves during sampling and treat them with Ethanol 70%.
- Use **sterile disposable** sampling containers.
- Prevent contact between the sampling containers and the valves' surfaces.
- Be sure that no foreign matter or liquids fall into the sampling containers.
- Perform the sample collection as fast as possible to minimize environmental contamination of the sample.
- **Use open flame to treat the surrounding air while sampling.**

➤ To take a microbial sampling:

Step 1. Open the Sampling valve and allow water to flow for 2-3 minutes. Then close the valve.

Step 2. If the sampling valve contains a cover at the top, open its cover and place it in a clean place.

Step 3. Spray all over the Sampling valve and inside it thoroughly with Ethanol 70%.

Step 4. Using a Bunsen burner, apply the flame comprehensively over the entire Sampling valve surface from its connection to the main pipe to its spout.

Step 5. Spray the cap thoroughly with Ethanol 70% and place it on the Sampling valve.

Step 6. Open the valve and allow the water to flow for 2 minutes. If you have applied the flame properly, vapor emerges from the spout. (If there is no vapor, application of the flame was insufficient. Repeat the previous step.)



Figure 9-5: Cleaning the Sampling Valve with Ethanol



ATTENTION!

Make sure never to touch the opening at the Sampling container's neck. While the Sampling container is open, do not turn the cap upside down.

Step 7. On the Sampling container, with one hand, remove its cap carefully, avoiding touching the inside of the cap with fingers or any other object. **Do not turn the cap upside down.**

Step 8. With your other hand position the Sampling valve under the water stream and take a water sample. Make sure never to touch the valve's opening or the opening at the container's neck. Fill the sampling container, leaving ample air space in bottle (at least 2.5 cm).

Step 9. Carefully replace the container's cap and tighten it to avoid cross contamination.

Step 10. Label the Sampling container with the following data:

- Sampling site
- Sampling point
- Sampler name
- Date the sample is taken

Step 11. Close the Sampling valve.

Step 12. Spray the sampling container thoroughly with Ethanol 70%.



Figure 9-6: Taking a Sample

Step 13. Store the water samples in a dark, refrigeration unit at a temperature of 2–8°C (35.6–46.4°F) until they are examined.

9.6.1 Microbial Analysis Guidelines

The microbial analysis is to be conducted according to the company guidelines which are used in the local microbiology laboratory.

- Atlantium's recommendation is to use the microbial filtration method for the testing.
- If a number of consecutive anomalous microbiological results are obtained in one of the sampling points, take the samples only after performing CIP with a disinfection chemical (peracetic acid is recommended). It is important to let the chemicals clean and sanitize the aseptic valves, as well by letting the cleaning solution flow through them. See *Cleaning In Place (CIP)* on page 109.
- In the event that system outlet counts are higher than the inlet counts, the results are to be considered as a sampling error.
- In the event that the microbial counts are not proportional in different dilutions, the results are to be considered as a laboratory error.
- For further guidance regarding microbial sampling and analysis, contact the Atlantium Application department.

10 System Maintenance

To keep the Atlantium system in peak form, preventive and periodic routine maintenance is essential. During commissioning, Atlantium trains your local staff how to do the tasks on a prescribed schedule or as needed.

The periodic routine maintenance schedule is:



WARNING!

- Unauthorized servicing or modification of this system in a manner not specified in this manual could expose personnel to potential electrical or other hazards.
- Improper use or adjustment of this system may invalidate the service warranty agreement.

-
- Monthly Maintenance Tasks:
 - Walk-Around Inspection Tour – see below
 - Cleaning In Place (CIP) (See page 109)
 - Checking the Electrical Cabinet Cooling Fans (See page 114)
 - Checking the Temperature Sensor Reading (See page 114)
 - Quarterly Maintenance Tasks:
 - Cleaning the Electrical Cabinet Air Filter (See page 139)
 - Quarterly/Yearly Maintenance Tasks:
 - Cleaning/Replacing the Quartz Tube and O-ring (See page 124)
 - Replacing the Electrical Cabinet Air Filters (See page 139)
 - Maintenance Tasks Performed as Needed:
 - Replacing a Lamp (See page 115)
 - Calibrating UVIS and UVT/UVIC Sensors (See page 128)
 - Replacing/Checking a UVT or UVI Sensor (See page 128)
 - Cleaning/Replacing a UVIS Mirror (See page 122)
 - Replacing/Editing System Parts in the **Control Module** (See page 134)
 - Cleaning/Replacing a UVIS Wave Guide (See page 121)
 - Testing System Lamps (See page 131)
 - Tuning the Atlantium System (See page 128)
 - Replacing a Ballast (Power Supply) (See page 137)
 - UV System Maintenance Tasks Rarely Required:
 - Cleaning/Replacing the Quartz Sleeve and O-ring (See page 124)
 - Replacing/Checking a UVT or UVIS Sensor (See page 128)
 - Replacing the Temperature Sensor (See page 130)

10.1 Walk-Around Inspection Tour



Walk-Around and inspect the Atlantium system once a month to check for water leaks, visible damage to the cables, or any other part of the system. Report any problem you find to an authorized technician.

➔ To perform a Walk-Around inspections:

- Step 1.** Walk around the Atlantium unit and inspect it. Look for leakage at the incoming and outgoing pipes.
- Step 2.** Inspect the cables running between the Unit and the Electrical Cabinet and Control Module for signs of fraying and any other visible damage.
- Step 3.** Inspect the Electrical Cabinet and Control Module for any signs of damage.
- Step 4.** Report any leakage or damage to an authorized technician immediately.

10.2 Cleaning In Place (CIP)



The CIP procedure consists of two parts:

- Cleaning to remove the accumulated particles that collect on the quartz tube over time
- Disinfectant cleaning to sterilize the quartz tube (not applicable to some Municipal installations)

Clean the Atlantium Unit's inner surfaces and quartz tube generally on a monthly basis. However, if the system performance is on target and the UVT Sensor readings are stable, you can clean using this process less frequently, depending on the specific chemistry of your water.

A CIP recirculation system is available from Atlantium as an optional CIP Kit. Connect the recirculation system to the CIP ports to create a closed-loop flow line that pumps a diluted solution through the Unit.

The estimated time for the CIP process is 1.5 hours.

Prepare the following:

- Protective chemical-safe goggles, clothing and gloves
- An external pump-driven recirculation system with a built-in 100 liter reservoir (26.5 US gallons) or Atlantium's CIP Kit including its accessories (See *CIP Ports and CIP Kits* on page 36.)
- Cleaning solution (See below)
- Disinfectant cleaning solution (not applicable to some Municipal installations) (See page 111)

10.2.1 Selecting the Correct Chemicals for CIP

It is **critically important** to select the correct cleaning and disinfecting chemical for the CIP process. It must be powerful enough to clean the inner surfaces of the Atlantium unit, while at the same time:

- Does not harm the components of the Unit that it comes into contact with (stainless steel 316, quartz and Viton® o-rings).
- Does not contaminate the product for which the water being treated by the system is used.

**WARNING!**

- The chemicals selected and the concentrations used must be approved for use by the facility's Quality Assurance and Safety officials, and must comply with the written protocols and procedures set in place by those departments.
- Use and handle in strict accordance with their manufacturer's instructions, product information sheets, and material safety data sheets (MSDS).
- Use of NaOH (Caustic Soda) may cause irreversible damage to the quartz. Use of this material requires an analysis of temperatures, concentration, frequency and duration and written approval from Atlantium's application engineer.

**WARNING!**

- It is the responsibility of the Atlantium system's owner/operator to use and apply the correct chemicals.
- The use of any other cleaning and disinfection chemicals requires Atlantium's approval.

**All chemicals must be obtained locally.**

- Atlantium does not supply chemicals – get them locally from commercial chemical suppliers.

Recommended Cleaning Solutions

The following chemicals have been tested and found to be very effective for cleaning and are recommended for use when cleaning the Atlantium system:

- Phosphoric acid (H_3PO_4) 2% diluted solution
- Nitric acid (HNO_3) 2% diluted solution
- HCl 2% diluted solution – recommended for metal based (rust-colored) deposits
- Sulfamic acid (H_3NSO_3) 1.5% diluted solution – recommended for calcium based deposits

**Recommended Disinfectant Solutions
(not applicable to some Municipal installations)**

(The Disinfectant CIP process is not applicable to some Municipal installations)

The following have been tested and found to be very effective for disinfection and are recommended for use when disinfecting the Atlantium system:

- Peracetic acid-based chemicals – 30 min. contact time, for example:
 - P-3 Oxonia Active 150 – 0.5% diluted solution according to manufacturer's recommendation
 - Divosan Forte – 0.5% diluted solution according to manufacturer's recommendation
- Hot water at 85°C (185°F) for 30 min.

10.2.2 Preparing the Chemical Solution

For both cleaning and disinfecting, the liquid in the reservoir must contain the desired percentage of the chemical additive (see above). Therefore, the purchased chemical solution must be diluted and added to the reservoir. Calculate the correct amount of chemical solution to be used for the CIP process.



- The concentrations refer to room temperature.
- The diluted solution values denote the required dilution of the chemical circulating through the Atlantium Unit.
- While calculating the diluted volumes, consider all volumes including the Unit, pipes and the recirculation tank.*

➤ To calculate the correct amount of chemical solution:

Step 1. Calculate the total volume you need for the CIP* by adding the water volumes in:

- The Atlantium unit volume (measured in liters/gallons) – Refer to the **Physical Specifications** Table in **Atlantium System Physical Specifications** on page 13
- The pipes between the Atlantium Unit and the **Inlet** and **Outlet Valves**
- The CIP recirculation system
- The hoses between the Unit and the recirculation system

Step 2. Calculate the amount of chemical additive to be added to the reservoir to achieve the desired dilution. Use this formula:

$$C = \frac{A * V_t}{B}$$

A – The target concentration for the liquid (chemical substance + water) in %

B – The concentration of the chemical substance to be added in %

C – Volume of the chemical substance to be added in liters/gallons

V_t – Volume of the water in the Unit + piping + CIP Reservoir (as described above*) in liters



CAUTION!

* If you intend to collect the solution into the CIP reservoir or other container, the total volume of the cleaning solution must not exceed the volume of the CIP reservoir or container.

Be sure to obtain from an engineer the correct calculation for the amount of chemicals to be used for the process you intend to use.

10.2.3 Performing the CIP Process

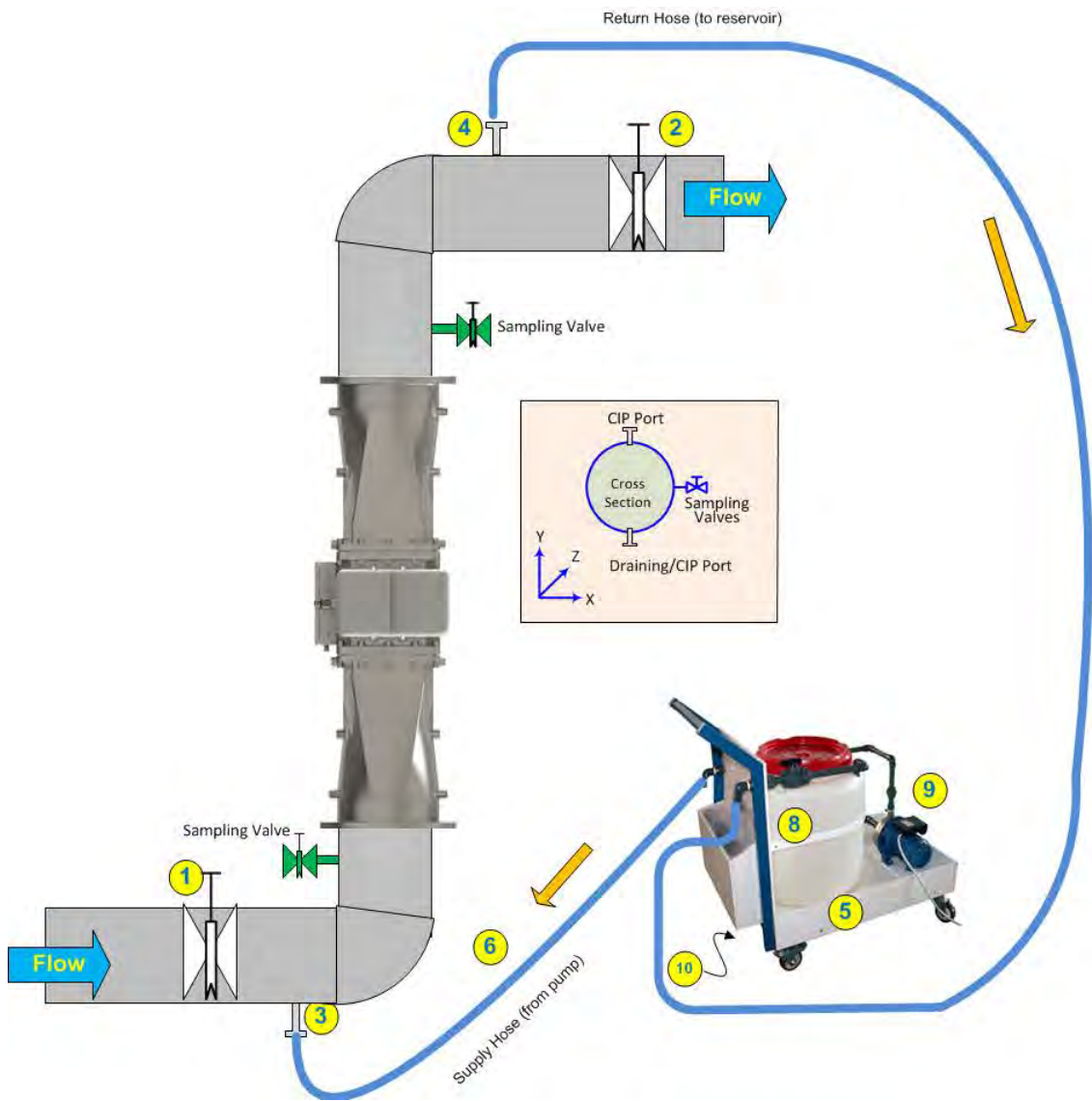


Figure 10-1: Recommended CIP Setup (generic)

➤ To Perform the CIP process:

- Step 1.** Shut down the Atlantium system, on the **Control Module**, by pressing the **Main ON/OFF Operation** button in the upper-right corner.
- Step 2.** On the Connection Box next to each lamp, turn the switch to the **Off** position.
- Step 3.** Close the Unit's inlet isolating valve (1) and outlet isolating valve (2).
- Step 4.** Remove the caps covering the CIP ports (3 & 4).
- Step 5.** Using the CIP cap's tri-clamp, connect the recirculation system's supply hose (6) to the CIP draining port (3) and the recirculation system return hose (7) to the CIP port (4).
- Step 6.** To fill the CIP reservoir (8) on CIP trolley (5) with water:
- **Filling option 1:**
 - Slightly open the Unit's inlet isolating valve (1) and open the CIP port on the outlet side (3) and allow the facility water to fill the recirculation system reservoir (8).
 - There is a full line indicator on the reservoir. Make sure the water in the reservoir (8) is up to the 100 liter (26.5 US gallons) level.
 - ◆ Close the Unit inlet isolating valve (1).
 - ◆ Close the CIP port (4).
 - **Filling option 2:**
 - Using an external water source, fill the recirculation system's reservoir (8) with 100 liters (26.5 US gallons) of room-temperature tap water (~25°C/75°F).
 - There is a full line indicator on the reservoir. Make sure the water in the recirculation system reservoir (8) is at the 100 liter (26.5 US gallons) level.
- Step 7.** Add the prepared chemical solution to the water in the reservoir. (See *Selecting the Correct Chemicals for CIP* on page 109.)
- Step 8.** Start the recirculation system's electric pump (9) and allow the solution to circulate through the Unit for **the period of time specified in the instructions for using the chemicals, but not less than 30 minutes**.
- Step 9.** As the solution is circulating through the Unit, open and close the sampling valves several times to make sure they also get flushed with CIP chemical solution.
- Step 10.** If the draining valve is separate from the CIP ports, open and close the draining valve several times as well.
- Step 11.** After the chemical solution has circulated through the Unit for a sufficient duration, the chemical solution must be drained and the system must be thoroughly rinsed while the Unit is operating.
- Step 12.** Turn on the Atlantium system at 100% power and wait until the system has stabilized and indicates that it is ready.
- Step 13.** If regulations and physical setup permit, connect the recirculation system drain valve (10) to the facility's sewage system. Close the inlet CIP port (3), stop the CIP recirculation pump (9) and open the CIP recirculation system drain valve (10). The water with the chemical solution begins to drain out.
- Step 14.** Alternatively, use the CIP reservoir to collect the chemical solution. Close the inlet CIP port (3), stop the CIP recirculation pump (9). Allow the chemical solution to drain entirely into the CIP reservoir. Dispose of the contents according to regulations.
- Step 15.** For installations with a recirculation system, to rinse the Atlantium system thoroughly, slightly open the isolating inlet valve (1). Allow the recirculation system to continue for another three cycles of water volume.
- Step 16.** To ensure that all of the chemicals are flushed out of the system, draw some water from the Sampling valve and test it using an appropriate indicator for the chemical used, such as measuring the pH (consult you chemical supplier).

Step 17. If you are planning to perform an additional CIP process using the disinfectant solution, refer to *Preparing the Chemical Solution* on page III and return to **Step 6**.

Or, to return to normal operating status, proceed to the next step:

Step 18. Open the isolating outlet valve (2) and close the CIP port on the outlet side (4).

Step 19. To avoid hammer effect if there is air trapped in the system, carefully open the inlet valve (1) completely and resume normal operation.

Step 20. Using the CIP cap's tri-clamp, disconnect the CIP recirculation hoses from CIP ports and close the CIP ports with their caps.

Step 21. Completely drain the CIP reservoir.

Step 22. Close the CIP recirculation system's drain valve (10).

Step 23. Roll up the CIP recirculation hoses and return them to their storage pocket on the CIP trolley (5).

Step 24. On the Connection Box next to each lamp, turn the switch to the **On** position.

Step 25. Turn on the Atlantium system by pressing the **Main ON/OFF Operation** button in the upper-right corner of the **Control Module** screen.

10.3 Checking the Cooling Fans



- Once a month the cooling fans of the Ballast Modules in the Electrical Cabinet must be checked to verify that the air is flowing out.
- Once a month the cooling fans in both lamp covers of the RZM unit must be checked to verify that the air is flowing out.
- Once a month both temperature sensors must be checked to verify that they are working properly.

➔ To check the cooling fans:

- Using your hand or a piece of paper, check that air is being blown out of the fans of the Ballast PS units and in the EC door.
- Similarly, check that air is being blown out of the fans in both lamp covers of the RZM unit.

10.4 Checking the Temperature Sensors' Readings

➔ To check the temperature sensors' readings:

Step 1. At the top of the screen, press  the **System Info** screen is displayed.

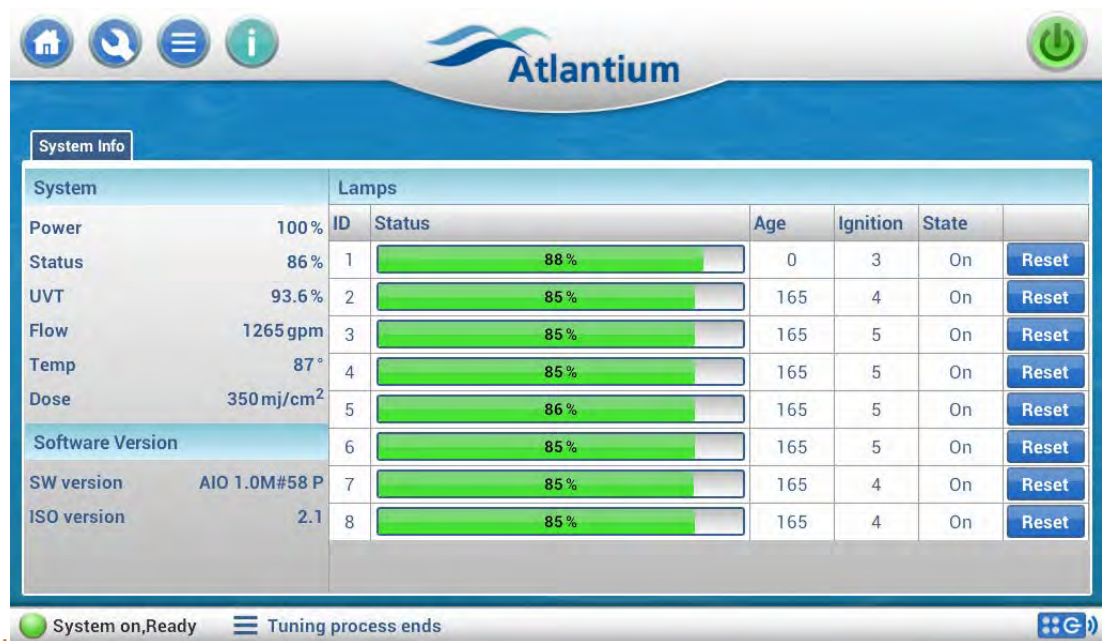


Figure 10-2: System Information Screen

- Step 2. In the middle-left, check the value for **Temp**.
- Step 3. Verify that the temperature readings are within acceptable temperature range.

10.5 Lamp & UVIS Assembly

The following Figure shows the Lamp Holder/Connector & UVIS Assembly, of which there are 8 in the RZM200-5 unit. You should refer to the Mechanical ICD documents for a detailed breakdown of the parts in this Assembly.

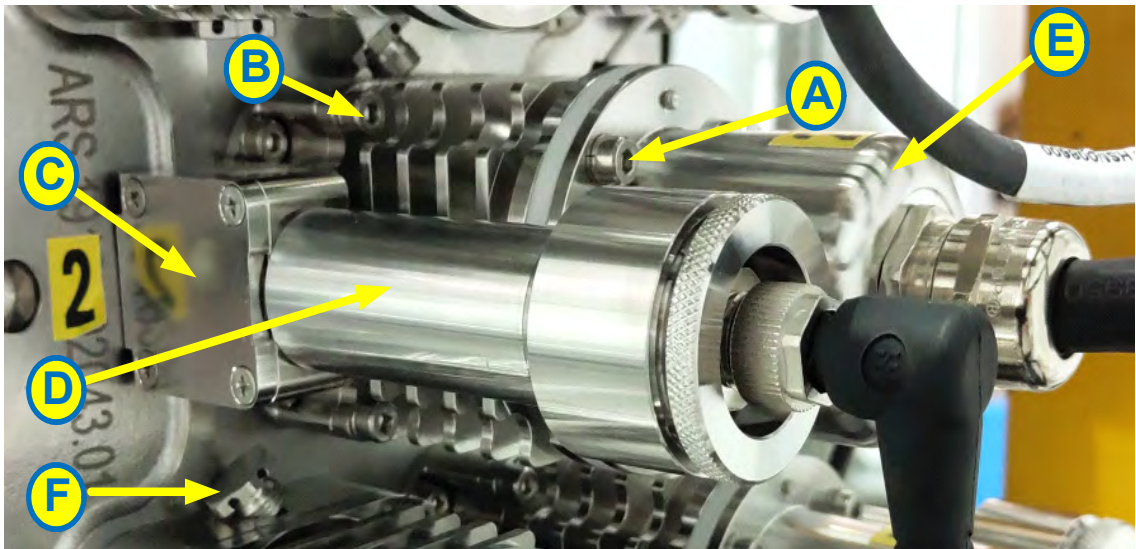


Figure 10-3: Lamp Holder/Connector & UVIS Assembly

Note the following components in this assembly:

Table 10-1: Lamp Holder/Connector & UVIS Assembly components

	Component	Details
A	Screw attaching lamp cover	1 of 4
B	Screw attaching sleeve holder	1 of 4
C	Cover of sensor mirror holder	Part of sleeve holder
D	Sensor assy	
E	Lamp cover	
F	Breather cap screws	

10.5.1 Replacing a Lamp



Performed
as needed

The UV lamp can be used as long as it provides the required dose and efficient service. You should monitor a lamp when its service status (displayed on CM) has dropped below 80%. A lamp should be replaced **ONLY** when necessary.

The entire Atlantium system must be shut down before any lamp can be replaced.

- Before replacing a UV lamp, shut down the entire Atlantium system.
- The UV lamp can be used as long as it provides the required dose and efficient service. (Generally, a lamp service status of 80% or more is considered efficient service.) Each lamp's performance is measured directly by a dedicated UV Intensity (UVI) Sensor. A lamp should be replaced when its status or performance declines and the lamp no longer provides the required performance at the appropriate electrical cost
- Once a lamp passes its rated hours, be alert to any changes in efficiency that could indicate the need to change the lamp
- To alert operator to the decline in lamp output, set a minimum UVI level in the **Control Module**. A warning is sent to the operator at both the user-set level and the internally coded minimum.

ATTENTION!



- The **RZM200-5** includes 5 lamps and each lamp has 2 connectors, one at either end. Therefore, there are a total of 10 lamp power cables to be connected!
 - Any mix up in connection may result in severe damage to the lamps and/or the ballast.
 - Each cable and lamp connector is labeled 1-5. When reconnecting, be sure that each connector is connected to a cable bearing the **SAME** number label 1-5.
-

R

CAUTION!



- After shut-down, allow the old lamp at least 10 minutes to cool down before starting replacement procedure.
- Read carefully the instructions provided in the new Lamp package.
- Wear white cotton gloves.
- Before installing a new lamp, verify that it is clean and free of grease. If not, rub it gently with the provided lens cloth. Never touch the lamp with your bare hands – hold it only by the end-connectors. Oily residues from fingerprints can damage the lamp.



- The UV lamps are designed with high internal positive pressure. Wear protective eye-wear while replacing a lamp. Pay strict attention to the safety warnings and precautions in the **Safety Overview** on page 40 in the front of this manual.
- In case of lamp breakage, refer to the Safety instructions in **Safety Overview** on page 40.

Prepare the following:

- Replacement lamp
- Lamp cleaning cloth (comes with the lamp)
- White cotton gloves
- Eye protection
- Appropriate Phillips screwdriver
- Caution signs
- Tweezers

➞ To shut down the entire system:

Step 1. Log in to system via Control Module (see page 74).

Step 2. On the top-right of the screen, press . The button turns gray. Wait approximately three minutes for the **Ready Lamps Off** signal to transmit.

Step 3. Open the **Control Module** using the attached key and locate the circuit breaker. Turn the circuit breaker to the **OFF** position.

Step 4. Open the Electrical Cabinet and turn the main circuit breaker to the **Off** position.

Step 5. Continue with the procedure, **To replace a lamp** below.

➞ To replace a lamp:

Step 1. Allow the lamps at least 10 minutes to cool down before starting.

Accessing the Lamp

Step 2. On the side of the RZM where the UVIS sensors are located, identify the lamp that is to be replaced.

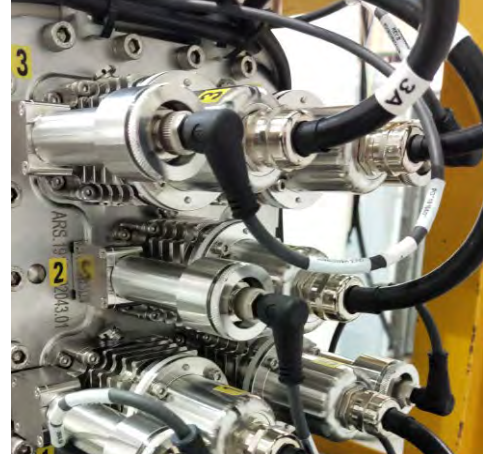


Figure 10-4: Select lamp by label

Step 3. Using an appropriate screwdriver, remove the four screws (B) that secure the lamp's connector assembly (A) and remove the connector assembly. Each screw has a washer.

Step 4. To prevent any damage to the lamp, make sure you pull the lamp's connector assembly (A) straight out and not at an angle. Gently set the connector assembly aside on a stable surface.

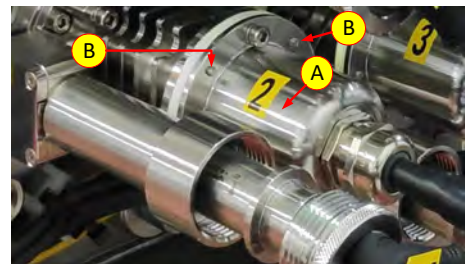


Figure 10-5: Removing the Lamp Connector Assembly

Step 5. Using an appropriate screwdriver, remove the four screws (D - two are showing) that secure the lamp's housing (C), and carefully remove the housing. Each screw has a washer.

Step 6. Using tweezers, carefully remove the lamp's stabilizer (E).

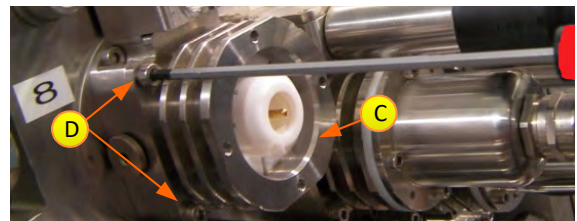


Figure 10-7: Removing the Lamp's Housing 1

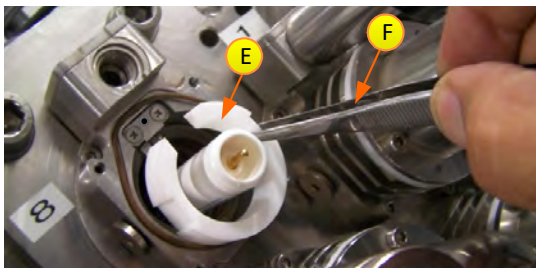


Figure 10-6: Removing the Lamp's Stabilizer

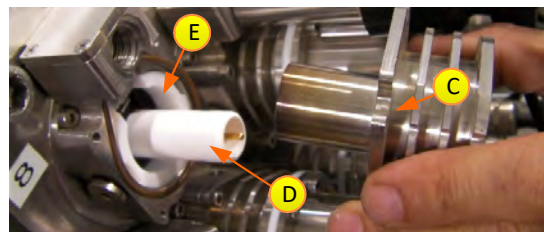


Figure 10-8: Removing the Lamp's Housing 2

Removing the Lamp

Figure 10-9 below shows the lamp and the wave guide after the lamp's housing is removed.

Step 7. Wearing appropriate cotton gloves and skin covering, carefully pull the lamp (H) straight out of the quartz tube in the Unit and set it aside on a stable surface.

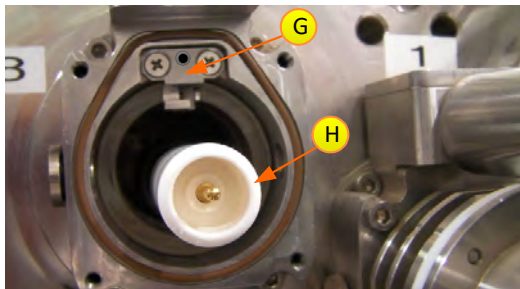


Figure 10-9: The Lamp and Wave Guide after the Lamp's Housing is Removed

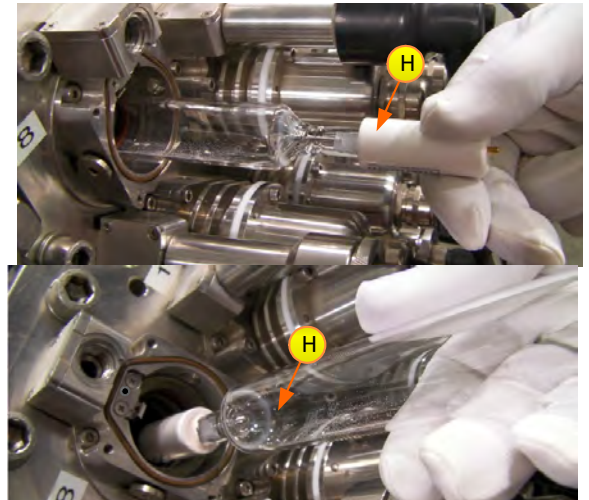


Figure 10-10: Removing the Lamp

Checking the Quartz Tube

Step 8. Check the inner surface of the quartz tube for residual moisture or high humidity. If any visible moisture exists, using a lens cleaning cloth, dry it.

Cleaning/Replacing the Wave Guide

Every time the lamp is replaced, the wave guide must be cleaned or, if needed, replaced.. See *Cleaning/Replacing the UV Intensity Wave Guide* on page 121.

Checking the O-rings

Step 9. Check that the lamp housing's O-ring is in place, clean, dry and undamaged. Replace if necessary.

Step 10. Unpack the new UV lamp from its vacuumed sealed package.

Step 11. Make a note of the serial number of the new lamp.

➔ Inserting the new lamp

Step 12. Wearing appropriate cotton gloves and skin covering, clean the new lamp using the cleaning pad provided with the lamp. Verify that no visible fingerprint or dust is on the lamp.

Step 13. Carefully insert the new lamp into the quartz tube in the Unit, all the way until it stops.

Step 14. Using tweezers, carefully insert the lamp's stabilizer (E).

Refastening the Lamp's Housing and Connector Assembly

Step 15. Insert the lamp's housing (C) that you removed in the beginning of this procedure.

Step 16. Ensure that the male contact point (4) on the end of the lamp has meshed with the female contact point (4) in the lamp's housing on the other side of the Unit.

Step 17. Using an appropriate Phillips screwdriver, fasten the four screws with their washers to secure it to the Unit.

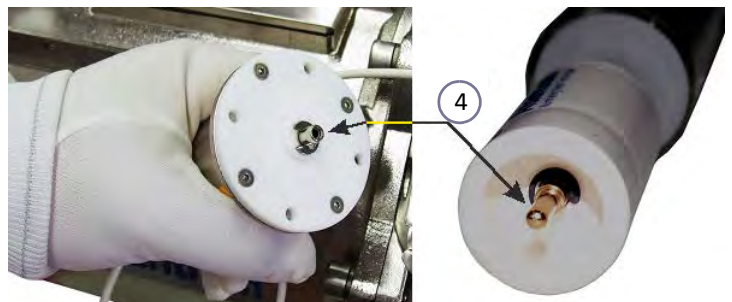


Figure 10-11: Lamp Connectors

Step 18. Insert the lamp's connector assembly (A) pushing it straight in and using an appropriate Phillips screwdriver, fasten the four screws (B) with their washers to secure it to the Unit.

➤ Reactivating the Lamp

Step 1. At the top of the screen, press . The **Monitor Tab** is displayed.



Lamp ID	Power %	Status %	UVT %	UVIS mA	UVTS mA	Lamp kW	Lamp V	Lamp A	Temp F	On Off	Lamp Enabled
1	100	54	98.6	16.5	12.5	4.20	400.00	10.50	N/C		☒
2	100	89	0.0	16.5	N/A	4.20	400.00	10.50	87.0		☒
3	100	89	0.0	16.5	N/A	4.20	400.00	10.50	N/C		☒
4	100	89	91.7	16.5	12.5	4.20	400.00	10.50	N/C		☒
5	100	89	0.0	16.5	N/A	4.20	400.00	10.50	N/C		☒
6	100	89	0.0	16.5	N/A	4.20	400.00	10.50	N/C		☒
7	100	89	0.0	16.5	N/A	4.20	400.00	10.50	N/C		☒
8	100	89	0.0	16.5	N/A	4.20	400.00	10.50	N/C		☒

System on, Ready Lamps On

Figure 10-12: Monitor Tab

Step 2. For each lamp in the system defined in the **Lamp ID** column, in the a corresponding **Lamp Enabled** column on the right, make a checkmark. The Lamp's **On/Off** icon turns gray. Check that the lamps are indeed on. Check that the values for the other columns are in normal range.

Step 3. At the top of the screen, press  the **Event Log** screen is displayed. Verify that an Event log item showing that the lamps have been ignited is displayed. The event may take a few seconds to appear on the screen.

Step 4. Activate the new lamp from the **Control Module**:

Step 5. Operate the system without the lamp's breather caps (see F in Figure 10-3 , on page 115) for one hour.

Step 6. Close all of the lamps' breather caps and remove the Caution floor signs.

➤ Recording Lamp's Serial Number

Step 7. Gather the following information about the old lamp you replaced:

- lamp's serial number
- system model
- the number of hours on the lamp usage clock when you changed the lamp
- the number of ignitions of the lamp
- the status of the lamp when you changed it

Step 8. Email this information to: support@atlantium.com.

Figure 10-13:

Observing Lamp Efficiency

Step 9. Observe the system's efficiency for 100 hours. If after this period there is a performance degradation of more than 5%, remove the UV intensity sensor and check for humidity. Clean if necessary. See the instructions on page 128.

Step 10. Perform **System Tuning** according the instructions on page 128.

10.5.2 Cleaning/Replacing the UV Intensity Wave Guide

For each UVIS sensor in the Atlantium Unit, there is an internal wave guide for directing UV light into the sensor. The UV Intensity Sensor's wave guide must be cleaned every time a lamp is replaced.

Prepare the following:

- Replacement UV Intensity Sensor's Wave Guide
- Wave Guide tool
- White cotton gloves
- Alcohol/isopropanol and lint-free lens cleaning paper
- Appropriate Phillips screwdrivers



CAUTION!



- Allow the old lamp at least 10 minutes to cool down before starting removal procedure.
- **UV EXPOSURE:** – Use Caution floor signs at a distance of 1m/40inch from the Atlantium Unit to warn passersby against possible UV light exposure and not to approach without proper eye and skin protection.

➔ To clean/replace the wave guide:

Accessing the Wave Guide to be Replaced

Step 1. On the side of the RZM where the UVIS sensors are located, identify the lamp whose wave guide is to be cleaned/replaced.

Follow the directions for removing a lamp. See **Lamp & UVIS Assembly** on page 115.

Removing the Wave Guide

Step 2. Screw the Wave Guide tool (L) onto the wave guide socket (J).

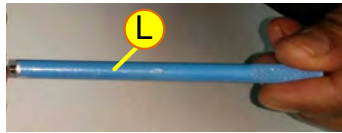


Figure 10-14: Wave Guide Tool

Step 3. While holding the Wave Guide tool (L) now attached to the wave guide (G), using an appropriate Phillips screwdriver, (M) unfasten each of the screws (K).

Step 4. With the Wave Guide tool (L), carefully pull out the Wave Guide.

Step 5. Unscrew the Wave Guide tool (L) and detach it from the Wave Guide (G).

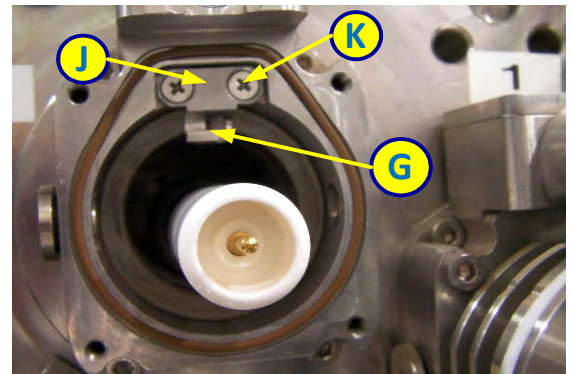


Figure 10-15: Wave Guide position



Figure 10-16: Wave Guide Tool and Wave Guide

Cleaning the Wave Guide

Step 6. While wearing white cotton gloves, carefully clean the wave guide with isopropanol and lint-free lens cleaning paper.

Step 7. Ensure that the wave guide is completely dry before returning it to its place.

Step 8. Examine the wave guide for marks and damage. Fingerprints or other stains may cause erroneous UV intensity sensor readings. If the wave guide's surface is damaged, it must be replaced.

Preparing the Replacement Wave Guide

Step 9. Screw the Wave Guide tool (L) onto the wave guide socket (J).

Step 10. With the Wave Guide tool (L), carefully insert the wave guide in place.

Step 11. While holding the Wave Guide tool (L) now attached to the wave guide (G), using an appropriate Phillips screwdriver, fasten each of the screws (K).

Step 12. Follow the directions to reinsert the lamp and refasten the lamp's housing and connector See **Inserting the new lamp** on page 119.

10.6 Cleaning/Replacing the UV Intensity Sensor Mirror

For each UVIS sensor in the Atlantium Unit, there is a small internal mirror for directing UV light into the sensor. The UV Intensity Sensor's Mirror must be cleaned every time the lamps are replaced.

Prepare the following:

- Replacement UV Intensity Sensor's Mirror
- Alcohol/isopropanol and lint-free lens cleaning paper
- Appropriate Phillips screwdrivers

➔ To clean/replace the mirror:.

Step 1. Locate the UVTIS sensor whose mirror is to be cleaned/replaced and using an appropriate Phillips screwdriver, remove the four screws that secure the square cover (A) of the sensor mirror holder. Each screw has a washer.

Step 2. Remove rubber gasket under the square cover. Replace if necessary and re-lube with silicone.



Figure 10-17: Screws (A) of mirror cover

Step 3. Carefully loosen the 2 Phillips screws (C) supporting the flange (B) which holds the mirror in place.

Step 4. Carefully remove the mirror.

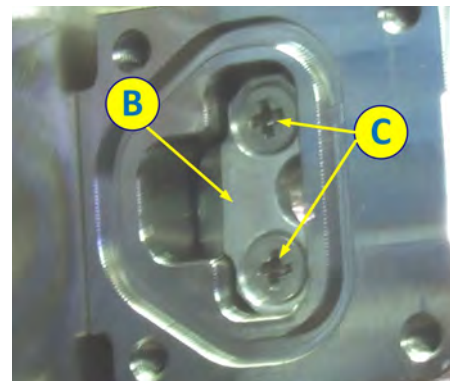


Figure 10-18: Mirror support flange

Step 5. Clean the reflective side of the mirror – the side facing into the Unit – with isopropanol and lint-free lens cleaning paper.

Step 6. Ensure that the mirror is completely dry before re-assembling the UV lamp.

Step 7. Examine the mirror's surface after cleaning and before you close the UV lamp' housing. Fingerprints or other stains may cause erroneous UV intensity sensor readings. If the mirror's surface is damaged, it must be replaced.



Figure 10-19: Cleaning the Mirror

Step 8. With the appropriate screwdriver, refasten the mirror, with its flange, in place.

Step 9. Using an appropriate Philips screwdriver, fasten the four screws with their washers to secure the square mirror cover back in place.

10.7 Cleaning/Replacing a Lamp's Quartz Sleeve and O-rings



An Atlantium Technician examines the quartz sleeves and O-rings during the yearly maintenance checks. However, in the rare event that you are asked to clean or replace a quartz sleeve, the procedure is provided here.

The quartz sleeve is cleaned as needed according to the UVT Sensor readings and when deposit collects on the quartz sleeve. **The quartz sleeve's rubber O-rings must be replaced when dismantling the quartz sleeve.**

Under normal environmental conditions, the quartz sleeve's rubber O-rings must be replaced once a year even if they pass visual inspection. For environments where the water temperature is above 70°C (160°F), the quartz sleeve's O-rings must be replaced quarterly, for which after the first year, O-rings may require replacement less frequently. For more information, consult your Atlantium representative.

Depending on the quality of the water being treated by the Atlantium Unit, the exterior surface of the quartz sleeve that houses a lamp may require periodic cleaning or replacement if after the CIP process the quartz sleeve is still not sufficiently clean.

Prepare the following:

- Two O-rings
- #4 Allen wrench
- RZM sleeve insert assembly (Sleeve Tool)
- Protective gloves
- Appropriate cleaning cloth. A mild, household cleaning solution
- Alcohol



Wear protective gloves. To protect the replacement quartz sleeve, do not touch the internal surface with your bare hands.

➤ To clean or replace the quartz sleeve:

Step 1. In the Main Control Application, set the system to Maintenance Mode. Follow the procedure in the section, *The Settings Configuration Screen* on page 75

Step 2. Open each of the EC-1/EC-2 RZM Electrical Cabinets and turn the main circuit breakers to the **Off** position.

Step 3. Using the Drain valve, drain the water from the Unit.

Step 4. Quartz sleeve removal requires that the lamp connector assemblies, lamp housings (on both ends of the lamp) and the lamp be removed. Follow the steps in the section, *Lamp & UVIS Assembly* on page 115 and remove the lamp connector assembly and lamp housing from the side of the Unit with the sensor, as well as the lamp, placing them in a secure location until they need to be reinstalled.



While pulling the lamp housing away, DO NOT tilt it and DO NOT pull it away from the lamp/sleeve axis at an angle.

Step 5. Using a #4 Allen wrench, remove the four Allen screws that secure the sleeve holder (A) opening them in a diagonal pattern (i.e. upper left, lower right, upper right, lower left) that secure the sleeve's holding ring to the Unit. Loosen one screw gradually (2 turns) on one side and do the same on the screw at the diagonal opposite side of the sleeve (2 turns).

Step 6. Continue to alternate until the sleeve holder is loose, and then remove the screws completely.

Step 7. On the other side of the quartz sleeve (the side without the sensor), remove repeat **Step 4.** to **Step 6.**

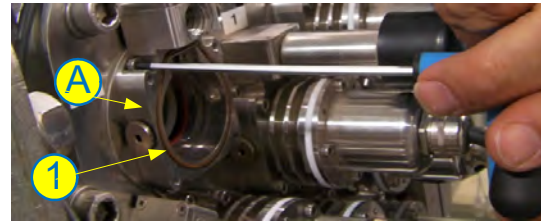


Figure 10-20: Removing the Sleeve Holder

Step 8. On one side of the quartz sleeve, remove the o-ring and insert the sleeve tool (B) all the way in.

Step 9. With the appropriate Allen wrench, fasten the four screws of the sleeve tool.

Step 10. Pull the handle so that the indicator moves from **In** to **Sleeve**, to **Out**. The Sleeve Insert Assembly pushes the sleeve out on the other side.



Figure 10-21: Pulling the Handle to Move the Indicator to Out



Figure 10-22: Sleeve Tool in case (top) and insertion



Figure 10-23: Fastening the Sleeve Tool's Screws

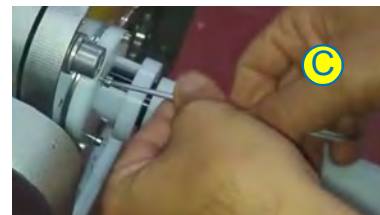


Figure 10-24: Sleeve is Pushed Out

Step 11. To prevent any damage to the sleeve, make sure you pull the sleeve straight out. Carefully pull out the quartz sleeve (F, G) and remove the O-ring (E)



Figure 10-25: Pull sleeve Out



While pulling the sleeve out, DO NOT tilt it and DO NOT pull it at an angle.

⇒ Cleaning the quartz sleeve

Step 12. If there is a deposit buildup on the sleeve's exterior surface, clean it with a mild, household cleaning solution.

Step 13. Clean the exterior and interior surfaces with alcohol (G). If the sleeve cannot be cleaned to a pristine condition, replace it with a new one.

Step 14. Ensure that there is no residual moisture or high humidity on the inner surface of the quartz sleeve. If any visible moisture exists, dry it before inserting it in place.

⇒ Returning the quartz sleeve

Step 15. Mount the new O-ring onto the quartz sleeve.

Step 16. On the Sleeve Tool, pull the handle so that the indicator moves from **Out** to **Sleeve**, to **In**.

Step 17. Holding the sleeve with the O-ring near you, insert the sleeve and make sure it moves over the Sleeve Tool into the Unit. Make sure it is centered.

Step 18. On the other side, guide the sleeve into place.



Figure 10-26: Cleaning the Quartz Sleeve

Step 19. Add the second new O-ring on the other side of the quartz sleeve.

Step 20. Unfasten and remove the Sleeve Insert Assembly.

Step 21. On each side, press in the O-ring (I) to make sure it is in place and secure around the sleeve.

Step 22. Install the quartz sleeve holder on the side without the UVIS mirror and close it loosely with 4 screws in a diagonal pattern (upper left, lower right, upper right, lower left). **Do not tighten the screws!!**



Figure 10-27: O-ring (generic)

Step 23. Install the quartz sleeve holder on the opposite side – the one with the UVIS and mirror.

Step 24. Using a #4 Allen wrench, close the screws in diagonal pattern and tighten them.

Step 25. Reinstall the lamp housings and lamp connector assemblies (on both ends of the lamp) and the lamp according to the directions in the section, **Lamp & UVIS Assembly** on page 115.

➔ Powering Up the System

Step 26. Open each of the EC-1/EC-2 RZM Electrical Cabinets and turn the main circuit breakers to the **ON** position.

Step 27. Refill the system with water and restart according to procedure on page 96.

10.8 Replacing/Checking a UVT Analyzer or UVIS Sensor

For some of the sensors, there is a disk covering the sensor with a pinhole in it. This pinhole is calibrated by Atlantium for the specific measurement point. **Do not** switch disks between sensors.

Prepare the following:

- Replacement UVIS sensor or appropriate type



Figure 10-28: Sensor with Cover

➔ To replace a sensor:

Step 1. To disconnect the sensor cable connector (A), unscrew the sensor's threaded locking nut.

Step 2. (B) from the bottom of the sensor housing

Step 3. Carefully pull the sensor (C) out of the housing .

Step 4. If a disk pinhole (D) is present it slides out.

Step 5. Unpack the new sensor (C) and the sensor disk with pinhole (if it is supplied).



Figure 10-29: Sensor Components

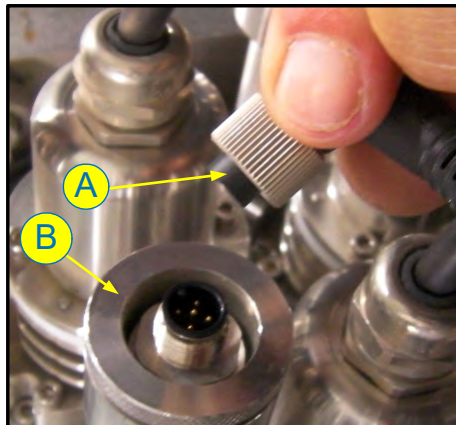


Figure 10-30: Replacing the Sensors 2



Figure 10-31: Replacing the Sensors 3

Step 6. To enable the system to track relevant data about the sensor, make a note of the serial number.

Step 7. Verify that the sensor lens (E) is clean. If needed, using a lens cloth to clean it.

Step 8. Insert the sensor (C) into the housing, and screw the threaded locking nut (B) back into the housing to secure the sensor.

Step 9. Reconnect the sensor (A) to the new sensor.

Step 10. In the **Control Module**, verify that the value for UVI for the lamp on which the sensor was replaced is within the range of **18-19.5mA**. For details on finding the Control Module screen with this information, see the section, **System Tuning** on page 132.



Figure 10-32: Replacing the Sensors 2 (Generic View 3)

10.8.1 Replacing the Temperature Sensor



In the event that the Temperature sensor requires replacement, this section details the procedure.



Follow your local regulations regarding whether a certified electrician may be required to perform this procedure.



Any person involved in handling the Connection Box card or its components is required to wear an electrostatic discharge (ESD) Wrist Strap.

Connect the ESD Wrist Strap to any bolt on the body of the Atlantium Unit or to the metal bracket under the Connection Box using a banana plug or an alligator clip.

Prepare the following:

- Replacement Temperature sensor
- #5 Allen wrench
- #3 Flat screwdriver
- #2 Phillips screwdriver

➤ To replace the Temperature sensor:

Step 11. In the Main Control Application, switch off all the lamps – follow the procedure in: ***To shut down the entire system:*** on page 117. Open each of the EC-1/EC-2 RZM Electrical Cabinets and turn the main circuit breakers to the **Off** position.

Step 12. Using the appropriate Allen wrench, open the Connection Box cover.

Step 13. Put on an electrostatic discharge (ESD) Wrist Strap and connect the strap to the body of the UV Unit **RZM200-5** as described in the note above.

Step 14. Verify with volt meter that there is no power in RZM I/O card and that no LED is lit on the card.

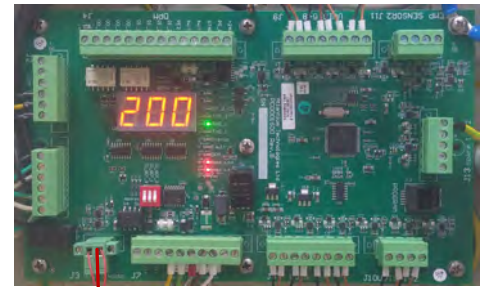


Figure 10-33: Location of the Temperature Sensor Connector

Step 15. Locate and disconnect the Temperature Sensor connector (A).

Step 16. Detach the wire from the terminal connector and remove the wire from the Connection Box via the gland.

Step 17. Using a #2 Phillips screwdriver, detach the temperature sensor (C) from the UV Unit **RZM200-5**.

Step 18. Attach the replacement Temperature sensor (B) to the UV Unit **RZM200-5**.

Step 19. Thread the wire through the gland and attach it to the terminal connector.

Step 20. Insert the connector (A) to its socket on the terminal block.

Step 21. Return the Connection Box cover and using the Allen wrench, fasten the screws. Tighten them in a diagonal pattern.

Step 22. Open each of the EC-1/EC-2 RZM Electrical Cabinets and turn the main circuit breakers to the **On** position.

Step 23. Start up the system according to procedure in ***Igniting the Lamps*** on page 96

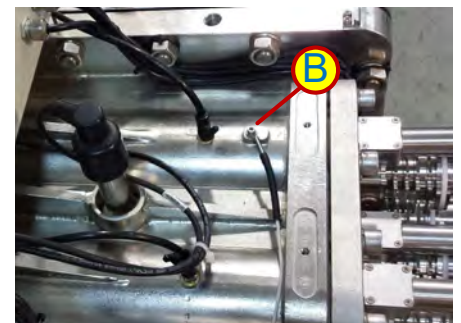


Figure 10-34: Location of the Temperature Sensor on the UV Unit **RZM200-5**

10.9 Testing System Lamps



During normal operations, an authorized facility technician may need to test the lamps via the **Tech Monitor** tab on the **Settings** screen of the **Control Unit**.

➤ To test the lamps:

Step 1. At the top of the screen, press . The **Monitor Tab** is displayed.

Step 2. Press the **Tech Monitor Tab**. The **Tech Monitor** screen appears.



Figure 10-35: Tech Monitor Tab

Step 3. For each lamp to be tested according to the **Lamp ID** column on the left, make a mark in the checkbox on the right.

Step 4. To adjust the power of the selected lamp(s), set the desired power value using the -/+ buttons. Check that the values for the other columns are in normal range.

Step 5. To turn off the selected lamp(s), press **Lamp Off**. The Lamp's **On/Off** icon turns gray. Check that the selected lamp is indeed off. Check that the values for the other columns are in normal range.

Step 6. To turn on the selected lamp(s), press **Lamp On**. The Lamp's **On/Off** icon turns green. Check that the selected lamp is indeed on. Check that the values for the other columns are in normal range.

When you leave the **Tech Monitor tab**, the lamps are returned to the settings of the system configuration.

10.10 System Tuning

System tuning is performed on these occasions:

- As the last step in system installation
- As the last step when lamps are changed and after 100 operating hours (as needed)
- Upon advice of an Atlantium Service Representative when the system is reset

➤ To tune the Atlantium system:

Step 1. Just prior to tuning, obtain a water sample from the sampling valve on the inlet side of the Atlantium Unit. Test the water sample in an external spectrophotometer. Write down the results. If a spectrophotometer is not available, record the current UVT value shown on the system's display.

Step 1. In the **Control Module**, on the top of the screen, press . The button turns gray. Wait 10 minutes for lamps to turn on and the system to stabilize.

Step 2. On the Atlantium Unit, locate each UVIS and UVT Intensity Sensors.

Step 3. Perform the following steps for each sensor (refer to the figures in the section, *Replacing/Checking a UVT Analyzer or UVIS Sensor* on page 128):

- a Unscrew the locking nut and gently pull out the sensor.
- b With a dry and clean lint free cloth, wipe the UVIS sensor's quartz window.
- c Reinsert the sensor and replace the locking nut.

Step 4. To turn on the Atlantium system, on the **Control Module** main screen, press . The button turns green. Wait approximately three minutes for the **Ready Lamps On** signal to transmit.

Step 5. At the top of the screen, press . The **Monitor Tab** is displayed.

Step 6. Press the **Tuning Tab**. The **Tuning** tab appears.



Figure 10-36: Tuning Tab (Actual numbers vary according to system configuration)

Step 7. Verify that the lamps are on and enabled (the number of lamps listed in the display is according to the number of lamps in your system configuration).

Step 8. For each lamp to be tuned, according to **Lamp ID** in column 1, make a mark in the appropriate checkbox in the last column on the right.

Step 9. Under **Status T0 %**, set the **Status T0 percentage** up to 100% using the **-/+** buttons.

Step 10. Wait a few minutes to allow the system to stabilize.

Step 11. To verify that the Sensor readings are stable, observe the reading for a few seconds to see that they remain the same numbers. For the UVIS sensor, see the **UVIS mA** column above. For the UVT Intensity sensor, see the **UVTS mA** column above. If the numbers are not stable, unscrew and remove the two breather caps on either side of the lamp. (See Figure 10-3 , on page 115.)

Step 12. Leave the caps open for 30 minutes to make sure no humidity is present in the lamp housing. Then replace the caps and check the sensor reading again.



When the Lamp's caps are removed, do not look directly into the opening without appropriate eye protection. The UV light emitted via the opening is a danger to the eyes.

Step 13. At the bottom of the screen, for **UVT To**, enter the current UVT value obtained from the spectrophotometer that you wrote down in **Step 17.** above (or the recorded current UVT value).

Step 14. To record the reason for tuning the system, At the bottom of the screen, for **Tuning Reason**, from the dropdown list, select the reason. The options are:

- System Integration
- Lamp changed
- Maintenance

Step 15. Press **Start Tuning**. A progress bar appears at the bottom of the screen. The process takes about 6 to 15 minutes.



If the system is not running, or if a lamp is not on, a popup message appears informing you that in order to start a tuning process, the system must be running.

Step 16. Verify that the UVT is the number you set previously (see the **UVTS mA** column) and that the Status on the Monitor tab is the same as the number you set in the **Status T0 %** column.

Step 17. In the **Control Module**, on the top of the screen, press . The button turns gray. Wait 10 minutes for lamps to turn on and the system to stabilize.

10.11 Replacing/Editing the System Components in the Control Module

The following system components may be replaced from time to time:

- Lamps
- UVIS
- UVTS
- Ballast (Module)s
- Cleaner
- Reference UVIS

If they are replaced, the new component needs to be recorded in the **Control Module**. If a record contains an error, it can be edited.

➔ To record a replacement system component:

Step 1. At the top of the screen, tap . The **Monitor Tab** is displayed.

Step 2. Tap the **System Parts tab**. The **System Parts** screen appears.

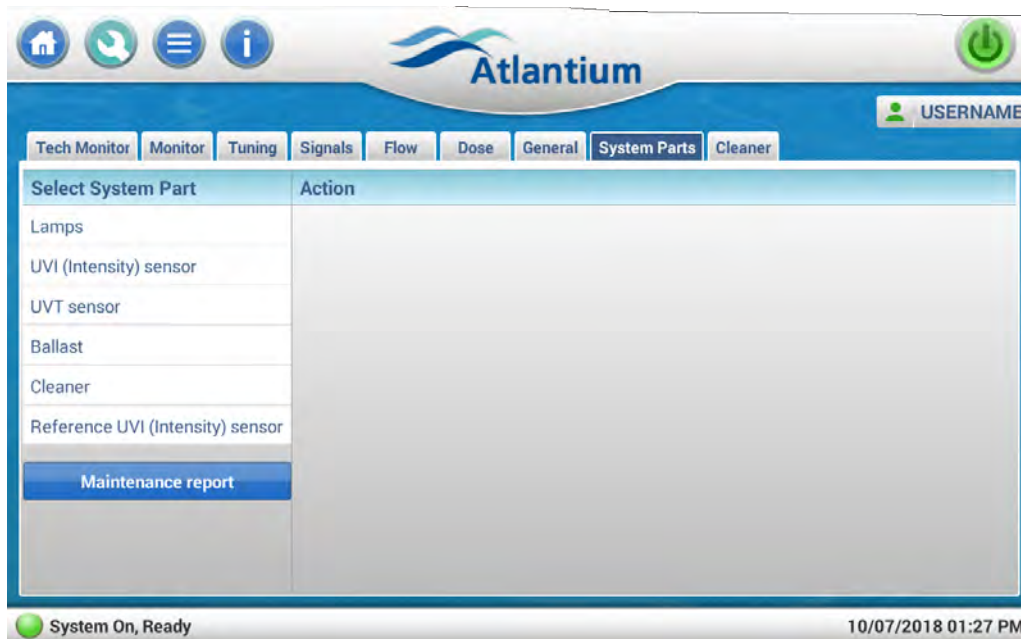


Figure 10-37: System Parts Screen – Lamps Selected

Step 3. Tap the desired system component. The options are:

- Lamps
- UVIS
- UVTS
- Ballasts
- Cleaner
- Reference UVIS

Under **Action**, the **Edit** and **Replace** buttons appear.

Step 4. Tap **Replace**. The field of the record appear.

Figure 10-38: System Parts- Lamps Replace Screen



The fields vary depending of the selected component.

- Step 5.** For **Unit**, if more than one Atlantium Unit is in the system configuration, select the Unit number.
- Step 6.** For **Choose Card**, from the dropdown list, select the corresponding Connection Box card (corresponding to the lamp associated with it) to which the component is connected.
- Step 7.** For **Serial Number**, enter the component's serial number.
- Step 8.** For **Replace Reason**, select the reason the component is being replaced. Option are:
- Expired
 - Malfunction
 - Other
- Step 9.** For **Age**, using the - button, reset the number of hours for the component's operating time to 0.
- Step 10.** Tap **Done**. The changes are saved in the database and the name of the logged in Operator appears in the Operator field.

10.11.1 Editing a System Parts Record

If you find you made an error in the record of the details of the system part, you can edit the record.

➔ To edit the system part record:

- Step 1.** Navigate to the System Parts screen as in the procedure above.
- Step 2.** Tap the desired system component.
Under **Action**, the **Edit** and **Replace** buttons appear.
- Step 3.** Tap **Edit**. The field of the record appear.

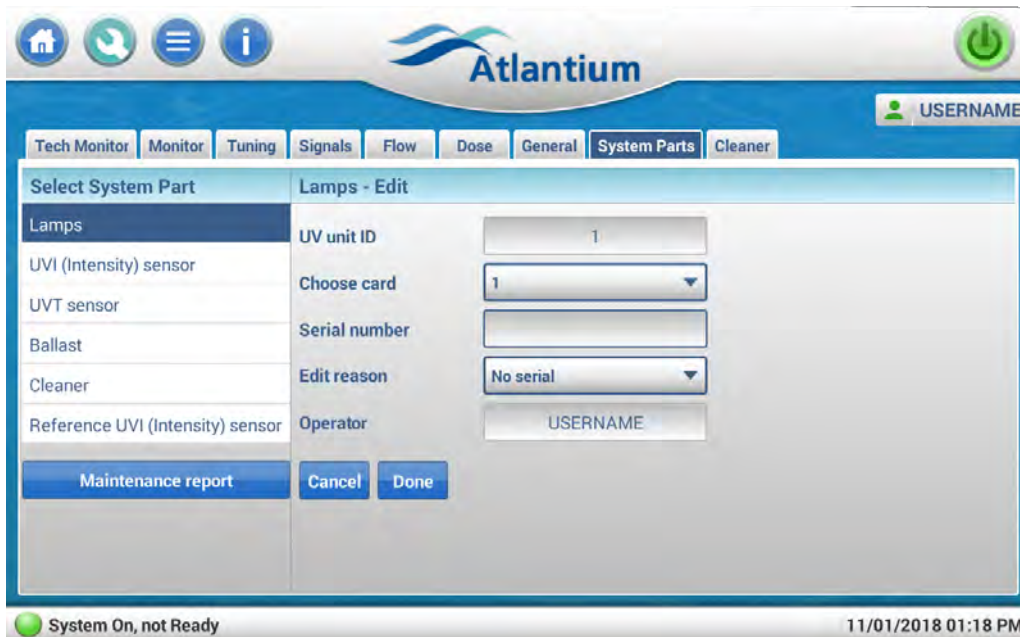


Figure 10-39: System Parts- Lamps Edit Screen

- Step 4.** Modify the fields as needed (see the procedure above).
- Step 5.** For Edit Reason, from the dropdown list, select the reason for editing. The options are:
- No Serial
 - Serial Mistake
 - Other
- Step 6.** Tap **Done**. The changes are saved in the database and the name of the logged in Operator appears in the Operator field.

10.12 Replacing a Ballast (Power Supply)



The entire Atlantium system must be shut down before replacing a Ballast (power supply).

Prepare the following:

- Replacement Ballast (Power Supply)
- Appropriate screwdrivers and wrench



The Ballast (Power Supply) is heavy. Be prepared to have to lift and carry this heavy object!

➤ To replace a Ballast (Power Supply):

Step 1. In the Main Control Application, follow the procedure in: *To shut down the entire system:* on page 117.

Step 2. Depending on which ballast needs to be replaced in the EC-1 or EC-2 Electrical Cabinet, turn the main circuit breakers to the **Off** position.

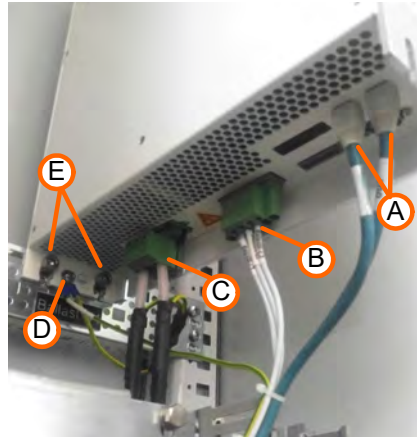


Figure 10-40: Cables to be Disconnected & Bolts to be removed or loosened

Step 3. Disconnect the cables marked **A**.

Step 4. Using an appropriate screwdriver, loosen the screws on both sides of the green connector marked **B** and pull off the connector.

Step 5. Do the same for the green connector marked **C**.

Step 6. Using an appropriate wrench, remove the nut holding the two ground wires marked **D** and remove the ground wires from the bolt.

Step 7. Using an appropriate wrench, loosen the bolts marked **E**.

Step 8. At the top of the Ballast, using an appropriate wrench, loosen the bolts marked **F**.

Step 9. Grasp the Ballast with both hands and push it up so that the mounting holes move over the bolts and pull it out. Place it on a stable surface.

Step 10. Lift the replacement Ballast with both hands and position it so that the mounting holes are over the bolts. Push it onto the bolts and over them.

Step 11. Using an appropriate wrench, tighten the bolts marked **E** and **F**.



Figure 10-41: A Ballast (Power Supply)

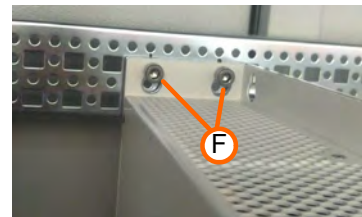


Figure 10-42: A Bolts at the top of the Ballast (Power Supply)

Configuring the Fieldbus Settings

Step 12. Using a small flat screwdriver, remove the cover marked **G**. The three Fieldbus Settings are exposed.

Step 13. Set the Fieldbus marked **H** to the number assigned to this Ballast as marked under the Ballast.

Step 14. Set the Fieldbus marked **I** to **0**.

Step 15. Set the Fieldbus marked **J** to **9**.

Step 16. Return the cover to its place and snap it in.

Step 17. Reconnect the ground wires marked **D** in **Figure 10-40** on page **138** and fasten the nut tightly.

Step 18. Reconnect the ground wires marked **D** in **Figure 10-40** on page **138** and fasten the nut tightly.

Step 19. Reconnect the green connectors marked **B** and **C** respectively in **Figure 10-40** on page **138** and fasten the screws on each side.

Step 20. Reconnect the wires marked **A**.



Figure 10-43: Fieldbus Settings

Step 21. In the EC-1 RZM Electrical Cabinet or EC-2 RZM Electrical Cabinet, turn the main circuit breakers to the **On** position.

Step 22. In the Main Control Application, follow the procedure in the section, **Reactivating the Lamp** on page **120**, to reactivate all the lamps. **Record the maintenance tasks in the Ballast Water Record Book.**

10.13 Cleaning/Replacing the Electrical Cabinet Air Filters



Every three months (on a quarterly basis), the Electrical Cabinet air filters are to be cleaned or replaced as needed. The air filters must be replaced once a year regardless of their state.

The Electrical Cabinet door includes 6 air filters. Two filters (A) are located on the upper part and four air filters (B) are located on the outer side of the fans on the lower part of the door.



Figure 10-44: Door of EC

➔ To replace an Electric Cabinet filter:

Step 1. Using a small, flat screwdriver on the lower side of the plastic grill, press in and disengage the locking tab. Pull the grill off of its frame.

Step 2. Remove the filter from the grill (B).

Step 3. Use the old filter as a template to cut a new one from your stock of new filter material.

- Alternatively, wash the filter under water with a mild detergent and a soft brush.
- Let the filter dry. **Make sure it is completely dry before reinstalling it.**

Step 4. Making sure that the arrows (C) on the side of the filter are pointing as shown in **Figure 10-46**, insert the clean filter in the grill and replace the grill into the door panel.

Step 5. Close the grill and push it in until it snaps into place.

Step 6. Repeat this procedure for each of the air filters.



Figure 10-45: Accessing the Air Filter

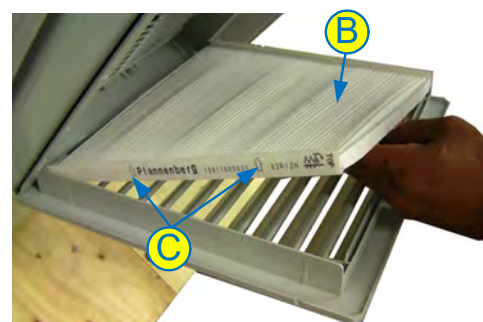


Figure 10-46: Accessing the Air Filter

11 Troubleshooting

11.1 Viewing System Errors

The sections headers in this chapter are comprised of message headers that appear as error messages on the Event log screen of the **Control Module**.

ID	Description	Date	Time
1	Dose above maximum defined level [73]	01/23/2019	09:26:09 AM
20	System ignition stage completed [72]	01/23/2019	09:25:39 AM
61	System ready [71]	01/23/2019	09:25:38 AM
18	System tuned On automatically: power restored. [70]	01/23/2019	09:25:21 AM
101	Communication restored [69]	01/23/2019	09:25:21 AM
65	Controller turned On [68]	01/23/2019	09:25:21 AM
1	Dose above maximum defined level [67]	01/22/2019	04:27:44 PM
20	System ignition stage completed [66]	01/22/2019	04:27:14 PM

Figure 11-1: Event log screen

11.2 Troubleshooting issues

The temperature is too high on page 152

Ballast communication error on page 158

Temperature is critically high! on page 152

The UV Lamp unexpectedly shut down on page 157

The UV Lamp unexpectedly shut down on page 157

The rate of the water flow is too low / too high on page 154

CRC error on page 157

UVT is too low! on page 154

UV Lamp ignition failed! on page 155

Control Module power failure on page 156

Each event/alarm provides the following additional information:

- System power

- System dose
- System flow
- System status
- System UVT
- System Temperature

See *Lamp Failure codes in Technician screen 3* on page [77](#).

Table 11-1: Voltage Error Codes and Troubleshooting procedures

Table 11-2: Lamp voltage value when lamp is off = Error Code	Table 11-3: Lamp turn off reason	Table 11-4: Probable Cause	Table 11-5: Troubleshooting
0 (± 24)	Unknown	Input voltage at one of the three phases is low	1 At the output side of the ballast's Circuit Breaker/fuses, check that the input voltage of the ballast is in the proper range at all three phases. Make sure that the ballast harness and input wires are properly secured tightly.
			2 If the voltage is within range, and the wires are secured tightly, try to reignite the lamp. If the lamp reignites, wait 10 minutes and see if the ballast is working properly.
		Internal fault at the ballast	3 If the problem reappears, or the lamp does not ignite, replace the ballast.
			Problem in all ballasts at the same time: 1 Try to locate the point where the voltage is low in the input phases feeding the ballasts. 2 If all of the voltage lines are in range, check for the source of the input voltage sags and stabilize the voltage.
50 (± 24)	Lamp nominal voltage too low	input voltage is too high	1 Check that the input voltage of the ballast is in the proper range and if yes try to reignite lamp. If the lamp reignites wait 10 minutes and see if the ballast is working properly.

Table 11-1: Voltage Error Codes and Troubleshooting procedures

Table 11-2: Lamp voltage value when lamp is off = Error Code	Table 11-3: Lamp turn off reason	Table 11-4: Probable Cause	Table 11-5: Troubleshooting
-		faulty ballast	2 If problem appears again or lamp does not start properly replace the ballast.
		faulty lamp	3 If the problem is not fixed replace the lamp.
100 (± 24)	Lamp nominal voltage too high	faulty lamp	If the problem is not fixed replace the lamp.
150 (± 24)	Device cooling *	Room's ambient temperature is above 45°C The ballast fan is faulty or faulty ballast	1 Check the room temperature. If it is too high, reduce the temperature via external means.
			2 Reignite the lamp and check that the ballast external fan is working. If not, try to clean it or replace it. If it is still not working replace the ballast
200 (± 24)	Lamp Internal failure / Phase lost	One phase of the supply is down due to a ballast CB / Fuse fault or wire disconnection at the input of the ballast	Problem in one ballast

Table 11-1: Voltage Error Codes and Troubleshooting procedures

Table 11-2: Lamp voltage value when lamp is off = Error Code	Table 11-3: Lamp turn off reason	Table 11-4: Probable Cause	Table 11-5: Troubleshooting
250 (± 24)	Lamp earth short circuit	One of the lamp's wires is touching the ground	1 Disconnect the faulty ballast from the supply by switching off its switch. Disconnect the Ballast wire connector and check using a DVM if there is short circuit between one of the lamp phases to the ground. If a short circuit is detected, locate the short circuited segment (junction box to ballast or junction box to lamp) and replace it.
300 (± 24)	Lamp not ignited / gone out	Ballast cable fault	1 Inspect the ballast cable & connectors between the electrical junction box and ballast.
		Faulty ballast/ Faulty lamp	2 Switch the cables between the ballast of the faulty lamp and the nearby ballast of a working lamp. If lamp doesn't work, replace lamp. If lamp works, replace the Ballast.
		Lamp connector	3 Check the lamp connector and if damaged, replace it and the lamp.

Table 11-1: Voltage Error Codes and Troubleshooting procedures

Table 11-2: Lamp voltage value when lamp is off = Error Code	Table 11-3: Lamp turn off reason	Table 11-4: Probable Cause	Table 11-5: Troubleshooting
350 (± 24)	Lamp short circuit	Cable fault	1 Disconnect the ballast and check that there is no short circuit between lamps wires
		Faulty ballast	2 Replace the ballast and check again. If the new ballast works, it means the cause was an internal ballast short circuit. Reduce the number of lamp ignitions per ballast to not more than six ignitions per day.
		Water inside the lamp sleeve	3 Replace the sleeve's O-rings
400 (± 24)	LAMP X MAINS UNDERVOLTAGE	Input voltage of the ballast is low due to a ballast CB fault or wire disconnection at the input of the ballast	Problem in one ballast
		Low voltage or loss connection	1 At the output side of the ballast's Fuses (inside the Junction Box) , check that the input voltage is according to the allowed range at all three phases. Make sure that the ballast harness and input wires are properly secured tightly.
	Phase failure mains voltage		2 If the voltage is valid and the wires are secured tightly, try to reignite lamp. If the lamp reignites, wait 10 minutes and see if the ballast is working properly.
		Faulty ballast	3 If problem appears again, or the lamp does not ignite, replace the ballast.
			Problem in all ballasts at the same time:

Table 11-1: Voltage Error Codes and Troubleshooting procedures

Table 11-2: Lamp voltage value when lamp is off = Error Code	Table 11-3: Lamp turn off reason	Table 11-4: Probable Cause	Table 11-5: Troubleshooting
		Low voltage	1 Try to locate the point where the voltage is low in the input phases feeding the ballasts.
			2 If all of the voltage lines are in range, check for the source of the input voltage sags and stabilize the voltage.

Table 11-1: Voltage Error Codes and Troubleshooting procedures

Table 11-2: Lamp voltage value when lamp is off = Error Code	Table 11-3: Lamp turn off reason	Table 11-4: Probable Cause	Table 11-5: Troubleshooting
450 (± 24)	Mains over-voltage	Input voltage is too high (above 565V) or unbalanced voltage supply	Problem in one ballast
		Voltage out of range	1 At the output side of the ballast's Fuses (inside the Junction Box) check that the input voltage of the ballast is in the proper range at all three phases.
			2 If the voltage is valid and the wires are fixed, try to reignite the lamp. If the lamp reignites, wait 10 minutes and see if the ballast is working properly.
		Faulty ballast	3 If the problem appears again at just one ballast, or the lamp does not start properly, replace the ballast.
			Problem in all ballasts at the same time:
		Unbalanced voltage supply	1 Verify that the the voltage supply is balanced (the three phases to PE voltages are equal $\pm 10\%$). If not create balanced supply by isolation transformer
		Voltage out of range	2 Check for the source of the input high voltage and stabilize the voltage.

Table 11-1: Voltage Error Codes and Troubleshooting procedures

Table 11-2: L amp voltage value when lamp is off = Error Code	Table 11-3: Lamp turn off reason	Table 11-4: Probable Cause	Table 11-5: Troubleshooting
500 (± 24)	Manual switch off		NA
600 (± 50)	Loss of ballast's internal parameters***		NA

* Voltage of lamp is low while the current to the lamp is at the maximum allowed value – in this state, the required power cannot be delivered to the lamp.

** The voltage required by the lamp cannot be created by the ballast. The current going to the lamp is zero.

*** In case of loss of internal ballast parameters, consult with Atlantium service personnel for instructions on how to solve this issue.

attention to the following safety notes when reviewing the troubleshooting issues:



The solution to the problem might involve exposure to UV light.

The system generates ultraviolet (UV) light within the Atlantium Unit, which can cause serious eye damage or blindness if you stare at it directly when it is working.

Do not look directly into the lamp enclosure during system operation, or while examining or servicing the system's internal components when the system is operating, or when a lamp's breather caps are open.



The solution to the problem might involve working in a high-voltage environment.

Before replacing a UV lamp or other components, and during any maintenance requiring lamps to be turned off, **turn the Electrical Cabinet BM main circuit breaker to the Off position.**

Place a sign on the relevant switch and **Control Module** alerting others **NOT** to touch the switch and touch screen during maintenance so that no one can inadvertently turn it on while maintenance is in progress.



The solution to the problem might involve working with hot surface.

Do not touch the UV lamp with your bare hands. Lamps can reach a temperature of 1,000°C under operating conditions.

Wait at least 10 minutes until the lamp is cooled down before replacing it.

Use appropriate protective gloves both to protect your hands and to avoid skin oils that leave fingerprints and/or harm the UV lamp.

11.3 The dose is above the maximum dose

Event Index	Control Module displays this warning message
1	The dose is above the maximum defined in the system settings

	Possible Cause	Solution
1	The current flow rate is too low. An upstream process has obstructed the flow rate.	Check the upstream piping and the components installed there, such as the pumps, filters.
2	The Max Dose parameter is set too low.	In the Control Module , raise the Max Dose setting. See <i>Configuring Dose Settings</i> on page 84.
3	The system is set for a constant power that is too high given the current water flow.	In the Control Module , change the Operation Mode from Power to Dose setting. See <i>Configuring Dose Settings</i> on page 84.

11.4 The UV dose is too low

Event Index	Control Module displays this warning message
2	Dose too low! Below minimum!

!

	Possible Cause	Solution
1	The Water flow rate is too high.	Check the water flow on the Control Module's main screen. If it is higher than normal, decrease the rate of the water flow through the system. See <i>Configuring the Flow Settings</i> on page 71.
2	UVT has declined because of chemicals (chlorine, etc.) or solids (sand, silt, etc.) in the water flow.	Carry out cleaning procedures. See <i>Cleaning In Place (CIP)</i> on page 109. Try to improve the water condition. Check on the status of upstream filtering equipment and upstream chemical processes.
3	UV lamp efficiency has dropped.	 Check the UVIS sensors and clean them if necessary. Check the lamp's efficiency reading. If it is low, replace the lamp. See <i>Lamp & UVIS Assembly</i> on page 115.
4	UVT has declined because of chemicals (chlorine, etc.) or solids (sand, silt, etc.) in the water flow.	Carry out cleaning procedures. See <i>Cleaning In Place (CIP)</i> on page 109. Try to improve the water condition. Check on the status of upstream filtering equipment and upstream chemical processes.
5	The UVT sensor has malfunctioned.	Switch between the UVT and UVIS sensors and see if the reading increased. If the reading is increased, switch the sensors back and replace the UVT sensor. See <i>Replacing/Checking a UVT Analyzer or UVIS Sensor</i> on page 128.
6	The system is operating on Power mode instead of Dose mode .	If the system settings are in Power mode, in the Control Module , access the Settings screen and change the Operation Mode from Power to Dose setting. Or, change the Power mode settings to allow higher power to respond to operating conditions. See <i>Configuring the Operation Mode</i> on page 70.

11.5 The temperature is too high

Event Index	Control Module displays this warning message
4	The lamp's temperature getting too high – check water flow

	Possible Cause	Solution
1	The Max temperature value is too low.	Check the system's Max Temperature settings. See <i>Configuring the General & Connection Settings</i> on page 87.
2	The water flow has significantly decreased.	Check the flow reading in the Control Module to see if it has deviated from normal flow. Check the flow via the flow meter. Check the water source for a problem.
3	There are air bubbles or the system is not fully filled with water.	Verify this by looking through the Atlantium Unit's view port.
4	You did not install a designated air release valve at the highest point on the inlet side of the system.	Release the air by installing a designated air release valve at the highest point on the inlet side of the system.

11.6 Temperature is critically high!

Event Index	Control Module displays this warning message
5	The lamp's temperature is too high and the unit will shut down

	Possible Cause	Solution
1	If this message appeared right after a BM Communication Error message (and both messages refer to the same lamp) and the temperature display on the monitor is 165°C/329°F, then most likely the communication between the Control Module and the Connection box is lost.	Follow the solutions offered in <i>CRC error</i> on page 157.
2	The temperature reading dropped to 0°C / 32°F due to a short-circuit in the Temperature sensor or Connection Box Card	 Replace the temperature sensor. See <i>Replacing the Temperature Sensor</i> on page 130. If the problem still persists replace the Connection Box card. Contact your Atlantium Service representative.
3	The Max temperature value is too low.	Check the system's Max Temperature settings. See <i>Configuring the Technician Settings</i> on page 74.

	Possible Cause	Solution	
4	There is no water flow.	Check the flow reading in the Control Module's main screen to see if it has deviated from normal flow. Check the flow via the flow meter. Check the water source for a problem.	
5	There are air bubbles or the system is not filled with water to capacity.	Verify this by looking through the Atlantium Unit's view port.	
6	You did not install a designated air release valve at the highest point on the inlet side of the system.	Release the air by installing designated air release valve at the highest point on the inlet side of the system.	
7	The Temperature sensor is faulty.	 Replace the temperature sensor. See Replacing the Temperature Sensor on page 130.	

11.7 Temperature sensor fault

Event Index	Control Module displays this warning message
6	One of the temperature sensors has malfunctioned or the communication with the Connection Box card was lost.

	Possible Cause	Solution
1	If this message appeared right after a BM Communication Error message (and both messages refer to the same lamp) and the temperature display on the monitor is 0°C/32°F, then most likely the communication between the Control Module and the Connection box is lost.	Follow the solutions offered in CRC error on page 157.
2	The temperature reading dropped to 0°C / 32°F due to a short-circuit in the Temperature sensor, or Connection Box Card	 Replace the temperature sensor. See Replacing the Temperature Sensor on page 130. If the problem still persists replace the Connection Box card. Contact your Atlantium Service representative.

11.8 The rate of the water flow is too low / too high

Event Index	Control Module displays this warning message
10	The water flow rate is higher than the maximum flow rate defined in the system settings. Check the water flow. System turned off automatically - water flow too low/high

	Possible Cause	Solution
1	The water flow rate into the system is too high.	Decrease the rate of water flow from the facility to the Atlantium Unit.
2	Maximum allowable flow rate parameter is too low.	Increase the setting of the Maximum allowable flow rate parameter to the maximum allowable flow rate.

11.9 UVT is too low!

Event Index	Control Module displays this warning message
11	The UVT is lower than the minimum UVT defined in the system settings.

	Possible Cause	Solution
1	Minimum allowable UVT parameter is too high.	Decrease the setting of the Minimum allowable UVT parameter to the minimum allowable UVT.
2	The UVT sensor's lens is fouled.	 Pull out the UVT sensor and see if the sensor's lens is foul or damaged. Clean the lens with a dry lens cloth. If it is damaged, replace the sensor with a new one and perform System Tuning according.
3	Deposit has collected in the UV chamber or lamp's quartz sleeve.	Look through the Atlantium Unit's view port. If the color of the light is in a shade of green or yellow, it probably indicate there is deposit present. Perform CIP. <i>Cleaning In Place (CIP)</i> on page 109
4	There are air bubbles or the system is not filled with water to capacity.	Verify this by looking through the Atlantium Unit's view port.

	Possible Cause	Solution	
5	You did not install a designated air release valve at the highest point on the inlet side of the system.	Install air release valve at the highest point on the inlet side of the system.	
6	There is a leak inside the UV chamber		Look through the Atlantium Unit's view port and check if there is condensation on the inner side of the view port and/or if there is water collected between the quartz tube sleeve and the stainless steel underneath. if you see this, contact Atlantium support for further instructions.
7	The UVT sensor has malfunctioned.		Switch between the UVT and UVIS sensors and see if the reading increased. If the reading is increased, switch the sensors back and replace the UVT sensor. See page See <i>Replacing/Checking a UVT Analyzer or UVIS Sensor</i> on page 128.

11.10 UV Lamp ignition failed!

Event Index	Control Module displays this warning message
14	Lamp ignition failed Check if it needs to be replaced, or if there is a faulty connection or ballast malfunction.



For Ballast version C, check the Voltage value error code. See *The dose is above the maximum dose* on page 150.

	Possible Cause	Solution	
1	The system is set to the wrong model.	In the Control Module , access Technician screen #2 . Verify that the system is set to the correct model, number of lamps and branches. See <i>Configuring the Technician Settings</i> on page 74.	
2	One or more connectors are burnt out.		Turn off the system's main power source On each Connection Box, turn the switch to the Off position. Disconnect and thoroughly inspect the Ballast connectors. IF replacement is needed, contact Atlantium Technical Support.
3	The power supply inside the Ballast is faulty.	 	In a system with more than one lamp, if possible, switch between the Ballast connectors in order to verify the Ballast is faulty, if so replace the Ballast . In a system with only one lamp, replace the Ballast a new one. See <i>Connecting the Cables to the Electrical Cabinet</i> on page 61.

	Possible Cause	Solution	
4	A lamp is faulty/burned out.	 	In a system with more than one lamp, switch between the lamps in order to verify the lamp is faulty, if so replace the lamp. In a system with only one lamp, replace the lamp with a new one. See <i>Replacing a UV Lamp</i> on page 139.
5	The ID number of the Connection Box card does not match ID number of the Ballast .		On each Connection Box, open the top cover and locate the 3-digit display (as shown on the right). Check that the set number is identical to the number marked on corresponding Ballast . If not identical, use the switch to change it to the correct number. 
6	One or more of the fuses in the Connection Box is burned out.		Replace the burned out fuse in the Connection Box. Contact Atlantium support for further instructions.
7	The Connection Box card is faulty.		On the Connection Box, open the top cover and check that the 3-digit display and LEDs are lit. If they are dark, replace the Connection Box card. Contact Atlantium support for further instructions.

11.13 Control Module power failure

Event Index	
39	The system turned off due to Control Module power failure.

	Possible Cause	Solution	
1	No power is feeding into the Connection Box.		Open the Connection boxes and verify that all Connection Boxes have power. See Connection Box Connections Diagram provided.
2	A wire is loose.		With a small flat screwdriver go over all the communication screws in the Control Module and Connections Box s and make sure all wires are located properly in their place and are well tightened. <i>Control Module Connections</i> on page 56
3	A jumper is missing.	A communication jumper might need to be installed, contact Atlantium Support for further instruction.	
4	Ground is not connected to the Control Module's power inlet.		A certified electrician is required to verify that the Control Module is properly grounded. <i>Control Module Connections</i> on page 56
5	There is a Hardware problem in the Control Module's I/O (ISO) or in the Connection Box.	Contact Atlantium support for further instructions.	

11.14 The UV Lamp unexpectedly shut down

Event Index	Control Module displays this warning message
43	A lamp shutdown unexpectedly.



For Ballast version C, check the Voltage value error code. See *The dose is above the maximum dose* on page 150.

	Possible Cause	Solution
1	A short power failure (up to 15 seconds) to the Atlantium Unit only, while the Control Module still had power.	After 3 minutes, turn the lamps back on using the On/Off button on the Control Module's main screen.
2	Ballast, lamp, connection failure	Check for faulty connection, ballast or lamp.



This situation can happen only in installations where the **Control Module** and the Atlantium Unit have separate power sources.

11.15 CRC error

Event Index	Event
44	Cyclic Redundancy Check error. Check if the parameters and counters have been reset to factory defaults. If so, contact Atlantium Technical Support.

	Possible Cause	Solution
1	While writing parameters into the Control Module's flash storage, the system encountered a problem due to an unexpected power failure or software upgrade.	Reenter all system parameters from scratch. Contact Atlantium support for further instructions. See <i>Configuring the General & Connection Settings</i> on page 87.



For Ballast version C, check the Voltage value error code. See *The dose is above the maximum dose* on page 150.

11.16 Ballast communication error

Event Index	Control Module displays this warning message
46	There is lost communication between one of the Connection Boxes and the Control Module . The system is shutting down.
69	

	Possible Cause		Solution
1	No power is feeding into the Electrical Cabinet.		Open the Electrical Cabinet and verify that all Connection Boxes have power.
2	A wire is loose.		With a small flat screwdriver go over all the communication screws in the Control Module and the Electrical CabinetBM and make sure all wires are located properly in their place and are well tightened. Control Module Connections on page 56
3	A jumper is missing.		A communication jumper might need to be installed, contact Atlantium Support for further instruction.
34	Ground is not connected to the Control Module's power inlet.		A certified electrician is required to verify that the Control Module is properly grounded. Control Module Connections on page 56
45	There is a Hardware problem in the Control Module's I/O (ISO) or in the Connection Box.		Contact Atlantium support for further instructions.

11.17 Tuning process aborted by system

Event Index	Control Module displays this warning message
100	Tuning process aborted by system

	Possible Cause	Solution
1	Communications fault	Check the communications cables and cable connections between the Control Module and other equipment. Restart the Control Module . If the problem persists, contact Atlantium Technical Support.

Appendix A Modbus Communication Protocol

Modbus is an application layer messaging protocol for client/server communication between devices connected on different types of buses or networks.

Modbus is often used to connect a supervisory computer with a Remote Terminal Unit (RTU) in supervisory control and data acquisition systems.

For information on how to configure the communication properties, see *Modbus Communication Connections* on page 67.

➤ Physical connection to Modbus

Physical connection to Modbus shall be performed by a technician. Ask your Atlantium representative for details.

➤ To set Modbus connection parameters in Control Module:

Step 1. Set Modbus address to 1

Step 2. Set baud rate to 19200

The installation technician will set the IP address.

A.1 Modbus Registry Map

The table on the following pages provides the information you need to help you configure your network to properly communicate with the Atlantium system.

ATTENTION



- Modbus registers are meant to be used for specific commands in a command algorithm. Write only once to any register and await response.
- Registers 194-224 are 'Read' Only
- All of the registers have a length of Unsigned 16 bits, even the Coils Bits
- To read input from a register, use Function Code 4 (Read input registers)
- To write output to a register, use Function Code 6 (Write output single register).

Table A-1: Modbus Registry Map (*) R = Read Only RW = Read/Write

Register address	Register name	Divide	Type	Comment	Units	Read/Write
0	Min Dose	:10		Modbus-Up to 65530 AIO-Up to 9999	mJ/cm ²	R
1	Max Dose	:1		Modbus-Up to 65530 AIO-Up to 9999	mJ/cm ²	R
2	Set For	:10		Modbus-Up to 65530 AIO-Up to 9999	mJ/cm ²	R
3	System Unit format		Coil Bit	0 - Imperial, 1 - Metric		R

Table A-1: Modbus Registry Map (*) R = Read Only RW = Read/Write

Register address	Register name	Divide	Type	Comment	Units	Read/Write
4	Flow in FS mode	:1		Min. Val = 1 , Max.Val = 30000 - Display according to format set on HMI (m³/h or gal/min)	m³/h	R/W
5	Min Flow	:100		Min. Val = 1 , Max.Val = 300 - Display according to format set on HMI (m³/h or gal/min)	m³/h	R/W
6	Max Flow	:1		Min. Val = 1 , Max.Val = 30000 - Display according to format set on HMI (m³/h or gal/min)	m³/h	2.1.1-B43
7	Operation Mode		Coil Bit	0: Power Mode, 1: Dose Mode		R/W
8	System Lamps status	:1		Average of all lamps status	%	R
204	Dose	:1		Actual dose	mJ/cm²	R
205	System On/Off		Coil Bit	0 - System OFF, 1 - System ON		R/W
227	UVT	:100		Actual UVT	%	R
228	Power	:1		Actual power	%	R
229	Flow	:1		Actual flow vlaue - Display according to format set on HMI (m³/h or gpm) - Converted from gpm to m³/h while register 3 =0	m³/h	R

Table A-1: Modbus Registry Map (*) R = Read Only RW = Read/Write

Register address	Register name	Divide	Type	Comment	Units	Read/Write
194	System Temperature	:100		Converted from °F to °C while register 3 =0	°C	R
32	Set Power	:1		Set the power for power mode	%	R/W
Lamp1						
35	UVIS	:100			4-20 mA	R
36	UVTS A	:100			4-20 mA	R
37	UVTS B	:100			4-20 mA	R
38	Lamp Current	:100			Amper	R
39	Lamp Voltage	:10			Voltage	R
40	Lamp On		Coil Bit	1 - Lamp is ON , 0 - Lamp is OFF		R
42	Lamp UVT A / UVT	:100		While LW checked, register will show UVT lamp	%	R
43	Lamp Status	:100			%	R
44	Lamp UVT B / LW	:100		While LW checked, register will show LW lamp	%	R

Table A-1: Modbus Registry Map (*) R = Read Only RW = Read/Write

Register address	Register name	Divide	Type	Comment	Units	Read/Write
46	Lamp Age	:1			Hour	R
47	Lamp Ignitions	:1				R
51	Lamp Enable		Coil Bit	0 - Disable, 1 - Enable		R/W
52	Turn On/Off		Coil Bit	1 - Turn ON , 0 - Turn OFF		R/W
53	Lamp state			0 -> lamp OFF 1 -> Lamp Ignition, 2 -> Lamp On		R
Lamp 2						
54	UVIS	:100			4-20 mA	R
55	UVTS A	:100			4-20 mA	R
56	UVTS B	:100			4-20 mA	R
57	Lamp Current	:100			Amper	R
58	Lamp Voltage	:10			Voltage	R
59	Lamp On		Coil Bit	1 - Lamp is ON , 0 - Lamp is OFF		R

Table A-1: Modbus Registry Map (*) R = Read Only RW = Read/Write

Register address	Register name	Divide	Type	Comment	Units	Read/Write
61	Lamp UVT A / UVT	:100		While LW checked, register will show UVT lamp	%	R
62	Lamp Status	:100			%	R
63	Lamp UVT B / LW	:100		While LW checked, register will show LW lamp	%	R
65	Lamp Age	:1			Hour	R
66	Lamp Ignitions	:1				R
70	Lamp Enable		Coil Bit	0 - Disable, 1 - Enable		R/W
71	Turn On/Off		Coil Bit	1 - Turn ON , 0 - Turn OFF		R/W
72	Lamp state			0 -> lamp OFF 1 -> Lamp Ignition, 2 -> Lamp On		R
Lamp 3						
73	UVIS	:100			4-20 mA	R
74	UVTS A	:100			4-20 mA	R
75	UVTS B	:100			4-20 mA	R

Table A-1: Modbus Registry Map (*) R = Read Only RW = Read/Write

Register address	Register name	Divide	Type	Comment	Units	Read/Write
76	Lamp Current	:100			Amper	R
77	Lamp Voltage	:10			Voltage	R
78	Lamp On		Coil Bit	1 - Lamp is ON , 0 - Lamp is OFF		R
80	Lamp UVT A / UVT	:100		While LW checked, register will show UVT lamp	%	R
81	Lamp Status	:100			%	R
82	Lamp UVT B / LW	:100		While LW checked, register will show LW lamp	%	R
84	Lamp Age	:1			Hour	R
85	Lamp Ignitions	:1				R
89	Lamp Enable		Coil Bit	0 - Disable, 1 - Enable		R/W
90	Turn On/Off		Coil Bit	1 - Turn ON , 0 - Turn OFF		R/W
91	Lamp state			0 -> lamp OFF 1 -> Lamp Ignition, 2 -> Lamp On		R
Lamp 4						

Table A-1: Modbus Registry Map (*) R = Read Only RW = Read/Write

Register address	Register name	Divide	Type	Comment	Units	Read/Write
92	UVIS	:100			4-20 mA	R
93	UVTS A	:100			4-20 mA	R
94	UVTS B	:100			4-20 mA	R
95	Lamp Current	:100			Amper	R
96	Lamp Voltage	:10			Voltage	R
97	Lamp On		Coil Bit	1 - Lamp is ON , 0 - Lamp is OFF		R
99	Lamp UVT A	:100			%	R
100	Lamp Status	:100			%	R
101	Lamp UVT B	:100			%	R
103	Lamp Age	:1			Hour	R
104	Lamp Ignitions	:1				R
108	Lamp Enable		Coil Bit	0 - Disable, 1 - Enable		R/W
109	Turn On/Off		Coil Bit	1 - Turn ON , 0 - Turn OFF		R/W

Table A-1: Modbus Registry Map (*) R = Read Only RW = Read/Write

Register address	Register name	Divide	Type	Comment	Units	Read/Write
110	Lamp state			0 -> lamp OFF 1 -> Lamp Ignition, 2 -> Lamp On		R
Lamp 5						
111	UVIS	:100			4-20 mA	R
112	UVTS A	:100			4-20 mA	R
113	UVTS B	:100			4-20 mA	R
114	Lamp Current	:100			Amper	R
115	Lamp Voltage	:10			Voltage	R
116	Lamp On		Coil Bit	1 - Lamp is ON , 0 - Lamp is OFF		R
118	Lamp UVT A	:100			%	R
119	Lamp Status	:100			%	R
120	Lamp UVT B	:100			%	R
122	Lamp Age	:1			Hour	R
123	Lamp Ignitions	:1				R

Table A-1: Modbus Registry Map (*) R = Read Only RW = Read/Write

Register address	Register name	Divide	Type	Comment	Units	Read/Write
127	Lamp Enable		Coil Bit	0 - Disable, 1 - Enable		R/W
128	Turn On/Off		Coil Bit	1 - Turn ON, 0 - Turn OFF		R/W
129	Lamp state			0 -> lamp OFF 1 -> Lamp Ignition, 2 -> Lamp On		R
Lamp 6						
130	UVIS	:100			4-20 mA	R
131	UVTS A	:100			4-20 mA	R
132	UVTS B	:100			4-20 mA	R
133	Lamp Current	:100			Amper	R
134	Lamp Voltage	:10			Voltage	R
135	Lamp On		Coil Bit	1 - Lamp is ON, 0 - Lamp is OFF		R
137	Lamp UVT A	:100			%	R
138	Lamp Status	:100			%	R
139	Lamp UVT B	:100			%	R

Table A-1: Modbus Registry Map (*) R = Read Only RW = Read/Write

Register address	Register name	Divide	Type	Comment	Units	Read/Write
141	Lamp Age	:1			Hour	R
142	Lamp Ignitions	:1				R
146	Lamp Enable		Coil Bit	0 - Disable, 1 - Enable		R/W
147	Turn On/Off		Coil Bit	1 - Turn ON , 0 - Turn OFF		R/W
148	Lamp state			0 -> lamp OFF 1 -> Lamp Ignition, 2 -> Lamp On		R
Lamp 7						
149	UVIS	:100			4-20 mA	R
150	UVTS A	:100			4-20 mA	R
151	UVTS B	:100			4-20 mA	R
152	Lamp Current	:100			Amper	R
153	Lamp Voltage	:10			Voltage	R
154	Lamp On		Coil Bit	1 - Lamp is ON , 0 - Lamp is OFF		R
156	Lamp UVT A	:100			%	R

Table A-1: Modbus Registry Map (*) R = Read Only RW = Read/Write

Register address	Register name	Divide	Type	Comment	Units	Read/Write
157	Lamp Status	:100			%	R
158	Lamp UVT B	:100			%	R
160	Lamp Age	:1			Hour	R
161	Lamp Ignitions	:1				R
165	Lamp Enable		Coil Bit	0 - Disable, 1 - Enable		R/W
166	Turn On/Off		Coil Bit	1 - Turn ON , 0 - Turn OFF		R/W
167	Lamp state			0 -> lamp OFF 1 -> Lamp Ignition, 2 -> Lamp On		R
Lamp 8						
168	UVIS	:100			4-20 mA	R
169	UVTS A	:100			4-20 mA	R
170	UVTS B	:100			4-20 mA	R
171	Lamp Current	:100			Amper	R
172	Lamp Voltage	:10			Voltage	R

Table A-1: Modbus Registry Map (*) R = Read Only RW = Read/Write

Register address	Register name	Divide	Type	Comment	Units	Read/Write
173	Lamp On		Coil Bit	1 - Lamp is ON , 0 - Lamp is OFF		R
175	Lamp UVT A	: 100			%	R
176	Lamp Status	: 100			%	R
177	Lamp UVT B	: 100			%	R
179	Lamp Age	: 1			Hour	R
180	Lamp Ignitions	: 1				R
184	Lamp Enable		Coil Bit	0 - Disable, 1 - Enable		R/W
185	Turn On/Off		Coil Bit	1 - Turn ON , 0 - Turn OFF		R/W
186	Lamp state			0 -> lamp OFF 1 -> Lamp Ignition, 2 -> Lamp On		R
Valve						
200	Bypass Valve		Coil Bit	1 - Valve ON , 0 - Valve OFF		R
201	Inlet Valve		Coil Bit	1 - Valve ON , 0 - Valve OFF		R
202	Outlet Valve/ Ready		Coil Bit	1 - Valve ON , 0 - Valve OFF		R

Table A-1: Modbus Registry Map (*) R = Read Only RW = Read/Write

Register address	Register name	Divide	Type	Comment	Units	Read/Write
Alarms						
203	General Alarm		Coil Bit	1 - Valve ON , 0 - Valve OFF		R
206	Over dose		Coil Bit	1 - Valve ON , 0 - Valve OFF		R
207	Min Dose #2		Coil Bit	1 - Valve ON , 0 - Valve OFF		R
208	High Temp #4		Coil Bit	1 - Valve ON , 0 - Valve OFF		R
209	Over Temp #5		Coil Bit	1 - Valve ON , 0 - Valve OFF		R
210	Shut Down Alarm		Coil Bit	1 - Valve ON , 0 - Valve OFF		R
211	Low UVT		Coil Bit	1 - Valve ON , 0 - Valve OFF		R
212	Lost Comm		Coil Bit	1 - Valve ON , 0 - Valve OFF		R
213	Under flow		Coil Bit	1 - Valve ON , 0 - Valve OFF		R
214	High Flow #10		Coil Bit	1 - Valve ON , 0 - Valve OFF		R
224	Under dose set #41		Coil Bit	1 - Valve ON , 0 - Valve OFF		R

Appendix B System Messages

The table below details the message that can appear on the **Control Module's** screen or in the Event logs. A number of parameters are User Defined and include setting a value and upon reaching, exceeding or falling below it, an Action is set to be triggered. The Action is also configurable. The options are:

- **None** - The event is not considered an error, so no corrective action is taken. No warning message is issued.
- **Caution** - The caution alerts you to an anomaly based on the user preferences.
- **Warning** - One or more warning messages are issued and it triggers a pop-up window message and a General Alarm (if configured, see *Configuring the General Alarm* on page 73).
- **Shut-Down** - In case of a serious error or dangerous condition, the system shuts down and triggers a pop-up window message and a General Alarm (if configured, see *Configuring the General Alarm* on page 73).

The other parameters are system defined. The type column in the table below details the type of parameter.

Table B-1: **Control Module's** Messages

Key	Type	Event Log	Line Message/Popup Title	Popup Message
1	The value is User Defined	Over Max. Dose	The dose is above the maximum dose	The dose is above the maximum defined in the system settings.
2	The value is User Defined (The default is None)	Dose Too Low	The UV dose is too low	Dose too low! Below minimum!
4	Warning - The default temperature threshold is 5°C (9°F) below what is defined for Over Temp	Temp. Too High	Temperature is Too High	The lamp's temperature getting too high -- check water flow.
5	Shutdown - - The default temperature threshold is User Defined	Over Temp	Temperature is Critically High!	The lamp's temperature is too high and the unit will shut down.
6	Shutdown - Hardware malfunction	Temperature Sensor Fault	Temperature Sensor Fault	One of the temperature sensors has malfunctioned or the communication with the Connection Box card was lost.
9	Shutdown ---The Minimum Flow is User Defined	Low Flow	The rate of the water flow is too low	The water flow rate is lower than the minimum flow rate. Check the water flow.
10	The value is User Defined	High Flow	The rate of the water flow is too high	The water flow rate is higher than the maximum flow rate allowed. Check the water flow.
11	The value is User Defined	Transmission Low	UVT is too low!	UVT is below the required value

Table B-1: **Control Module's** Messages

Key	Type	Event Log	Line Message/Popup Title	Popup Message
14	Warning	Lamp Ignition	UV Lamp Ignition Failed!	The UV lamp has failed to ignite due to an electronic fault. The UV lamp may be burnt out or a wire may be disconnected.
16	Appears in the Event Log only - pops up from additional info from Event Log	User On		The Control Module screen or an external signal turned on the system.
17	Appears in the Event Log only - pops up from additional info from Event Log	User Off		The Control Module screen or an external signal shut down the system.
18	Appears in the Event Log only - pops up from additional info from Event Log	Auto On		System power was restored or flow raise above minimum. The lamps reignite.
19	Appears in the Event Log only - pops up from additional info from Event Log	Auto Off		Flow rate below user low flow setting; automatic shutdown.
20	Appears in the Event Log only - pops up from additional info from Event Log	Lamps On		System Ready. All the enabled lamps are on.
24	Caution	Dose On Spec	Dose On Spec	The UV dose being delivered by the system is above the minimum validated value
25	Warning	Dose Off Spec	Dose Off Spec	The UV dose being delivered by the system is below the minimum validated value
26	Caution	UVT On Spec	UVT On Spec	The UVT value is above the minimum validated UVT
27	Warning	UVT Off Spec	UVT Off Spec	The UVT value is below the minimum validated UVT
28	Caution	Flow On Spec	Flow On Spec	The flow rate is below the maximum validated flow
29	Warning	Flow Off Spec	Flow Off Spec	The flow rate is above the maximum validated flow
33	Warning	UVT 4th calibration failed	UVT Analyzer 4th calibration check failed	UVT Analyzer calibration check failed 4 times in a row. System is off-spec
34	The value is User Defined (Default as None)	Validated Min UVT	Validated Min UVT	UVT is below Validated Min UVT
35	The value is User Defined (Default as None)	Validated Max Flow	Validated Max Flow	Flow is above Validated Max Flow

Table B-1: **Control Module's** Messages

Key	Type	Event Log	Line Message/Popup Title	Popup Message
39	Appears in the Event Log only - pops up from additional info from Event Log	Power Failure	Power Failure	The system was turned off due to power failure
41	The value is User Defined	UV Dose	The UV Dose is below the set level	The UV Dose is below the Operate At dose
42	Warning - Sensor needs to be replaced or there may be a loose wire	Temp. Sensor Fault	Temperature Sensor Fault	Fault on lamp temperature sensor
43	Shutdown	Lamp Shutdown	The UV lamp unexpectedly shutdown	Lamp shutdown
44	Appears in the Event Log only - pops up from additional info from Event Log	CRC Error	CRC Error	System error parameters and counters might be reset to factory defaults. Please check
45	Shutdown	PS Fault	Power supply fault	Ballast fault [value displayed], error code: [number displayed].
46	Warning	Communication Error	Communication Error	Lost communication with the Ballast Module
49	Appears in the Event Log only - pops up from additional info from Event Log	UVIS calib info	UVIS calib info	UVIS calib lamp [value displayed]
50	Appears in the Event Log only - pops up from additional info from Event Log	UVT calib info	UVT calib info	UVT calib lamp [value displayed]
53	Appears in the Event Log only - pops up from additional info from Event Log	Settings Changed	Settings changed	Settings changed [value displayed]
54	Caution	UVIS is calibrated successfully	UVIS is calibrated successfully	UVIS is calibrated successfully
55	Caution	UVIS calibration failed	UVIS Calibration Failed	UVIS calibration failed, Sensor is out of calibration. Correction Factor is implemented
56	Caution	UVT successful calibration check	UVT successful calibration check	UVT Analyzer successful calibration check
57	Caution	UVT check failed	UVT Analyzer check failed	UVT Analyzer check failed; Auto recalibration was done

Appendix C Checking the System Parameters

You can use the checklist below to keep a record of your checks on Atlantium system parameters. **Print this page.**

Table C-1: Atlantium System Parameters Checklist

[illegible]

Appendix D Glossary of Terms

Term	Definition
Auxiliaries	Valves, flow devices, etc. essential to the workings of the Atlantium system but not directly a part of the Atlantium system.
Ballast	The power supply that translates the electrical power to UV power.
Connection Box	The box that sits on top of each lamp unit and holds the circuit board for the Electrical Cabinet or Ballast Module electrical configurations. Not relevant for Ballast Module/Cable-style.
Control Module	Control Module that controls operations and dose measurement. Comes in two versions: Standard and Premium. The Standard has a monochromatic screen and several key features. The Premium has a color screen can accommodate a full complement of reporting and trending features.
External On/Off	This enables an external source (SCADA, PLC, Process) to turn the Atlantium system on and off.
IP56	Protected from limited dust ingress. Protected from high pressure water jets from any direction
<i>Electrical configuration styles: Atlantium provides a choice of electrical configurations.</i>	
Electrical Cabinet-style (EC)	EC – Electrical Cabinet.
Ballast Module	Waterproof case for the ballast which connects via a consolidated plug from the circuit board which sits on top of each lamp unit.
General Alarm	This enables the Atlantium unit to send an alarm to a buzzer, cell phone or other destination.
Inlet/Outlet Valves	Isolating Valves on the Inlet/outlet sides.
Lamp Chamber	The chamber which holds the lamp.
Lifting Support	The lifting element which can also be used to help support the Unit.
Pinhole	This is a small disc that is fitted for each sensor port to ensure that the right amount of light penetrates.
Portable CIP System	The portable unit Atlantium sells that connects to the Unit to clean and disinfect it without disassembling. It includes several kits – a valve kit, an accessories kit and the CIP kit itself.
Quartz Chamber	The quartz pipe in which the UV exposure occurs.
System Ready Signal	The signal the Atlantium unit provides to tell the production system that the Atlantium System is ready to supply properly treated water.

Term	Definition
System Tuning	This is process is performed when key components are replaced or changed to synchronize them.
Triangle Support Brackets	RZ-Series only- brackets that can be used to support and balance a unit.
UV Intensity Sensor	The sensor that measures the UV intensity in the Lamp chamber and feeds the real time dose calculation.
UVT Analyzer	The sensor that measures the lamp UV intensity through the water. Together with the UV Intensity sensor and the Control Module, it analyzes, tracks and reports the water quality.

Appendix E Consumables and Spares



The Part Numbers and Descriptions of the Consumables & Spare Parts shown in these tables are subject to change.

Table E-1: Spares list for RZM200-5

Part Number	Description	QTY
Included and required		
MPE011300	O-RING 2-326 V0680-70 FOR RZM SLEEVE	10
OPM000099	UV LAMP 400mm 6.9kW LONG NECK	10
Recommended spare parts		
OPM000205	QUARTZ SLEEVE,RZM200-5	1
PS0000009	UV BALLAST, 6.9KW, 380-480V	1
AP0001600	RZM/EP600 UV UNIT CARD ASSY	1
SAM000600	MIRROR ASSY RZM	2
FNL002001	FAN, 24VDC, FOR RZM LAMP HOUSING	1
SAM000800	WAVE GUIDE ASSY	1
HSL010950	PT100 TEMPERATURE SENSOR, 1.1m	1
EL0000279	FUSE 2A, TIME-LAG, 5X20mm	10
EL0000280	FUSE 4A, TIME-LAG, 5X20mm	10